

MIDDLESEX SCHOOL
MATHEMATICS DEPARTMENT



Mathematics Curriculum

Math 12 through Math 49

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Math 12 Curriculum

Intermediate Algebra | Middlesex School Mathematics

COURSE AT A GLANCE

Review Content

REVIEW

Primary resources: [1.1 Use the Language of Algebra](#); [1.3 Fractions](#); [1.5 Properties of Real Numbers](#)
Supplemental resources: [1.2 Integers](#)

UNIT LEARNING OBJECTIVES

- I can evaluate numerical expressions using the order of operations.
- I can perform operations with signed integers and fractions.
- I can simplify an algebraic expression by combining like terms.
- I can use the distributive property to rewrite an expression.
- I can determine the prime factorization of a whole number.

Linear Equations and Applications

REQUIRED

Primary resources: [2.1 Use a General Strategy to Solve Linear Equations](#); [2.3 Solve a Formula for a Specific Variable](#); [2.2 Use a Problem Solving Strategy](#); [2.4 Solve Mixture and Uniform Motion Applications](#)
Supplemental resources: [2.2 Use a Problem Solving Strategy](#)

UNIT LEARNING OBJECTIVES

- I can solve a linear equation and present equivalent steps clearly.
- I can solve a linear equation containing fractions.
- I can rearrange a formula to isolate a specified variable.
- I can translate a verbal relationship into a linear equation.
- I can solve and interpret a number, mixture, or motion problem using a linear equation.

Inequalities and Absolute-Value Equations

REQUIRED

Primary resources: [2.5 Solve Linear Inequalities](#); [2.6 Solve Compound Inequalities](#); [2.7 Solve Absolute Value Inequalities](#)

UNIT LEARNING OBJECTIVES

- I can solve and graph a one-variable linear inequality.
- I can solve a compound inequality involving AND.
- I can solve a compound inequality involving OR.
- I can express an inequality solution using interval notation.
- I can solve an absolute-value equation and check its solutions.

Lines and Introductory Functions

REQUIRED

Primary resources: [3.2 Slope of a Line](#); [3.1 Graph Linear Equations in Two Variables](#); [3.3 Find the Equation of a Line](#); [3.5 Relations and Functions](#)

UNIT LEARNING OBJECTIVES

- I can determine and interpret the slope of a line.
- I can graph a line from an equation in a common form.
- I can write an equation of a line from given information.
- I can determine whether two lines are parallel, perpendicular, or neither.
- I can evaluate a function using function notation.
- I can determine whether a relation is a function using the vertical line test or input-output data.
- I can determine domain and range from a graph or table.
- I can interpret domain and range in context.

Systems of Linear Equations

REQUIRED

Primary resources: [4.1 Solve Systems of Linear Equations with Two Variables](#); [4.2 Solve Applications with Systems of Equations](#)

UNIT LEARNING OBJECTIVES

- I can interpret the solution of a linear system as an intersection point.
- I can solve a two-variable linear system by graphing.
- I can solve a two-variable linear system by substitution.
- I can solve a two-variable linear system by elimination.
- I can create and solve a two-variable linear system for an application.

Polynomial Operations and Basic Factoring

REQUIRED

Primary resources: [5.2 Properties of Exponents and Scientific Notation](#); [5.1 Add and Subtract Polynomials](#); [5.3 Multiply Polynomials](#); [5.4 Dividing Polynomials](#); [6.1 Greatest Common Factor and Factor by Grouping](#)

UNIT LEARNING OBJECTIVES

- I can simplify expressions using integer exponent rules.
- I can add and subtract polynomials.
- I can multiply polynomials, including binomials.
- I can divide a polynomial by a monomial.
- I can factor a greatest common factor from a polynomial.

Extension Content

EXTENSION

Primary resources: [Intermediate Algebra 2e 2.7 Solve Absolute Value Inequalities](#); [Intermediate Algebra 2e 4.7 Graphing Systems of Linear Inequalities](#); [Intermediate Algebra 2e 6.2 Factor Trinomials](#); [Precalculus 2e 1.2 Domain and Range](#)

UNIT LEARNING OBJECTIVES

- I can solve and graph an absolute-value inequality.
- I can graph a system of linear inequalities and identify its solution region.
- I can factor a quadratic trinomial.
- I can identify a relationship as discrete or continuous.

IN-DEPTH CURRICULUM

SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Review Content

REVIEW

OpenStax coverage: **Primary resources:** [1.1 Use the Language of Algebra](#); [1.3 Fractions](#); [1.5 Properties of Real Numbers](#)

Supplemental resources: [1.2 Integers](#)

M12-REV-001

I can evaluate numerical expressions using the order of operations.

OpenStax: **Primary resources:** [1.1 Use the Language of Algebra](#)

SUPPORTING SKILLS



Apply the order of operations to a numerical expression.

SK-NUM-ORDER-OPERATIONS · inherited from middle school

M12-REV-002

I can perform operations with signed integers and fractions.

OpenStax: **Primary resources:** [1.3 Fractions](#)

Supplemental resources: [1.2 Integers](#)

SUPPORTING SKILLS



Add, subtract, multiply, and divide fractions.

SK-NUM-FRACTION-OPERATE · inherited from middle school



Add, subtract, multiply, and divide signed integers.

SK-NUM-INTEGGER-OPERATE · inherited from middle school

M12-REV-003

I can simplify an algebraic expression by combining like terms.

OpenStax: **Primary resources:** [1.1 Use the Language of Algebra](#)

SUPPORTING SKILLS

- R** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- R** Identify and combine like terms.
SK-ALG-LIKE-TERMS · inherited from middle school

M12-REV-004

I can use the distributive property to rewrite an expression.

OpenStax: **Primary resources:** [1.5 Properties of Real Numbers](#)

SUPPORTING SKILLS

- R** Apply the distributive property in either direction.
SK-ALG-DISTRIBUTE · inherited from middle school

M12-REV-005

I can determine the prime factorization of a whole number.

OpenStax: **Primary resources:** [1.1 Use the Language of Algebra](#)

SUPPORTING SKILLS

- R** Express a whole number as a product of prime factors.
SK-NUM-PRIME-FACTOR · inherited from middle school

Linear Equations and Applications

REQUIRED

OpenStax coverage: **Primary resources:** [2.1 Use a General Strategy to Solve Linear Equations](#); [2.3 Solve a Formula for a Specific Variable](#); [2.2 Use a Problem Solving Strategy](#); [2.4 Solve Mixture and Uniform Motion Applications](#)
Supplemental resources: [2.2 Use a Problem Solving Strategy](#)

M12-EQN-001

I can solve a linear equation and present equivalent steps clearly.

OpenStax: **Primary resources:** [2.1 Use a General Strategy to Solve Linear Equations](#)

SUPPORTING SKILLS

- I** Check a proposed solution in the original equation.
SK-ALG-CHECK-SOLUTION
- A** Apply the distributive property in either direction.
SK-ALG-DISTRIBUTE · inherited from middle school
- I** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS
- A** Identify and combine like terms.
SK-ALG-LIKE-TERMS · inherited from middle school

M12-EQN-002

I can solve a linear equation containing fractions.

OpenStax: **Primary resources:** [2.1 Use a General Strategy to Solve Linear Equations](#)

SUPPORTING SKILLS

- A** Check a proposed solution in the original equation.
SK-ALG-CHECK-SOLUTION · first introduced in Math 12
- I** Clear fractions from an equation using a common denominator.
SK-ALG-CLEAR-FRACTIONS
- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- A** Add, subtract, multiply, and divide fractions.
SK-NUM-FRACTION-OPERATE · inherited from middle school

M12-EQN-003

I can rearrange a formula to isolate a specified variable.

OpenStax: **Primary resources:** [2.3 Solve a Formula for a Specific Variable](#)

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I** Isolate a specified variable in an equation or formula.
SK-ALG-VARIABLE-ISOLATE

M12-EQN-004

I can translate a verbal relationship into a linear equation.

OpenStax: **Primary resources:** [2.2 Use a Problem Solving Strategy](#)

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I** Translate quantities and relationships into an equation.
SK-ALG-TRANSLATE-EQUATION

I can solve and interpret a number, mixture, or motion problem using a linear equation.

OpenStax: **Primary resources:** [2.4 Solve Mixture and Uniform Motion Applications](#)

Supplemental resources: [2.2 Use a Problem Solving Strategy](#)

SUPPORTING SKILLS

- A** Check a proposed solution in the original equation.
SK-ALG-CHECK-SOLUTION · first introduced in Math 12
- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- D** Translate quantities and relationships into an equation.
SK-ALG-TRANSLATE-EQUATION · first introduced in Math 12

Inequalities and Absolute-Value Equations

REQUIRED

OpenStax coverage: **Primary resources:** [2.5 Solve Linear Inequalities](#); [2.6 Solve Compound Inequalities](#); [2.7 Solve Absolute Value Inequalities](#)

M12-INE-001

I can solve and graph a one-variable linear inequality.

OpenStax: **Primary resources:** [2.5 Solve Linear Inequalities](#)

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I** Represent an inequality solution on a number line.
SK-INE-NUMBER-LINE
- I** Reverse an inequality when multiplying or dividing by a negative value.
SK-INE-REVERSE-SIGN

M12-INE-002

I can solve a compound inequality involving AND.

OpenStax: **Primary resources:** [2.6 Solve Compound Inequalities](#)

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I** Interpret AND as the intersection of two inequality solution sets.
SK-INE-COMPOUND-AND
- A** Represent an inequality solution on a number line.
SK-INE-NUMBER-LINE · first introduced in Math 12
- A** Reverse an inequality when multiplying or dividing by a negative value.
SK-INE-REVERSE-SIGN · first introduced in Math 12

M12-INE-003

I can solve a compound inequality involving OR.

OpenStax: **Primary resources:** [2.6 Solve Compound Inequalities](#)

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I** Interpret OR as the union of two inequality solution sets.
SK-INE-COMPOUND-OR
- A** Represent an inequality solution on a number line.
SK-INE-NUMBER-LINE · first introduced in Math 12
- A** Reverse an inequality when multiplying or dividing by a negative value.
SK-INE-REVERSE-SIGN · first introduced in Math 12

M12-INE-004

I can express an inequality solution using interval notation.

OpenStax: **Primary resources:** [2.6 Solve Compound Inequalities](#)

SUPPORTING SKILLS

- A** Represent an inequality solution on a number line.
SK-INE-NUMBER-LINE · first introduced in Math 12
- I** Read inequality symbols and state the relationship each symbol expresses.
SK-NOT-INEQUALITY-READ
- I** Translate between inequality and interval notation.
SK-NOT-INTERVAL

M12-INE-005

I can solve an absolute-value equation and check its solutions.

OpenStax: **Primary resources:** [2.7 Solve Absolute Value Inequalities](#)

SUPPORTING SKILLS

- I** Rewrite an absolute-value equation as two cases.
SK-ABS-SPLIT-EQUATION
- A** Check a proposed solution in the original equation.
SK-ALG-CHECK-SOLUTION · first introduced in Math 12
- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12

Lines and Introductory Functions

REQUIRED

OpenStax coverage: **Primary resources:** [3.2 Slope of a Line](#); [3.1 Graph Linear Equations in Two Variables](#); [3.3 Find the Equation of a Line](#); [3.5 Relations and Functions](#)

M12-LIN-001

I can determine and interpret the slope of a line.

OpenStax: **Primary resources:** [3.2 Slope of a Line](#)

SUPPORTING SKILLS

- I** Determine slope from a graph.
SK-LIN-SLOPE-GRAPH
- I** Calculate slope from two points.
SK-LIN-SLOPE-POINTS
- I** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS

M12-LIN-002

I can graph a line from an equation in a common form.

OpenStax: **Primary resources:** [3.1 Graph Linear Equations in Two Variables](#)

SUPPORTING SKILLS

- I** Convert among common forms of a linear equation.
SK-LIN-FORM-CONVERT
- I** Use slope to generate additional points on a line.
SK-LIN-GRAPH-SLOPE
- I** Plot an intercept accurately on a coordinate plane.
SK-LIN-PLOT-INTERCEPT

M12-LIN-003

I can write an equation of a line from given information.

OpenStax: **Primary resources:** [3.3 Find the Equation of a Line](#)

SUPPORTING SKILLS

- A** Isolate a specified variable in an equation or formula.
SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- D** Convert among common forms of a linear equation.
SK-LIN-FORM-CONVERT · first introduced in Math 12
- A** Calculate slope from two points.
SK-LIN-SLOPE-POINTS · first introduced in Math 12

M12-LIN-004

I can determine whether two lines are parallel, perpendicular, or neither.

OpenStax: **Primary resources:** [3.3 Find the Equation of a Line](#)

SUPPORTING SKILLS

- I** Recognize that distinct parallel lines have equal slopes.
SK-LIN-PARALLEL-SLOPE
- I** Determine the negative reciprocal of a nonzero slope.
SK-LIN-PERPENDICULAR-SLOPE

M12-LIN-005

I can evaluate a function using function notation.

OpenStax: **Primary resources:** [3.5 Relations and Functions](#)

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT
- I** Identify the input and output represented by function notation.
SK-FUNC-NOTATION-READ
- I** Recall and correctly use the terms relation, function, input, output, domain, and range.
SK-FUNC-VOCABULARY

M12-LIN-006

I can determine whether a relation is a function using the vertical line test or input-output data.

OpenStax: **Primary resources:** [3.5 Relations and Functions](#)

SUPPORTING SKILLS

- D** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- I** Apply the vertical line test to a graph.
SK-FUNC-VERTICAL-LINE
- A** Recall and correctly use the terms relation, function, input, output, domain, and range.
SK-FUNC-VOCABULARY · first introduced in Math 12

M12-LIN-008

I can determine domain and range from a graph or table.

OpenStax: **Primary resources:** [3.5 Relations and Functions](#)

SUPPORTING SKILLS

- I** Read domain and range from a graph.
SK-FUNC-DOMAIN-RANGE-GRAPH
- I** Read domain and range from a table.
SK-FUNC-DOMAIN-RANGE-TABLE

M12-LIN-009

I can interpret domain and range in context.

OpenStax: **Primary resources:** [3.5 Relations and Functions](#)

SUPPORTING SKILLS

- I** Restrict domain to values meaningful in a context.
SK-FUNC-CONTEXT-DOMAIN
- I** Interpret the range of a function in a context.
SK-FUNC-CONTEXT-RANGE
- I** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN

Systems of Linear Equations

REQUIRED

OpenStax coverage: **Primary resources:** [4.1 Solve Systems of Linear Equations with Two Variables](#); [4.2 Solve Applications with Systems of Equations](#)

M12-SYS-001

I can interpret the solution of a linear system as an intersection point.

OpenStax: **Primary resources:** [4.1 Solve Systems of Linear Equations with Two Variables](#)

SUPPORTING SKILLS

- I** Locate and interpret the intersection of two lines.

SK-SYS-GRAPH-INTERSECTION

M12-SYS-002

I can solve a two-variable linear system by graphing.

OpenStax: **Primary resources:** [4.1 Solve Systems of Linear Equations with Two Variables](#)

SUPPORTING SKILLS

- A** Use slope to generate additional points on a line.

SK-LIN-GRAPH-SLOPE · first introduced in Math 12

- A** Plot an intercept accurately on a coordinate plane.

SK-LIN-PLOT-INTERCEPT · first introduced in Math 12

- I** Check an ordered pair in both equations of a system.

SK-SYS-CHECK

- D** Locate and interpret the intersection of two lines.

SK-SYS-GRAPH-INTERSECTION · first introduced in Math 12

M12-SYS-003

I can solve a two-variable linear system by substitution.

OpenStax: **Primary resources:** [4.1 Solve Systems of Linear Equations with Two Variables](#)

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.

SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12

- A** Check an ordered pair in both equations of a system.

SK-SYS-CHECK · first introduced in Math 12

- I** Substitute an equivalent expression for a variable.

SK-SYS-SUBSTITUTE

M12-SYS-004

I can solve a two-variable linear system by elimination.

OpenStax: **Primary resources:** [4.1 Solve Systems of Linear Equations with Two Variables](#)

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.

SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12

- A** Check an ordered pair in both equations of a system.

SK-SYS-CHECK · first introduced in Math 12

- I** Add equations to eliminate a variable.

SK-SYS-ELIMINATE

I can create and solve a two-variable linear system for an application.

OpenStax: **Primary resources:** [4.2 Solve Applications with Systems of Equations](#)

SUPPORTING SKILLS

- A** Check an ordered pair in both equations of a system.
SK-SYS-CHECK · first introduced in Math 12
- A** Add equations to eliminate a variable.
SK-SYS-ELIMINATE · first introduced in Math 12
- I** Define two variables and write two equations for a context.
SK-SYS-MODEL
- A** Substitute an equivalent expression for a variable.
SK-SYS-SUBSTITUTE · first introduced in Math 12

Polynomial Operations and Basic Factoring

REQUIRED

OpenStax coverage: **Primary resources:** [5.2 Properties of Exponents and Scientific Notation](#); [5.1 Add and Subtract Polynomials](#); [5.3 Multiply Polynomials](#); [5.4 Dividing Polynomials](#); [6.1 Greatest Common Factor and Factor by Grouping](#)

M12-POL-001

I can simplify expressions using integer exponent rules.

OpenStax: **Primary resources:** [5.2 Properties of Exponents and Scientific Notation](#)

SUPPORTING SKILLS

- I** Apply the power rule for exponents.
SK-EXP-POWER
- I** Apply the product rule for exponents.
SK-EXP-PRODUCT
- I** Apply the quotient rule for exponents.
SK-EXP-QUOTIENT

M12-POL-002

I can add and subtract polynomials.

OpenStax: **Primary resources:** [5.1 Add and Subtract Polynomials](#)

SUPPORTING SKILLS

- A** Identify and combine like terms.
SK-ALG-LIKE-TERMS · inherited from middle school
- I** Identify like polynomial terms by variable part and exponent.
SK-POLY-LIKE-TERMS
- I** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY

M12-POL-003

I can multiply polynomials, including binomials.

OpenStax: **Primary resources:** [5.3 Multiply Polynomials](#)

SUPPORTING SKILLS

- A** Apply the distributive property in either direction.
SK-ALG-DISTRIBUTE · inherited from middle school
- I** Multiply two binomials using distribution.
SK-POLY-BINOMIAL-MULTIPLY
- I** Multiply a monomial by a polynomial.
SK-POLY-DISTRIBUTE
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M12-POL-004

I can divide a polynomial by a monomial.

OpenStax: **Primary resources:** [5.4 Dividing Polynomials](#)

SUPPORTING SKILLS

- A** Apply the quotient rule for exponents.
SK-EXP-QUOTIENT · first introduced in Math 12
- I** Divide each term of a polynomial by a monomial.
SK-POLY-DIVIDE-MONOMIAL
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M12-POL-005

I can factor a greatest common factor from a polynomial.

OpenStax: **Primary resources:** [6.1 Greatest Common Factor and Factor by Grouping](#)

SUPPORTING SKILLS

- A** Apply the distributive property in either direction.
SK-ALG-DISTRIBUTE · inherited from middle school
- I** Factor the greatest common monomial factor from a polynomial.
SK-FACTOR-GCF
- I** Determine the numerical and variable greatest common factor of polynomial terms.
SK-POLY-GCF
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

Extension Content

EXTENSION

OpenStax coverage: **Primary resources:** [Intermediate Algebra 2e 2.7 Solve Absolute Value Inequalities](#); [Intermediate Algebra 2e 4.7 Graphing Systems of Linear Inequalities](#); [Intermediate Algebra 2e 6.2 Factor Trinomials](#); [Precalculus 2e 1.2 Domain and Range](#)

M12-EXT-001

I can solve and graph an absolute-value inequality.

OpenStax: **Primary resources:** [2.7 Solve Absolute Value Inequalities](#)

SUPPORTING SKILLS

- I** Rewrite an absolute-value inequality as a compound inequality.
SK-ABS-SPLIT-INEQUALITY
- A** Interpret AND as the intersection of two inequality solution sets.
SK-INE-COMPOUND-AND · first introduced in Math 12
- A** Interpret OR as the union of two inequality solution sets.
SK-INE-COMPOUND-OR · first introduced in Math 12
- A** Represent an inequality solution on a number line.
SK-INE-NUMBER-LINE · first introduced in Math 12

M12-EXT-002

I can graph a system of linear inequalities and identify its solution region.

OpenStax: **Primary resources:** [4.7 Graphing Systems of Linear Inequalities](#)

SUPPORTING SKILLS

- I** Graph the boundary line of a linear inequality with the correct line type.
SK-INE-GRAPH-BOUNDARY
- I** Identify the overlapping solution region of multiple linear inequalities.
SK-INE-SYSTEM-REGION
- A** Use slope to generate additional points on a line.
SK-LIN-GRAPH-SLOPE · first introduced in Math 12

M12-EXT-003

I can factor a quadratic trinomial.

OpenStax: **Primary resources:** [6.2 Factor Trinomials](#)

SUPPORTING SKILLS

- I** Verify a factorization by multiplication.
SK-FACTOR-CHECK
- I** Factor a quadratic trinomial with a nonunit leading coefficient.
SK-FACTOR-TRINOMIAL-GENERAL
- I** Factor a monic quadratic trinomial.
SK-FACTOR-TRINOMIAL-MONIC

M12-EXT-004

I can identify a relationship as discrete or continuous.

OpenStax: **Primary resources:** [1.2 Domain and Range](#)

SUPPORTING SKILLS

- I** Distinguish discrete inputs from values varying continuously over an interval.
SK-FUNC-DISCRETE-CONTINUOUS

Math 21 Curriculum

Algebra and its Functions | Middlesex School Mathematics

COURSE AT A GLANCE

Review Content

REVIEW

Primary resources: [Intermediate Algebra 2e 3.2 Slope of a Line](#); [Intermediate Algebra 2e 3.1 Graph Linear Equations in Two Variables](#); [College Algebra 2e 4.1 Linear Functions](#); [Intermediate Algebra 2e 4.1 Solve Systems of Linear Equations with Two Variables](#); [Intermediate Algebra 2e 5.2 Properties of Exponents and Scientific Notation](#)

UNIT LEARNING OBJECTIVES

- I can determine and interpret the slope of a line.
- I can graph a line from an equation and write an equation from given information.
- I can apply a linear function to a context.
- I can identify the solution of a linear system as an intersection point.
- I can solve a simple two-variable linear system using an appropriate method.
- I can simplify an expression using basic positive-integer exponent rules.

Functions and Function Interpretation

REQUIRED

Primary resources: [Intermediate Algebra 2e 3.5 Relations and Functions](#); [College Algebra 2e 3.3 Rates of Change and Behavior of Graphs](#)
Supplemental resources: [Intermediate Algebra 2e 3.6 Graphs of Functions](#); [College Algebra 2e 3.1 Functions and Function Notation](#); [College Algebra 2e 4.1 Linear Functions](#)

UNIT LEARNING OBJECTIVES

- I can evaluate a function from a formula, graph, or table.
- I can determine whether a relation is a function.
- I can determine domain and range from a graph or table.
- I can solve for an input that produces a specified output.
- I can interpret inputs, outputs, domain, and range in context.
- I can calculate and interpret average rate of change.
- I can compare functions represented by formulas, graphs, or tables.

Polynomials and Factoring

REQUIRED

Primary resources: [5.1 Add and Subtract Polynomials](#); [5.3 Multiply Polynomials](#); [6.1 Greatest Common Factor and Factor by Grouping](#); [6.2 Factor Trinomials](#); [6.3 Factor Special Products](#); [6.5 Polynomial Equations](#)

UNIT LEARNING OBJECTIVES

- I can add and subtract polynomials.
- I can multiply polynomials.
- I can factor out a greatest common factor.
- I can factor a quadratic trinomial.
- I can factor a difference of squares and other required special cases.
- I can use factoring to determine the zeros of a polynomial.

Rational Expressions and Equations

REQUIRED

Primary resources: [7.1 Multiply and Divide Rational Expressions](#); [7.2 Add and Subtract Rational Expressions](#); [7.4 Solve Rational Equations](#)

UNIT LEARNING OBJECTIVES

- I can state excluded values for a rational expression.
- I can simplify a rational expression while preserving its restrictions.
- I can multiply and divide rational expressions.
- I can add and subtract rational expressions using a common denominator.
- I can solve a rational equation and reject extraneous solutions.

Exponents and Radicals

REQUIRED

Primary resources: [Intermediate Algebra 2e 8.3 Simplify Rational Exponents](#); [Intermediate Algebra 2e 8.2 Simplify Radical Expressions](#); [Intermediate Algebra 2e 8.6 Solve Radical Equations](#)
Supplemental resources: [College Algebra 2e 2.6 Other Types of Equations](#)

UNIT LEARNING OBJECTIVES

- I can simplify expressions containing negative exponents.
- I can rewrite expressions between radical and rational-exponent forms.
- I can simplify radical expressions.
- I can apply exponent rules to expressions with integer and rational exponents.
- I can solve a basic equation involving a radical or rational exponent and check for extraneous solutions.

Quadratic Functions and Equations

REQUIRED

Primary resources: [9.6 Graph Quadratic Functions Using Properties](#); [9.7 Graph Quadratic Functions Using Transformations](#); [6.5 Polynomial Equations](#); [9.1 Solve Quadratic Equations Using the Square Root Property](#); [9.2 Solve Quadratic Equations by Completing the Square](#); [9.3 Solve Quadratic Equations Using the Quadratic Formula](#); [9.5 Solve Applications of Quadratic Equations](#)
Supplemental resources: [9.6 Graph Quadratic Functions Using Properties](#); [6.5 Polynomial Equations](#)

UNIT LEARNING OBJECTIVES

- I can identify key features of a quadratic function from its graph or equation.
- I can connect standard, factored, and vertex forms to features of a parabola.
- I can solve a quadratic equation by factoring.
- I can solve a quadratic equation using the square-root property.
- I can solve a quadratic equation by completing the square.
- I can solve a quadratic equation using the quadratic formula.
- I can choose an efficient method for solving a quadratic equation.
- I can graph a quadratic function using its vertex, intercepts, and symmetry.
- I can create and interpret a quadratic model for a projectile, area, or optimization problem.

Extension Content

EXTENSION

Primary resources: [8.8 Use the Complex Number System](#); [9.3 Solve Quadratic Equations Using the Quadratic Formula](#)

UNIT LEARNING OBJECTIVES

- I can perform basic arithmetic with complex numbers.
- I can express the nonreal solutions of a quadratic equation using complex numbers.

IN-DEPTH CURRICULUM

SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Review Content

REVIEW

OpenStax coverage: **Primary resources:** [Intermediate Algebra 2e 3.2 Slope of a Line](#); [Intermediate Algebra 2e 3.1 Graph Linear Equations in Two Variables](#); [College Algebra 2e 4.1 Linear Functions](#); [Intermediate Algebra 2e 4.1 Solve Systems of Linear Equations with Two Variables](#); [Intermediate Algebra 2e 5.2 Properties of Exponents and Scientific Notation](#)

M21-REV-001

I can determine and interpret the slope of a line.

OpenStax: **Primary resources:** [3.2 Slope of a Line](#)

SUPPORTING SKILLS

- Determine slope from a graph.
SK-LIN-SLOPE-GRAPH · first introduced in Math 12
- Calculate slope from two points.
SK-LIN-SLOPE-POINTS · first introduced in Math 12
- Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

M21-REV-002

I can graph a line from an equation and write an equation from given information.

OpenStax: **Primary resources:** [3.1 Graph Linear Equations in Two Variables](#)

SUPPORTING SKILLS

- Convert among common forms of a linear equation.
SK-LIN-FORM-CONVERT · first introduced in Math 12
- Use slope to generate additional points on a line.
SK-LIN-GRAPH-SLOPE · first introduced in Math 12
- Plot an intercept accurately on a coordinate plane.
SK-LIN-PLOT-INTERCEPT · first introduced in Math 12
- Calculate slope from two points.
SK-LIN-SLOPE-POINTS · first introduced in Math 12

M21-REV-003

I can apply a linear function to a context.

OpenStax: **Primary resources:** [4.1 Linear Functions](#)

SUPPORTING SKILLS

- I** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET
- R** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

M21-REV-004

I can identify the solution of a linear system as an intersection point.

OpenStax: **Primary resources:** [4.1 Solve Systems of Linear Equations with Two Variables](#)

SUPPORTING SKILLS

- R** Locate and interpret the intersection of two lines.
SK-SYS-GRAPH-INTERSECTION · first introduced in Math 12

M21-REV-005

I can solve a simple two-variable linear system using an appropriate method.

OpenStax: **Primary resources:** [4.1 Solve Systems of Linear Equations with Two Variables](#)

SUPPORTING SKILLS

- R** Check an ordered pair in both equations of a system.
SK-SYS-CHECK · first introduced in Math 12
- R** Add equations to eliminate a variable.
SK-SYS-ELIMINATE · first introduced in Math 12
- R** Locate and interpret the intersection of two lines.
SK-SYS-GRAPH-INTERSECTION · first introduced in Math 12
- R** Substitute an equivalent expression for a variable.
SK-SYS-SUBSTITUTE · first introduced in Math 12

M21-REV-006

I can simplify an expression using basic positive-integer exponent rules.

OpenStax: **Primary resources:** [5.2 Properties of Exponents and Scientific Notation](#)

SUPPORTING SKILLS

- R** Apply the power rule for exponents.
SK-EXP-POWER · first introduced in Math 12
- R** Apply the product rule for exponents.
SK-EXP-PRODUCT · first introduced in Math 12
- R** Apply the quotient rule for exponents.
SK-EXP-QUOTIENT · first introduced in Math 12

Functions and Function Interpretation

REQUIRED

OpenStax coverage: **Primary resources:** [Intermediate Algebra 2e 3.5 Relations and Functions](#); [College Algebra 2e 3.3 Rates of Change and Behavior of Graphs](#)

Supplemental resources: [Intermediate Algebra 2e 3.6 Graphs of Functions](#); [College Algebra 2e 3.1 Functions and Function Notation](#); [College Algebra 2e 4.1 Linear Functions](#)

M21-FUN-001

I can evaluate a function from a formula, graph, or table.

OpenStax: **Primary resources:** [3.5 Relations and Functions](#)

Supplemental resources: [3.6 Graphs of Functions](#)

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · first introduced in middle school
 - D** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
 - A** Identify the input and output represented by function notation.
SK-FUNC-NOTATION-READ · first introduced in Math 12
-

M21-FUN-002

I can determine whether a relation is a function.

OpenStax: **Primary resources:** [3.5 Relations and Functions](#)

SUPPORTING SKILLS

- D** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
 - D** Apply the vertical line test to a graph.
SK-FUNC-VERTICAL-LINE · first introduced in Math 12
-

M21-FUN-003

I can determine domain and range from a graph or table.

OpenStax: **Primary resources:** [3.5 Relations and Functions](#)

SUPPORTING SKILLS

- D** Read domain and range from a graph.
SK-FUNC-DOMAIN-RANGE-GRAPH · first introduced in Math 12
 - D** Read domain and range from a table.
SK-FUNC-DOMAIN-RANGE-TABLE · first introduced in Math 12
-

M21-FUN-004

I can solve for an input that produces a specified output.

OpenStax: **Primary resources:** [3.5 Relations and Functions](#)

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I** Solve for an input associated with a specified output.
SK-FUNC-SOLVE-INPUT

M21-FUN-005

I can interpret inputs, outputs, domain, and range in context.

OpenStax: **Primary resources:** [3.5 Relations and Functions](#)

SUPPORTING SKILLS

- D** Restrict domain to values meaningful in a context.
SK-FUNC-CONTEXT-DOMAIN · first introduced in Math 12
- D** Interpret the range of a function in a context.
SK-FUNC-CONTEXT-RANGE · first introduced in Math 12
- D** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12

M21-FUN-006

I can calculate and interpret average rate of change.

OpenStax: **Primary resources:** [3.3 Rates of Change and Behavior of Graphs](#)

SUPPORTING SKILLS

- I** Calculate average rate of change over an interval.
SK-FUNC-AROC
- I** Interpret average rate of change with contextual units.
SK-FUNC-AROC-INTERPRET
- D** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

M21-FUN-007

I can compare functions represented by formulas, graphs, or tables.

OpenStax: **Primary resources:** [Intermediate Algebra 2e 3.5 Relations and Functions](#)

Supplemental resources: [College Algebra 2e 3.1 Functions and Function Notation](#); [College Algebra 2e 4.1 Linear Functions](#)

SUPPORTING SKILLS

- I** Compare functions shown in different representations.
SK-FUNC-COMPARE-REP
- I** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH
- I** Use tabular patterns to predict graph shape and behavior.
SK-REP-TABLE-GRAPH

Polynomials and Factoring

REQUIRED

OpenStax coverage: **Primary resources:** [5.1 Add and Subtract Polynomials](#); [5.3 Multiply Polynomials](#); [6.1 Greatest Common Factor and Factor by Grouping](#); [6.2 Factor Trinomials](#); [6.3 Factor Special Products](#); [6.5 Polynomial Equations](#)

M21-POL-001

I can add and subtract polynomials.

OpenStax: **Primary resources:** [5.1 Add and Subtract Polynomials](#)

SUPPORTING SKILLS

- A** Identify and combine like terms.
SK-ALG-LIKE-TERMS · inherited from middle school
- D** Identify like polynomial terms by variable part and exponent.
SK-POLY-LIKE-TERMS · first introduced in Math 12
- R** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-002

I can multiply polynomials.

OpenStax: **Primary resources:** [5.3 Multiply Polynomials](#)

SUPPORTING SKILLS

- A** Apply the distributive property in either direction.
SK-ALG-DISTRIBUTE · inherited from middle school
- D** Multiply two binomials using distribution.
SK-POLY-BINOMIAL-MULTIPLY · first introduced in Math 12
- D** Multiply a monomial by a polynomial.
SK-POLY-DISTRIBUTE · first introduced in Math 12
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-003

I can factor out a greatest common factor.

OpenStax: **Primary resources:** [6.1 Greatest Common Factor and Factor by Grouping](#)

SUPPORTING SKILLS

- R** Verify a factorization by multiplication.
SK-FACTOR-CHECK · first introduced in Math 12
- D** Factor the greatest common monomial factor from a polynomial.
SK-FACTOR-GCF · first introduced in Math 12
- A** Determine the numerical and variable greatest common factor of polynomial terms.
SK-POLY-GCF · first introduced in Math 12
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-004

I can factor a quadratic trinomial.

OpenStax: **Primary resources:** [6.2 Factor Trinomials](#)

SUPPORTING SKILLS

- A** Verify a factorization by multiplication.
SK-FACTOR-CHECK · first introduced in Math 12
- R** Factor a quadratic trinomial with a nonunit leading coefficient.
SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- R** Factor a monic quadratic trinomial.
SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-005

I can factor a difference of squares and other required special cases.

OpenStax: **Primary resources:** [6.3 Factor Special Products](#)

SUPPORTING SKILLS

- A** Verify a factorization by multiplication.
SK-FACTOR-CHECK · first introduced in Math 12
- I** Factor a difference of two squares.
SK-FACTOR-DIFF-SQUARES
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-006

I can use factoring to determine the zeros of a polynomial.

OpenStax: **Primary resources:** [6.5 Polynomial Equations](#)

SUPPORTING SKILLS

- A** Factor a difference of two squares.
SK-FACTOR-DIFF-SQUARES · first introduced in Math 21
- A** Factor a quadratic trinomial with a nonunit leading coefficient.
SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- A** Factor a monic quadratic trinomial.
SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- I** Apply the zero-product property.
SK-FACTOR-ZERO-PRODUCT
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

Rational Expressions and Equations

REQUIRED

OpenStax coverage: **Primary resources:** [7.1 Multiply and Divide Rational Expressions](#); [7.2 Add and Subtract Rational Expressions](#); [7.4 Solve Rational Equations](#)

M21-RAT-001

I can state excluded values for a rational expression.

OpenStax: **Primary resources:** [7.1 Multiply and Divide Rational Expressions](#)

SUPPORTING SKILLS

- I** Determine excluded values from an original denominator.
SK-RAT-RESTRICTIONS
- I** Recall and correctly use rational-expression terms including numerator, denominator, restriction, and excluded value.
SK-RAT-VOCABULARY

M21-RAT-002

I can simplify a rational expression while preserving its restrictions.

OpenStax: **Primary resources:** [7.1 Multiply and Divide Rational Expressions](#)

SUPPORTING SKILLS

- I** Cancel common factors rather than individual terms.
SK-RAT-CANCEL-FACTORS
- I** Factor numerators and denominators completely.
SK-RAT-FACTOR
- A** Determine excluded values from an original denominator.
SK-RAT-RESTRICTIONS · first introduced in Math 21
- A** Recall and correctly use rational-expression terms including numerator, denominator, restriction, and excluded value.
SK-RAT-VOCABULARY · first introduced in Math 21

M21-RAT-003

I can multiply and divide rational expressions.

OpenStax: **Primary resources:** [7.1 Multiply and Divide Rational Expressions](#)

SUPPORTING SKILLS

- A** Add, subtract, multiply, and divide fractions.
SK-NUM-FRACTION-OPERATE · inherited from middle school
- A** Cancel common factors rather than individual terms.
SK-RAT-CANCEL-FACTORS · first introduced in Math 21
- A** Factor numerators and denominators completely.
SK-RAT-FACTOR · first introduced in Math 21
- A** Determine excluded values from an original denominator.
SK-RAT-RESTRICTIONS · first introduced in Math 21

M21-RAT-004

I can add and subtract rational expressions using a common denominator.

OpenStax: **Primary resources:** [7.2 Add and Subtract Rational Expressions](#)

SUPPORTING SKILLS

- A** Add, subtract, multiply, and divide fractions.
SK-NUM-FRACTION-OPERATE · inherited from middle school
- I** Combine rational expressions over a common denominator.
SK-RAT-COMBINE
- I** Determine the least common denominator of rational expressions.
SK-RAT-LCD

M21-RAT-005

I can solve a rational equation and reject extraneous solutions.

OpenStax: **Primary resources:** [7.4 Solve Rational Equations](#)

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I** Clear denominators from a rational equation.
SK-RAT-CLEAR-DENOMINATORS
- I** Check candidate solutions against denominator restrictions.
SK-RAT-EXTRANEIOUS
- A** Determine excluded values from an original denominator.
SK-RAT-RESTRICTIONS · first introduced in Math 21

Exponents and Radicals

REQUIRED

OpenStax coverage: **Primary resources:** [Intermediate Algebra 2e 8.3 Simplify Rational Exponents](#); [Intermediate Algebra 2e 8.2 Simplify Radical Expressions](#); [Intermediate Algebra 2e 8.6 Solve Radical Equations](#)
Supplemental resources: [College Algebra 2e 2.6 Other Types of Equations](#)

M21-EXP-001

I can simplify expressions containing negative exponents.

OpenStax: **Primary resources:** [8.3 Simplify Rational Exponents](#)

SUPPORTING SKILLS

- I** Rewrite a negative exponent using a reciprocal.
SK-EXP-NEGATIVE
- A** Apply the product rule for exponents.
SK-EXP-PRODUCT · first introduced in Math 12
- A** Apply the quotient rule for exponents.
SK-EXP-QUOTIENT · first introduced in Math 12

M21-EXP-002

I can rewrite expressions between radical and rational-exponent forms.

OpenStax: **Primary resources:** [8.3 Simplify Rational Exponents](#)

SUPPORTING SKILLS

- I** Interpret numerator and denominator roles in a rational exponent.
SK-EXP-RATIONAL
- I** Convert between radical and rational-exponent notation.
SK-RAD-CONVERT
- I** Read radical and rational-exponent notation and identify the index, radicand, base, numerator, and denominator.
SK-RAD-NOTATION-READ

M21-EXP-003

I can simplify radical expressions.

OpenStax: **Primary resources:** [8.2 Simplify Radical Expressions](#)

SUPPORTING SKILLS

- A** Express a whole number as a product of prime factors.
SK-NUM-PRIME-FACTOR · inherited from middle school
- I** Extract perfect-power factors from a radical.
SK-RAD-FACTOR-PERFECT

I can apply exponent rules to expressions with integer and rational exponents.

OpenStax: **Primary resources:** [8.3 Simplify Rational Exponents](#)

SUPPORTING SKILLS

- A** Rewrite a negative exponent using a reciprocal.
SK-EXP-NEGATIVE · first introduced in Math 21
- D** Apply the power rule for exponents.
SK-EXP-POWER · first introduced in Math 12
- D** Apply the product rule for exponents.
SK-EXP-PRODUCT · first introduced in Math 12
- D** Apply the quotient rule for exponents.
SK-EXP-QUOTIENT · first introduced in Math 12
- A** Interpret numerator and denominator roles in a rational exponent.
SK-EXP-RATIONAL · first introduced in Math 21
- I** Apply the zero-exponent rule.
SK-EXP-ZERO

I can solve a basic equation involving a radical or rational exponent and check for extraneous solutions.

OpenStax: **Primary resources:** [Intermediate Algebra 2e 8.6 Solve Radical Equations](#)

Supplemental resources: [College Algebra 2e 2.6 Other Types of Equations](#)

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I** Require an even-root radicand to be nonnegative when determining domain.
SK-FUNC-DOMAIN-EVEN-ROOT
- I** Check a candidate solution for extraneous results.
SK-RAD-CHECK
- I** Isolate a radical expression before solving.
SK-RAD-ISOLATE

Quadratic Functions and Equations

REQUIRED

OpenStax coverage: **Primary resources:** [9.6 Graph Quadratic Functions Using Properties](#); [9.7 Graph Quadratic Functions Using Transformations](#); [6.5 Polynomial Equations](#); [9.1 Solve Quadratic Equations Using the Square Root Property](#); [9.2 Solve Quadratic Equations by Completing the Square](#); [9.3 Solve Quadratic Equations Using the Quadratic Formula](#); [9.5 Solve Applications of Quadratic Equations](#)
Supplemental resources: [9.6 Graph Quadratic Functions Using Properties](#); [6.5 Polynomial Equations](#)

M21-QUAD-001

I can identify key features of a quadratic function from its graph or equation.

OpenStax: **Primary resources:** [9.6 Graph Quadratic Functions Using Properties](#)

SUPPORTING SKILLS

- I** Identify standard, factored, and vertex forms.
SK-QUAD-FORM-IDENTIFY
- I** Determine x- and y-intercepts of a quadratic function.
SK-QUAD-INTERCEPTS
- I** Determine the vertex and axis of symmetry.
SK-QUAD-VERTEX
- I** Recall and correctly use quadratic-function terms including parabola, vertex, axis of symmetry, intercept, zero, and solution.
SK-QUAD-VOCABULARY

M21-QUAD-002

I can connect standard, factored, and vertex forms to features of a parabola.

OpenStax: **Primary resources:** [9.7 Graph Quadratic Functions Using Transformations](#)

Supplemental resources: [9.6 Graph Quadratic Functions Using Properties](#); [6.5 Polynomial Equations](#)

SUPPORTING SKILLS

- I** Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.
SK-QUAD-FORM-CONVERT
- D** Identify standard, factored, and vertex forms.
SK-QUAD-FORM-IDENTIFY · first introduced in Math 21
- A** Determine x- and y-intercepts of a quadratic function.
SK-QUAD-INTERCEPTS · first introduced in Math 21
- A** Determine the vertex and axis of symmetry.
SK-QUAD-VERTEX · first introduced in Math 21
- D** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21

M21-QUAD-003

I can solve a quadratic equation by factoring.

OpenStax: **Primary resources:** [6.5 Polynomial Equations](#)

SUPPORTING SKILLS

- A** Factor a quadratic trinomial with a nonunit leading coefficient.
SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- A** Factor a monic quadratic trinomial.
SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- D** Apply the zero-product property.
SK-FACTOR-ZERO-PRODUCT · first introduced in Math 21
- I** Determine real and complex zeros using an appropriate method.
SK-POLY-ZEROS

M21-QUAD-004

I can solve a quadratic equation using the square-root property.

OpenStax: **Primary resources:** [9.1 Solve Quadratic Equations Using the Square Root Property](#)

SUPPORTING SKILLS

- I** Apply the square-root property with both signs.
SK-QUAD-SQUARE-ROOT
- A** Extract perfect-power factors from a radical.
SK-RAD-FACTOR-PERFECT · first introduced in Math 21

M21-QUAD-005

I can solve a quadratic equation by completing the square.

OpenStax: **Primary resources:** [9.2 Solve Quadratic Equations by Completing the Square](#)

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I** Create a perfect-square trinomial by completing the square.
SK-QUAD-COMPLETE-SQUARE

M21-QUAD-006

I can solve a quadratic equation using the quadratic formula.

OpenStax: **Primary resources:** [9.3 Solve Quadratic Equations Using the Quadratic Formula](#)

SUPPORTING SKILLS

- I** Use the discriminant to predict the number and type of solutions.
SK-QUAD-DISCRIMINANT
- I** Substitute coefficients accurately into the quadratic formula.
SK-QUAD-FORMULA
- A** Extract perfect-power factors from a radical.
SK-RAD-FACTOR-PERFECT · first introduced in Math 21

M21-QUAD-007

I can choose an efficient method for solving a quadratic equation.

OpenStax: **Primary resources:** [9.3 Solve Quadratic Equations Using the Quadratic Formula](#)

SUPPORTING SKILLS

- A** Apply the zero-product property.
SK-FACTOR-ZERO-PRODUCT · first introduced in Math 21
- A** Create a perfect-square trinomial by completing the square.
SK-QUAD-COMPLETE-SQUARE · first introduced in Math 21
- A** Substitute coefficients accurately into the quadratic formula.
SK-QUAD-FORMULA · first introduced in Math 21
- I** Select an efficient quadratic-solving method from equation structure.
SK-QUAD-METHOD-SELECT
- A** Apply the square-root property with both signs.
SK-QUAD-SQUARE-ROOT · first introduced in Math 21

I can graph a quadratic function using its vertex, intercepts, and symmetry.

OpenStax: **Primary resources:** [9.6 Graph Quadratic Functions Using Properties](#)

SUPPORTING SKILLS

- A** Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.
SK-QUAD-FORM-CONVERT · first introduced in Math 21
- D** Determine x- and y-intercepts of a quadratic function.
SK-QUAD-INTERCEPTS · first introduced in Math 21
- D** Determine the vertex and axis of symmetry.
SK-QUAD-VERTEX · first introduced in Math 21
- D** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21

I can create and interpret a quadratic model for a projectile, area, or optimization problem.

OpenStax: **Primary resources:** [9.5 Solve Applications of Quadratic Equations](#)

SUPPORTING SKILLS

- D** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- I** Interpret vertex and zeros in an application.
SK-QUAD-MODEL-FEATURES
- A** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

Extension Content

EXTENSION

OpenStax coverage: **Primary resources:** [8.8 Use the Complex Number System](#); [9.3 Solve Quadratic Equations Using the Quadratic Formula](#)

M21-EXT-001

I can perform basic arithmetic with complex numbers.

OpenStax: **Primary resources:** [8.8 Use the Complex Number System](#)

SUPPORTING SKILLS

- I** Simplify integer powers of the imaginary unit.
SK-CPLX-I-POWERS
- I** Add, subtract, and multiply complex numbers.
SK-CPLX-OPERATE

M21-EXT-002

I can express the nonreal solutions of a quadratic equation using complex numbers.

OpenStax: **Primary resources:** [9.3 Solve Quadratic Equations Using the Quadratic Formula](#)

SUPPORTING SKILLS

- A** Simplify integer powers of the imaginary unit.
SK-CPLX-I-POWERS · first introduced in Math 21
- I** Express a quadratic solution with a negative discriminant in complex form.
SK-CPLX-QUAD-SOLUTION
- A** Use the discriminant to predict the number and type of solutions.
SK-QUAD-DISCRIMINANT · first introduced in Math 21

Math 22 Curriculum

Geometry | Middlesex School Mathematics | Resource: Math 22 Geometry Packet

COURSE AT A GLANCE

Review Content

REVIEW

No mapped OpenStax section

UNIT LEARNING OBJECTIVES

- I can solve a linear equation arising from a geometric relationship.
- I can solve a proportion and interpret the resulting scale factor.
- I can simplify an exact square root and approximate it when needed.
- I can calculate distance, midpoint, and slope on the coordinate plane.
- I can substitute measurements into a formula with consistent units.

Geometry Basics

REQUIRED

1.1 Points, Lines, Planes; 1.2 Segments, Measurement, Addition, Midpoints, Bisectors; 1.3 Angles, Measurement, Addition, Bisecting, Pairs; 1.4 Reasoning; Angle/Line Pairs; Parallel and Perpendicular; 1.5 Parallel-Line Converses and Proofs

UNIT LEARNING OBJECTIVES

- I can name and represent points, lines, planes, segments, and rays using correct notation.
- I can identify collinear, coplanar, intersecting, and concurrent figures.
- I can calculate segment lengths using betweenness and the segment-addition postulate.
- I can determine a midpoint or use a segment bisector relationship.
- I can classify and measure angles.
- I can calculate angle measures using angle addition and angle bisectors.
- I can identify and use complementary, supplementary, vertical, and linear-pair relationships.
- I can distinguish a definition, postulate, theorem, converse, and counterexample.
- I can determine angle relationships formed by parallel lines and a transversal.
- I can use angle relationships to determine whether lines are parallel.
- I can solve problems involving parallel, perpendicular, vertical, or horizontal lines.
- I can construct a logical geometric argument from stated facts and accepted results.

Triangle Congruence and Proof

REQUIRED

2.1 Triangles, Sides, and Angles; 2.2 Congruent Polygons; SAS; 2.3 SSS and HL; 2.5 ASA and AAS; 2.4 Isosceles and Equilateral Triangles

UNIT LEARNING OBJECTIVES

- I can classify triangles by sides and angles.
- I can use triangle angle-sum and exterior-angle relationships.
- I can write a congruence statement with corresponding vertices in the correct order.
- I can determine whether triangles are congruent by SAS.
- I can determine whether triangles are congruent by SSS.
- I can determine whether right triangles are congruent by HL.
- I can determine whether triangles are congruent by ASA or AAS.
- I can apply isosceles and equilateral triangle theorems and their converses.
- I can use corresponding parts of congruent triangles after proving congruence.
- I can write a structured proof involving triangle congruence.

Quadrilaterals, Perimeter, and Area

REQUIRED

3.1 Polygon Angles, Perimeter, and Area; 3.2 Parallelograms; 3.3 Rhombi, Rectangles, Squares; 3.4 Trapezoids; 3.5 Kites and Composite Polygons

UNIT LEARNING OBJECTIVES

- I can calculate interior or exterior angle measures of a polygon.
- I can calculate perimeter and area of common polygons.
- I can apply properties of parallelograms to determine unknown measures.
- I can determine whether a quadrilateral is a parallelogram.
- I can apply properties of rectangles, rhombi, and squares.

Quadrilaterals, Perimeter, and Area (Continued)

REQUIRED

Similarity and Proportional Reasoning

REQUIRED

4.1 Dilations and Similarity; 4.2 Similar Polygons, Perimeter, and Area; 4.3 Similarity by AA; 4.4 Similarity by SSS and SAS; 4.5 Proportionality Theorems

UNIT LEARNING OBJECTIVES - CONTINUED

- I can classify a quadrilateral from its stated or marked properties.
- I can apply trapezoid and trapezoid-midsegment relationships.
- I can apply properties of kites.
- I can calculate the perimeter or area of a composite polygon.
- I can justify a quadrilateral conclusion using definitions and theorems.

UNIT LEARNING OBJECTIVES

- I can determine image lengths and coordinates under a dilation.
- I can identify the scale factor of a dilation or similarity relationship.
- I can write a similarity statement with corresponding vertices in the correct order.
- I can use proportional corresponding sides to solve similar-polygon problems.
- I can relate similarity scale factor to perimeter and area scale factors.
- I can prove triangles similar by AA, SSS, or SAS.
- I can apply triangle proportionality theorems and their converses.
- I can solve and justify an indirect-measurement or other similarity application.

Right Triangles

REQUIRED

5.1 Pythagorean Theorem and Triangle Inequalities; 5.2 Special Right Triangles; 5.3 HL Similarity and Right-Triangle Similarity; 5.4 Sine and Cosine; 5.5 Tangent and Solving Right Triangles

UNIT LEARNING OBJECTIVES

- I can apply the Pythagorean theorem to determine a missing side.
- I can use the converse of the Pythagorean theorem to classify a triangle.
- I can apply triangle-inequality relationships.
- I can determine exact side lengths in 45-45-90 and 30-60-90 triangles.
- I can apply right-triangle similarity relationships.
- I can use geometric-mean relationships created by an altitude to the hypotenuse.
- I can select sine, cosine, or tangent for a right-triangle problem.
- I can determine a missing side or angle of a right triangle.
- I can solve a contextual right-triangle problem.
- I can justify a right-triangle solution using an appropriate theorem or ratio.

Circles

REQUIRED

6.1 Circle Vocabulary, Circumference, Arc Length; 6.2 Circle, Sector, and Segment Area; 6.3 Angle, Arc, and Chord Relationships; 6.4 Tangent, Secant, and Chord Relationships; 6.5 Tangent/Secant/Chord Angle, Arc, and Segment Relationships

UNIT LEARNING OBJECTIVES

- I can identify and name radii, diameters, chords, secants, tangents, arcs, sectors, and segments.
- I can calculate circumference and arc length.
- I can calculate the area of a circle, sector, or segment.
- I can determine angle and arc measures involving central and inscribed angles.
- I can apply relationships among chords, arcs, and distances from the center.
- I can apply tangent-radius and tangent-segment relationships.
- I can determine angle measures formed by chords, secants, or tangents.
- I can determine segment lengths formed by intersecting chords.
- I can determine segment lengths formed by secants or tangents.
- I can solve a multi-step circle problem and justify each relationship used.

IN-DEPTH CURRICULUM

SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Review Content

REVIEW

Resource coverage: No mapped OpenStax section

M22-REV-001

I can solve a linear equation arising from a geometric relationship.

Resource: No mapped OpenStax section

SUPPORTING SKILLS

-  Check a proposed solution in the original equation.
SK-ALG-CHECK-SOLUTION · first introduced in Math 12
-  Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
-  Translate quantities and relationships into an equation.
SK-ALG-TRANSLATE-EQUATION · first introduced in Math 12

M22-REV-002

I can solve a proportion and interpret the resulting scale factor.

Resource: No mapped OpenStax section

SUPPORTING SKILLS

-  Use cross multiplication when two ratios form a proportion.
SK-ALG-CROSS-MULTIPLY · inherited from middle school
-  Calculate and interpret a similarity or dilation scale factor.
SK-SIM-SCALE-FACTOR

M22-REV-003

I can simplify an exact square root and approximate it when needed.

Resource: No mapped OpenStax section

SUPPORTING SKILLS

- I** Distinguish exact values from measured or rounded approximations.

SK-GE0-EXACT-APPROX

- R** Extract perfect-power factors from a radical.

SK-RAD-FACTOR-PERFECT · first introduced in Math 21

M22-REV-004

I can calculate distance, midpoint, and slope on the coordinate plane.

Resource: No mapped OpenStax section

SUPPORTING SKILLS

- R** Calculate horizontal or vertical distance from the difference of coordinates.

SK-COORD-DISTANCE-AXIS · inherited horizontal/vertical coordinate distance

- I** Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula.

SK-COORD-DISTANCE-PYTHAGOREAN

- R** Calculate the midpoint of a segment from endpoint coordinates.

SK-COORD-MIDPOINT · inherited from earlier mathematics

- R** Calculate slope from two points.

SK-LIN-SLOPE-POINTS · first introduced in Math 12

M22-REV-005

I can substitute measurements into a formula with consistent units.

Resource: No mapped OpenStax section

SUPPORTING SKILLS

- I** Select and report appropriate linear, square, or angular units.

SK-GE0-UNITS

- A** Apply the order of operations to a numerical expression.

SK-NUM-ORDER-OPERATIONS · inherited from middle school

Geometry Basics

REQUIRED

Resource coverage: 1.1 Points, Lines, Planes; 1.2 Segments, Measurement, Addition, Midpoints, Bisectors; 1.3 Angles, Measurement, Addition, Bisecting, Pairs; 1.4 Reasoning; Angle/Line Pairs; Parallel and Perpendicular; 1.5 Parallel-Line Converses and Proofs

M22-BAS-001

I can name and represent points, lines, planes, segments, and rays using correct notation.

Resource: 1.1 Points, Lines, Planes

SUPPORTING SKILLS

- I** Add congruence, parallel, perpendicular, and angle markings to a diagram.
SK-GE0-DIAGRAM-MARK
- I** Recall definitions of points, lines, planes, segments, rays, angles, and intersections.
SK-GE0-FOUNDATIONAL-DEFINITIONS
- I** Read and write geometric notation correctly.
SK-GE0-NOTATION

M22-BAS-002

I can identify collinear, coplanar, intersecting, and concurrent figures.

Resource: 1.1 Points, Lines, Planes

SUPPORTING SKILLS

- I** Distinguish stated or marked facts from visual appearance.
SK-GE0-DIAGRAM-INFER
- A** Read and write geometric notation correctly.
SK-GE0-NOTATION · first introduced in Math 22

M22-BAS-003

I can calculate segment lengths using betweenness and the segment-addition postulate.

Resource: 1.2 Segments, Measurement, Addition, Midpoints, Bisectors

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- A** Translate quantities and relationships into an equation.
SK-ALG-TRANSLATE-EQUATION · first introduced in Math 12
- I** Translate segment addition into an equation.
SK-GE0-SEGMENT-ADDITION

M22-BAS-004

I can determine a midpoint or use a segment bisector relationship.

Resource: 1.2 Segments, Measurement, Addition, Midpoints, Bisectors

SUPPORTING SKILLS

- I** Use a bisector to create equal measures.
SK-GE0-BISECTOR
- I** Use a midpoint to create equal segment lengths.
SK-GE0-MIDPOINT
- A** Translate segment addition into an equation.
SK-GE0-SEGMENT-ADDITION · first introduced in Math 22

M22-BAS-005

I can classify and measure angles.

Resource: 1.3 Angles, Measurement, Addition, Bisecting, Pairs

SUPPORTING SKILLS

- I** Identify complementary, supplementary, vertical, and linear-pair relationships.
SK-ANG-PAIR-IDENTIFY
- I** Measure or calculate an angle accurately.
SK-GEO-ANGLE-MEASURE

M22-BAS-006

I can calculate angle measures using angle addition and angle bisectors.

Resource: 1.3 Angles, Measurement, Addition, Bisecting, Pairs

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I** Translate angle addition into an equation.
SK-GEO-ANGLE-ADDITION
- A** Use a bisector to create equal measures.
SK-GEO-BISECTOR · first introduced in Math 22

M22-BAS-007

I can identify and use complementary, supplementary, vertical, and linear-pair relationships.

Resource: 1.3 Angles, Measurement, Addition, Bisecting, Pairs; 1.4 Reasoning; Angle/Line Pairs; Parallel and Perpendicular

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- A** Translate quantities and relationships into an equation.
SK-ALG-TRANSLATE-EQUATION · first introduced in Math 12
- D** Identify complementary, supplementary, vertical, and linear-pair relationships.
SK-ANG-PAIR-IDENTIFY · first introduced in Math 22

M22-BAS-008

I can distinguish a definition, postulate, theorem, converse, and counterexample.

Resource: 1.1 Points, Lines, Planes; 1.4 Reasoning; Angle/Line Pairs; Parallel and Perpendicular

SUPPORTING SKILLS

- I** Recognize and apply the converse of a conditional statement.
SK-PROOF-CONVERSE
- I** Disprove a universal statement with a counterexample.
SK-PROOF-COUNTEREXAMPLE
- I** Apply a geometric definition as justification.
SK-PROOF-DEFINITION
- I** Select a theorem that connects known facts to a target statement.
SK-PROOF-THEOREM-SELECT

M22-BAS-009

I can determine angle relationships formed by parallel lines and a transversal.

Resource: 1.5 Parallel-Line Converses and Proofs

SUPPORTING SKILLS

- I** Use parallel-line angle relationships.
SK-ANG-PARALLEL-APPLY
- I** Identify corresponding, alternate, and same-side angle pairs.
SK-ANG-TRANSVERSAL-IDENTIFY

M22-BAS-010

I can use angle relationships to determine whether lines are parallel.

Resource: 1.5 Parallel-Line Converses and Proofs

SUPPORTING SKILLS

- I** Use angle relationships to prove lines parallel.
SK-ANG-PARALLEL-CONVERSE
- A** Identify corresponding, alternate, and same-side angle pairs.
SK-ANG-TRANSVERSAL-IDENTIFY · first introduced in Math 22

M22-BAS-011

I can solve problems involving parallel, perpendicular, vertical, or horizontal lines.

Resource: 1.4 Reasoning; Angle/Line Pairs; Parallel and Perpendicular

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- A** Use parallel-line angle relationships.
SK-ANG-PARALLEL-APPLY · first introduced in Math 22
- I** Use right-angle relationships created by perpendicular lines.
SK-LIN-PERPENDICULAR-ANGLE

M22-BAS-012

I can construct a logical geometric argument from stated facts and accepted results.

Resource: 1.4 Reasoning; Angle/Line Pairs; Parallel and Perpendicular; 1.5 Parallel-Line Converses and Proofs

SUPPORTING SKILLS

- I** Translate givens into diagram markings and statements.
SK-PROOF-GIVEN
- I** Organize statements and reasons into a valid proof sequence.
SK-PROOF-STRUCTURE
- A** Select a theorem that connects known facts to a target statement.
SK-PROOF-THEOREM-SELECT · first introduced in Math 22

Triangle Congruence and Proof

REQUIRED

Resource coverage: 2.1 Triangles, Sides, and Angles; 2.2 Congruent Polygons; SAS; 2.3 SSS and HL; 2.5 ASA and AAS; 2.4 Isosceles and Equilateral Triangles

M22-CON-001

I can classify triangles by sides and angles.

Resource: 2.1 Triangles, Sides, and Angles

SUPPORTING SKILLS

- A** Add congruence, parallel, perpendicular, and angle markings to a diagram.
SK-GE0-DIAGRAM-MARK · first introduced in Math 22
- I** Classify a triangle by side lengths and angle measures.
SK-TRI-CLASSIFY
- I** Recall and correctly use terminology and notation for triangle classification, congruence, and corresponding parts.
SK-TRI-CONGRUENCE-VOCAB

M22-CON-002

I can use triangle angle-sum and exterior-angle relationships.

Resource: 2.1 Triangles, Sides, and Angles

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I** Apply the triangle angle-sum theorem.
SK-TRI-ANGLE-SUM
- I** Apply the triangle exterior-angle theorem.
SK-TRI-EXTERIOR-ANGLE

M22-CON-003

I can write a congruence statement with corresponding vertices in the correct order.

Resource: 2.2 Congruent Polygons; SAS

SUPPORTING SKILLS

- I** Match corresponding vertices, sides, and angles in order.
SK-GE0-CORRESPONDENCE
- I** Write congruence statements in corresponding order.
SK-PROOF-CONGRUENCE-ORDER
- A** Recall and correctly use terminology and notation for triangle classification, congruence, and corresponding parts.
SK-TRI-CONGRUENCE-VOCAB · first introduced in Math 22

M22-CON-004

I can determine whether triangles are congruent by SAS.

Resource: 2.2 Congruent Polygons; SAS

SUPPORTING SKILLS

- A** Match corresponding vertices, sides, and angles in order.
SK-GE0-CORRESPONDENCE · first introduced in Math 22
- D** Select a theorem that connects known facts to a target statement.
SK-PROOF-THEOREM-SELECT · first introduced in Math 22
- I** Select SAS, SSS, HL, ASA, or AAS from marked information.
SK-TRI-CONGRUENCE-SELECT

M22-CON-005

I can determine whether triangles are congruent by SSS.

Resource: 2.3 SSS and HL

SUPPORTING SKILLS

- A** Match corresponding vertices, sides, and angles in order.
SK-GE0-CORRESPONDENCE · first introduced in Math 22
- D** Select SAS, SSS, HL, ASA, or AAS from marked information.
SK-TRI-CONGRUENCE-SELECT · first introduced in Math 22

M22-CON-006

I can determine whether right triangles are congruent by HL.

Resource: 2.3 SSS and HL

SUPPORTING SKILLS

- A** Use right-angle relationships created by perpendicular lines.
SK-LIN-PERPENDICULAR-ANGLE · first introduced in Math 22
- D** Select SAS, SSS, HL, ASA, or AAS from marked information.
SK-TRI-CONGRUENCE-SELECT · first introduced in Math 22

M22-CON-007

I can determine whether triangles are congruent by ASA or AAS.

Resource: 2.5 ASA and AAS

SUPPORTING SKILLS

- A** Apply the triangle angle-sum theorem.
SK-TRI-ANGLE-SUM · first introduced in Math 22
- D** Select SAS, SSS, HL, ASA, or AAS from marked information.
SK-TRI-CONGRUENCE-SELECT · first introduced in Math 22

M22-CON-008

I can apply isosceles and equilateral triangle theorems and their converses.

Resource: 2.4 Isosceles and Equilateral Triangles

SUPPORTING SKILLS

- D** Recognize and apply the converse of a conditional statement.
SK-PROOF-CONVERSE · first introduced in Math 22
- I** Apply isosceles-triangle theorems and converses.
SK-TRI-ISOSCELES

M22-CON-009

I can use corresponding parts of congruent triangles after proving congruence.

Resource: 2.2 Congruent Polygons; SAS; 2.5 ASA and AAS

SUPPORTING SKILLS

- A** Write congruence statements in corresponding order.
SK-PROOF-CONGRUENCE-ORDER · first introduced in Math 22
- I** Use corresponding parts only after triangle congruence is established.
SK-PROOF-CPCTC

I can write a structured proof involving triangle congruence.

Resource: 2.3 SSS and HL; 2.4 Isosceles and Equilateral Triangles; 2.5 ASA and AAS

SUPPORTING SKILLS

- A** Use corresponding parts only after triangle congruence is established.
SK-PROOF-CPCTC · first introduced in Math 22
- A** Translate givens into diagram markings and statements.
SK-PROOF-GIVEN · first introduced in Math 22
- D** Organize statements and reasons into a valid proof sequence.
SK-PROOF-STRUCTURE · first introduced in Math 22
- A** Select SAS, SSS, HL, ASA, or AAS from marked information.
SK-TRI-CONGRUENCE-SELECT · first introduced in Math 22

Quadrilaterals, Perimeter, and Area

REQUIRED

Resource coverage: 3.1 Polygon Angles, Perimeter, and Area; 3.2 Parallelograms; 3.3 Rhombi, Rectangles, Squares; 3.4 Trapezoids; 3.5 Kites and Composite Polygons

M22-QUAD-001

I can calculate interior or exterior angle measures of a polygon.

Resource: 3.1 Polygon Angles, Perimeter, and Area

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I** Calculate polygon interior or exterior angle sums.
SK-POLY-ANGLE-SUM

M22-QUAD-002

I can calculate perimeter and area of common polygons.

Resource: 3.1 Polygon Angles, Perimeter, and Area

SUPPORTING SKILLS

- I** Add boundary lengths without omitting or duplicating sides.
SK-GE0-PERIMETER
- D** Select and report appropriate linear, square, or angular units.
SK-GE0-UNITS · first introduced in Math 22
- A** Apply the order of operations to a numerical expression.
SK-NUM-ORDER-OPERATIONS · inherited from middle school

M22-QUAD-003

I can apply properties of parallelograms to determine unknown measures.

Resource: 3.2 Parallelograms

SUPPORTING SKILLS

- A** Translate quantities and relationships into an equation.
SK-ALG-TRANSLATE-EQUATION · first introduced in Math 12
- I** Apply parallelogram properties and tests.
SK-QUAD-PARALLELOGRAM

M22-QUAD-004

I can determine whether a quadrilateral is a parallelogram.

Resource: 3.2 Parallelograms

SUPPORTING SKILLS

- A** Recognize and apply the converse of a conditional statement.
SK-PROOF-CONVERSE · first introduced in Math 22
- D** Apply parallelogram properties and tests.
SK-QUAD-PARALLELOGRAM · first introduced in Math 22

M22-QUAD-005

I can apply properties of rectangles, rhombi, and squares.

Resource: 3.3 Rhombi, Rectangles, Squares

SUPPORTING SKILLS

- I** Use the inclusive hierarchy of quadrilateral classifications.
SK-QUAD-HIERARCHY
- I** Distinguish and apply rectangle, rhombus, and square properties.
SK-QUAD-RECT-RHOM-SQUARE

M22-QUAD-006

I can classify a quadrilateral from its stated or marked properties.

Resource: 3.3 Rhombi, Rectangles, Squares

SUPPORTING SKILLS

- A** Distinguish stated or marked facts from visual appearance.
SK-GE0-DIAGRAM-INFER · first introduced in Math 22
- I** Recall the definitions and defining properties of the major quadrilateral families.
SK-QUAD-DEFINITIONS
- D** Use the inclusive hierarchy of quadrilateral classifications.
SK-QUAD-HIERARCHY · first introduced in Math 22

M22-QUAD-007

I can apply trapezoid and trapezoid-midsegment relationships.

Resource: 3.4 Trapezoids

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I** Apply trapezoid and midsegment relationships.
SK-QUAD-TRAPEZOID

M22-QUAD-008

I can apply properties of kites.

Resource: 3.5 Kites and Composite Polygons

SUPPORTING SKILLS

- I** Apply kite properties.
SK-QUAD-KITE

M22-QUAD-009

I can calculate the perimeter or area of a composite polygon.

Resource: 3.5 Kites and Composite Polygons

SUPPORTING SKILLS

- I** Decompose a composite region into familiar shapes.
SK-GE0-AREA-DECOMPOSE
- D** Add boundary lengths without omitting or duplicating sides.
SK-GE0-PERIMETER · first introduced in Math 22
- A** Select and report appropriate linear, square, or angular units.
SK-GE0-UNITS · first introduced in Math 22

I can justify a quadrilateral conclusion using definitions and theorems.

Resource: 3.2 Parallelograms; 3.3 Rhombi, Rectangles, Squares

SUPPORTING SKILLS

- D** Apply a geometric definition as justification.
SK-PROOF-DEFINITION · first introduced in Math 22
- A** Organize statements and reasons into a valid proof sequence.
SK-PROOF-STRUCTURE · first introduced in Math 22
- D** Select a theorem that connects known facts to a target statement.
SK-PROOF-THEOREM-SELECT · first introduced in Math 22

Similarity and Proportional Reasoning

REQUIRED

Resource coverage: 4.1 Dilations and Similarity; 4.2 Similar Polygons, Perimeter, and Area; 4.3 Similarity by AA; 4.4 Similarity by SSS and SAS; 4.5 Proportionality Theorems

M22-SIM-001

I can determine image lengths and coordinates under a dilation.

Resource: 4.1 Dilations and Similarity

SUPPORTING SKILLS

- A** Match corresponding vertices, sides, and angles in order.
SK-GE0-CORRESPONDENCE · first introduced in Math 22
- D** Calculate and interpret a similarity or dilation scale factor.
SK-SIM-SCALE-FACTOR · first introduced in Math 22

M22-SIM-002

I can identify the scale factor of a dilation or similarity relationship.

Resource: 4.1 Dilations and Similarity

SUPPORTING SKILLS

- A** Use cross multiplication when two ratios form a proportion.
SK-ALG-CROSS-MULTIPLY · inherited from middle school
- D** Calculate and interpret a similarity or dilation scale factor.
SK-SIM-SCALE-FACTOR · first introduced in Math 22
- I** Recall and correctly use the terms similarity, dilation, scale factor, and corresponding parts.
SK-SIM-VOCAB

M22-SIM-003

I can write a similarity statement with corresponding vertices in the correct order.

Resource: 4.2 Similar Polygons, Perimeter, and Area

SUPPORTING SKILLS

- D** Match corresponding vertices, sides, and angles in order.
SK-GE0-CORRESPONDENCE · first introduced in Math 22
- I** Write similarity statements in corresponding order.
SK-PROOF-SIMILARITY-ORDER

M22-SIM-004

I can use proportional corresponding sides to solve similar-polygon problems.

Resource: 4.2 Similar Polygons, Perimeter, and Area

SUPPORTING SKILLS

- A** Use cross multiplication when two ratios form a proportion.
SK-ALG-CROSS-MULTIPLY · inherited from middle school
- A** Match corresponding vertices, sides, and angles in order.
SK-GE0-CORRESPONDENCE · first introduced in Math 22
- I** Set up proportions using corresponding sides.
SK-SIM-PROPORTION

M22-SIM-005

I can relate similarity scale factor to perimeter and area scale factors.

Resource: 4.2 Similar Polygons, Perimeter, and Area

SUPPORTING SKILLS

- I** Relate length scale factor to perimeter and area factors.
SK-SIM-PERIMETER-AREA
- A** Calculate and interpret a similarity or dilation scale factor.
SK-SIM-SCALE-FACTOR · first introduced in Math 22

M22-SIM-006

I can prove triangles similar by AA, SSS, or SAS.

Resource: 4.3 Similarity by AA; 4.4 Similarity by SSS and SAS

SUPPORTING SKILLS

- A** Write similarity statements in corresponding order.
SK-PROOF-SIMILARITY-ORDER · first introduced in Math 22
- D** Organize statements and reasons into a valid proof sequence.
SK-PROOF-STRUCTURE · first introduced in Math 22
- I** Recognize AA triangle similarity.
SK-SIM-AA
- I** Test SSS or SAS triangle similarity.
SK-SIM-SSS-SAS

M22-SIM-007

I can apply triangle proportionality theorems and their converses.

Resource: 4.5 Proportionality Theorems

SUPPORTING SKILLS

- A** Recognize and apply the converse of a conditional statement.
SK-PROOF-CONVERSE · first introduced in Math 22
- I** Apply triangle proportionality relationships.
SK-SIM-TRI-PROPORTIONALITY

M22-SIM-008

I can solve and justify an indirect-measurement or other similarity application.

Resource: 4.3 Similarity by AA; 4.4 Similarity by SSS and SAS

SUPPORTING SKILLS

- A** Select a theorem that connects known facts to a target statement.
SK-PROOF-THEOREM-SELECT · first introduced in Math 22
- D** Set up proportions using corresponding sides.
SK-SIM-PROPORTION · first introduced in Math 22
- A** Calculate and interpret a similarity or dilation scale factor.
SK-SIM-SCALE-FACTOR · first introduced in Math 22

Right Triangles

REQUIRED

Resource coverage: 5.1 Pythagorean Theorem and Triangle Inequalities; 5.2 Special Right Triangles; 5.3 HL Similarity and Right-Triangle Similarity; 5.4 Sine and Cosine; 5.5 Tangent and Solving Right Triangles

M22-RGT-001

I can apply the Pythagorean theorem to determine a missing side.

Resource: 5.1 Pythagorean Theorem and Triangle Inequalities

SUPPORTING SKILLS

- A** Extract perfect-power factors from a radical.
SK-RAD-FACTOR-PERFECT · first introduced in Math 21
- I** Apply the Pythagorean theorem or its converse.
SK-TRI-PYTHAGOREAN

M22-RGT-002

I can use the converse of the Pythagorean theorem to classify a triangle.

Resource: 5.1 Pythagorean Theorem and Triangle Inequalities

SUPPORTING SKILLS

- A** Recognize and apply the converse of a conditional statement.
SK-PROOF-CONVERSE · first introduced in Math 22
- D** Apply the Pythagorean theorem or its converse.
SK-TRI-PYTHAGOREAN · first introduced in Math 22

M22-RGT-003

I can apply triangle-inequality relationships.

Resource: 5.1 Pythagorean Theorem and Triangle Inequalities

SUPPORTING SKILLS

- A** Select a theorem that connects known facts to a target statement.
SK-PROOF-THEOREM-SELECT · first introduced in Math 22
- A** Classify a triangle by side lengths and angle measures.
SK-TRI-CLASSIFY · first introduced in Math 22

M22-RGT-004

I can determine exact side lengths in 45-45-90 and 30-60-90 triangles.

Resource: 5.2 Special Right Triangles

SUPPORTING SKILLS

- A** Distinguish exact values from measured or rounded approximations.
SK-GE0-EXACT-APPROX · first introduced in Math 22
- R** Extract perfect-power factors from a radical.
SK-RAD-FACTOR-PERFECT · first introduced in Math 21
- I** Recall special-right-triangle ratios.
SK-TRI-SPECIAL-RATIOS

M22-RGT-005

I can apply right-triangle similarity relationships.

Resource: 5.3 HL Similarity and Right-Triangle Similarity

SUPPORTING SKILLS

- A** Recognize AA triangle similarity.
SK-SIM-AA · first introduced in Math 22
- D** Set up proportions using corresponding sides.
SK-SIM-PROPORTION · first introduced in Math 22

M22-RGT-006

I can use geometric-mean relationships created by an altitude to the hypotenuse.

Resource: 5.3 HL Similarity and Right-Triangle Similarity

SUPPORTING SKILLS

- A** Set up proportions using corresponding sides.
SK-SIM-PROPORTION · first introduced in Math 22
- I** Apply geometric-mean relationships in a right triangle split by an altitude.
SK-TRI-GEOMETRIC-MEAN

M22-RGT-007

I can select sine, cosine, or tangent for a right-triangle problem.

Resource: 5.4 Sine and Cosine; 5.5 Tangent and Solving Right Triangles

SUPPORTING SKILLS

- A** Add congruence, parallel, perpendicular, and angle markings to a diagram.
SK-Geo-DIAGRAM-MARK · first introduced in Math 22
- I** Recall and correctly use the terms leg, hypotenuse, opposite side, adjacent side, and trigonometric ratio.
SK-TRI-RIGHT-VOCAB
- I** Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
SK-TRI-TRIG-RATIO
- I** Label sides relative to a selected angle.
SK-TRIG-TRIANGLE-LABEL

M22-RGT-008

I can determine a missing side or angle of a right triangle.

Resource: 5.4 Sine and Cosine; 5.5 Tangent and Solving Right Triangles

SUPPORTING SKILLS

- A** Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
SK-TRI-TRIG-RATIO · first introduced in Math 22
- I** Use an inverse trigonometric function to determine a missing angle.
SK-TRI-TRIG-SOLVE-ANGLE
- I** Use a trigonometric ratio to determine a missing side.
SK-TRI-TRIG-SOLVE-SIDE

I can solve a contextual right-triangle problem.

Resource: 5.4 Sine and Cosine; 5.5 Tangent and Solving Right Triangles

SUPPORTING SKILLS

- A** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- D** Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
SK-TRI-TRIG-RATIO · first introduced in Math 22
- A** Use an inverse trigonometric function to determine a missing angle.
SK-TRI-TRIG-SOLVE-ANGLE · first introduced in Math 22
- A** Use a trigonometric ratio to determine a missing side.
SK-TRI-TRIG-SOLVE-SIDE · first introduced in Math 22

I can justify a right-triangle solution using an appropriate theorem or ratio.

Resource: 5.4 Sine and Cosine; 5.5 Tangent and Solving Right Triangles

SUPPORTING SKILLS

- D** Select a theorem that connects known facts to a target statement.
SK-PROOF-THEOREM-SELECT · first introduced in Math 22
- A** Apply the Pythagorean theorem or its converse.
SK-TRI-PYTHAGOREAN · first introduced in Math 22
- A** Recall special-right-triangle ratios.
SK-TRI-SPECIAL-RATIOS · first introduced in Math 22
- A** Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
SK-TRI-TRIG-RATIO · first introduced in Math 22

Circles

REQUIRED

Resource coverage: 6.1 Circle Vocabulary, Circumference, Arc Length; 6.2 Circle, Sector, and Segment Area; 6.3 Angle, Arc, and Chord Relationships; 6.4 Tangent, Secant, and Chord Relationships; 6.5 Tangent/Secant/Chord Angle, Arc, and Segment Relationships

M22-CIR-001

I can identify and name radii, diameters, chords, secants, tangents, arcs, sectors, and segments.

Resource: 6.1 Circle Vocabulary, Circumference, Arc Length

SUPPORTING SKILLS

A Identify circle parts and relationships from a diagram.

SK-CIR-VOCAB · first introduced in Math 22

A Read and write geometric notation correctly.

SK-GEO-NOTATION · first introduced in Math 22

M22-CIR-002

I can calculate circumference and arc length.

Resource: 6.1 Circle Vocabulary, Circumference, Arc Length

SUPPORTING SKILLS

I Relate a central angle to a fraction of a circle.

SK-CIR-ARC-FRACTION

I Calculate arc length from radius and central angle.

SK-CIR-ARC-LENGTH

I Calculate circumference from radius or diameter.

SK-CIR-CIRCUMFERENCE

A Select and report appropriate linear, square, or angular units.

SK-GEO-UNITS · first introduced in Math 22

M22-CIR-003

I can calculate the area of a circle, sector, or segment.

Resource: 6.2 Circle, Sector, and Segment Area

SUPPORTING SKILLS

I Calculate the area of a circle.

SK-CIR-AREA

I Calculate sector area from radius and central angle.

SK-CIR-SECTOR-AREA

I Calculate segment area by subtracting a triangle from a sector.

SK-CIR-SEGMENT-AREA

A Select and report appropriate linear, square, or angular units.

SK-GEO-UNITS · first introduced in Math 22

M22-CIR-004

I can determine angle and arc measures involving central and inscribed angles.

Resource: 6.3 Angle, Arc, and Chord Relationships

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- D** Relate a central angle to a fraction of a circle.
SK-CIR-ARC-FRACTION · first introduced in Math 22
- I** Relate an inscribed angle to its intercepted arc.
SK-CIR-INSCRIBED-ANGLE

M22-CIR-005

I can apply relationships among chords, arcs, and distances from the center.

Resource: 6.3 Angle, Arc, and Chord Relationships; 6.4 Tangent, Secant, and Chord Relationships

SUPPORTING SKILLS

- I** Apply equal-chord, equal-arc, and center-distance relationships.
SK-CIR-CHORD-RELATION
- I** Identify circle parts and relationships from a diagram.
SK-CIR-VOCAB

M22-CIR-006

I can apply tangent-radius and tangent-segment relationships.

Resource: 6.4 Tangent, Secant, and Chord Relationships

SUPPORTING SKILLS

- I** Select and apply an intersecting-segment product relationship.
SK-CIR-SEGMENT-PRODUCT
- I** Use the perpendicular relationship between a radius and tangent.
SK-CIR-TANGENT-RADIUS

M22-CIR-007

I can determine angle measures formed by chords, secants, or tangents.

Resource: 6.5 Tangent/Secant/Chord Angle, Arc, and Segment Relationships

SUPPORTING SKILLS

- I** Select the correct angle formula for chords, secants, or tangents.
SK-CIR-ANGLE-RELATION
- A** Identify circle parts and relationships from a diagram.
SK-CIR-VOCAB · first introduced in Math 22

M22-CIR-008

I can determine segment lengths formed by intersecting chords.

Resource: 6.4 Tangent, Secant, and Chord Relationships; 6.5 Tangent/Secant/Chord Angle, Arc, and Segment Relationships

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I** Select and apply an intersecting-segment product relationship.
SK-CIR-SEGMENT-PRODUCT

M22-CIR-009

I can determine segment lengths formed by secants or tangents.

Resource: 6.5 Tangent/Secant/Chord Angle, Arc, and Segment Relationships

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- D** Select and apply an intersecting-segment product relationship.
SK-CIR-SEGMENT-PRODUCT · first introduced in Math 22

M22-CIR-010

I can solve a multi-step circle problem and justify each relationship used.

Resource: 6.5 Tangent/Secant/Chord Angle, Arc, and Segment Relationships

SUPPORTING SKILLS

- A** Select the correct angle formula for chords, secants, or tangents.
SK-CIR-ANGLE-RELATION · first introduced in Math 22
- A** Select and apply an intersecting-segment product relationship.
SK-CIR-SEGMENT-PRODUCT · first introduced in Math 22
- A** Identify circle parts and relationships from a diagram.
SK-CIR-VOCAB · first introduced in Math 22
- D** Select a theorem that connects known facts to a target statement.
SK-PROOF-THEOREM-SELECT · first introduced in Math 22

Math 31 Curriculum

Advanced Algebra | Middlesex School Mathematics

COURSE AT A GLANCE

Review Content

REVIEW

Primary resources: [2.2 Use a Problem Solving Strategy](#); [5.2 Properties of Exponents and Scientific Notation](#); [5.1 Add and Subtract Polynomials](#); [5.3 Multiply Polynomials](#)

UNIT LEARNING OBJECTIVES

I can calculate a percentage, percent increase, or percent decrease.

I can simplify an expression using exponent properties.

I can use polynomial vocabulary to distinguish terms, factors, coefficients, degree, and polynomial structure.

I can expand a product of two binomials.

Linear Functions and Function Notation

REQUIRED

Primary resources: [College Algebra 2e 3.1 Functions and Function Notation](#); [College Algebra 2e 3.3 Rates of Change and Behavior of Graphs](#); [College Algebra 2e 3.2 Domain and Range](#); [College Algebra 2e 4.1 Linear Functions](#); [Intermediate Algebra 2e 4.1 Solve Systems of Linear Equations with Two Variables](#)

UNIT LEARNING OBJECTIVES

I can interpret function notation as a statement about input and output.

I can determine whether a graph or table represents a function.

I can evaluate a function or solve for an input using a graph, table, or formula.

I can identify independent and dependent variables in a relationship.

I can calculate average rate of change from two points.

I can interpret interval notation and identify increasing or decreasing intervals.

I can determine total change from an average rate of change over an interval.

I can determine and interpret the units of an average rate of change.

I can identify a linear function from a table, graph, or formula.

I can interpret the slope and intercept of a linear model.

I can determine the intersection of two linear relationships graphically or algebraically.

I can write equations of horizontal, vertical, parallel, and perpendicular lines through a specified point.

Function Behavior and Structure

REQUIRED

Primary resources: [3.1 Functions and Function Notation](#); [3.2 Domain and Range](#); [3.5 Transformation of Functions](#); [3.4 Composition of Functions](#); [3.7 Inverse Functions](#); [3.3 Rates of Change and Behavior of Graphs](#)

UNIT LEARNING OBJECTIVES

I can determine whether a relation is a function.

I can evaluate a function from a formula, graph, or table.

I can determine domain and range from a formula, graph, or table.

I can determine algebraic domain restrictions caused by denominators and even roots.

I can graph a piecewise function from its formula.

I can write a piecewise formula that represents a graph.

I can graph a transformed function using shifts, stretches, compressions, and reflections.

I can describe how a transformation affects domain, range, or asymptotes.

I can evaluate a composition from formulas, graphs, or tables.

I can write a formula for a composition of two functions.

I can find an inverse function algebraically.

I can evaluate and interpret an inverse from a graph, formula, or table.

I can determine and interpret the units of an inverse function.

I can explain how the domain and range of a function relate to those of its inverse.

I can determine the range of a function by analyzing the domain of its inverse.

I can verify algebraically that two functions are inverses.

I can interpret the meaning and units of a composition.

I can classify a graph or table as concave up or concave down using changes in average rate of change.

Function Behavior and Structure (Continued)

REQUIRED

UNIT LEARNING OBJECTIVES - CONTINUED

I can determine whether a function is invertible using the horizontal line test.

I can graph an inverse function as a reflection across the line $y = x$.

Quadratic Functions and Equations

REQUIRED

Primary resources: [5.1 Quadratic Functions](#); [2.5 Quadratic Equations](#)
Supplemental resources: [5.3 Graphs of Polynomial Functions](#); [5.5 Zeros of Polynomial Functions](#)

UNIT LEARNING OBJECTIVES

I can identify the vertex, intercepts, axis of symmetry, and concavity of a quadratic function.

I can connect standard, factored, and vertex forms to features of a quadratic graph.

I can rewrite a quadratic function in a form that reveals a requested feature.

I can solve a quadratic equation by factoring or using the quadratic formula.

I can graph a quadratic function using its algebraic structure and key features.

I can construct a quadratic function from specified zeros, a vertex, or graphical information.

I can create and interpret a quadratic trajectory model.

Exponential Functions and Models

REQUIRED

Primary resources: [6.1 Exponential Functions](#); [6.2 Graphs of Exponential Functions](#)

UNIT LEARNING OBJECTIVES

I can distinguish linear change from exponential change using differences, ratios, or percent change.

I can identify initial value and growth or decay factor in an exponential model.

I can write an exponential function from a table, graph, or context.

I can evaluate and interpret an exponential model.

I can calculate growth under periodic compounding.

I can calculate growth under continuous compounding.

I can determine an exponential formula from two points or other sufficient data.

I can determine domain, range, intercepts, and asymptotes of an exponential function.

I can compare the growth rates of two exponential functions.

I can determine the percent growth or decay represented by a continuous exponential model.

Logarithmic Functions and Models

REQUIRED

Primary resources: [6.3 Logarithmic Functions](#); [6.5 Logarithmic Properties](#); [6.6 Exponential and Logarithmic Equations](#); [6.7 Exponential and Logarithmic Models](#); [6.4 Graphs of Logarithmic Functions](#)
Supplemental resources: [6.2 Graphs of Exponential Functions](#)

UNIT LEARNING OBJECTIVES

I can convert between logarithmic and exponential forms.

I can evaluate a logarithm using its exponential meaning.

I can expand or condense an expression using product, quotient, and power properties.

I can solve an exponential equation using logarithms.

I can solve a logarithmic equation using exponential form and check domain restrictions.

I can calculate and interpret doubling time or half-life.

I can determine domain, range, intercepts, and asymptotes of a logarithmic function.

I can explain how exponential and logarithmic functions undo one another.

I can solve a multi-step exponential or logarithmic equation.

I can determine an intersection of exponential functions graphically or algebraically.

I can rewrite a discrete exponential model as an equivalent continuous exponential model.

I can solve and interpret an exponential or logarithmic modeling problem.

Extension Content

EXTENSION

Primary resources: [3.6 Absolute Value Functions](#); [3.5 Transformation of Functions](#); [6.7 Exponential and Logarithmic Models](#)

UNIT LEARNING OBJECTIVES

I can analyze an absolute-value function using transformations.

I can determine even or odd symmetry from a graph or formula.

I can interpret a contextual logarithmic scale.

IN-DEPTH CURRICULUM

SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Review Content

REVIEW

OpenStax coverage: **Primary resources:** [2.2 Use a Problem Solving Strategy](#); [5.2 Properties of Exponents and Scientific Notation](#); [5.1 Add and Subtract Polynomials](#); [5.3 Multiply Polynomials](#)

M31-REV-001

I can calculate a percentage, percent increase, or percent decrease.

OpenStax: **Primary resources:** [2.2 Use a Problem Solving Strategy](#)

SUPPORTING SKILLS

- Calculate a percentage of a quantity or determine what percent one quantity is of another.
SK-NUM-PERCENT-CALCULATE · inherited from middle school
- Calculate and interpret percent increase or decrease.
SK-NUM-PERCENT-CHANGE · inherited from middle school

M31-REV-002

I can simplify an expression using exponent properties.

OpenStax: **Primary resources:** [5.2 Properties of Exponents and Scientific Notation](#)

SUPPORTING SKILLS

- Rewrite a negative exponent using a reciprocal.
SK-EXP-NEGATIVE · first introduced in Math 21
- Apply the power rule for exponents.
SK-EXP-POWER · first introduced in Math 12
- Apply the product rule for exponents.
SK-EXP-PRODUCT · first introduced in Math 12
- Apply the quotient rule for exponents.
SK-EXP-QUOTIENT · first introduced in Math 12
- Apply the zero-exponent rule.
SK-EXP-ZERO · first introduced in Math 21

M31-REV-003

I can use polynomial vocabulary to distinguish terms, factors, coefficients, degree, and polynomial structure.

OpenStax: Primary resources: [5.1 Add and Subtract Polynomials](#)

SUPPORTING SKILLS

- R** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M31-REV-004

I can expand a product of two binomials.

OpenStax: Primary resources: [5.3 Multiply Polynomials](#)

SUPPORTING SKILLS

- A** Apply the distributive property in either direction.
SK-ALG-DISTRIBUTE · inherited from middle school
- R** Multiply two binomials using distribution.
SK-POLY-BINOMIAL-MULTIPLY · first introduced in Math 12

Linear Functions and Function Notation

REQUIRED

OpenStax coverage: **Primary resources:** [College Algebra 2e 3.1 Functions and Function Notation](#); [College Algebra 2e 3.3 Rates of Change and Behavior of Graphs](#); [College Algebra 2e 3.2 Domain and Range](#); [College Algebra 2e 4.1 Linear Functions](#); [Intermediate Algebra 2e 4.1 Solve Systems of Linear Equations with Two Variables](#)

M31-LIN-001

I can interpret function notation as a statement about input and output.

OpenStax: **Primary resources:** [3.1 Functions and Function Notation](#)

SUPPORTING SKILLS

- D** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- A** Identify the input and output represented by function notation.
SK-FUNC-NOTATION-READ · first introduced in Math 12
- I** Use function notation to express and solve an input-output question.
SK-FUNC-NOTATION-SOLVE

M31-LIN-002

I can determine whether a graph or table represents a function.

OpenStax: **Primary resources:** [3.1 Functions and Function Notation](#)

SUPPORTING SKILLS

- D** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D** Apply the vertical line test to a graph.
SK-FUNC-VERTICAL-LINE · first introduced in Math 12

M31-LIN-003

I can evaluate a function or solve for an input using a graph, table, or formula.

OpenStax: **Primary resources:** [3.1 Functions and Function Notation](#)

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- D** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- A** Use function notation to express and solve an input-output question.
SK-FUNC-NOTATION-SOLVE · first introduced in Math 31
- D** Solve for an input associated with a specified output.
SK-FUNC-SOLVE-INPUT · first introduced in Math 21

M31-LIN-004

I can identify independent and dependent variables in a relationship.

OpenStax: **Primary resources:** [3.1 Functions and Function Notation](#)

SUPPORTING SKILLS

- I** Identify independent and dependent variables.
SK-FUNC-INDEPENDENT-DEPENDENT
- A** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12

M31-LIN-005

I can calculate average rate of change from two points.

OpenStax: **Primary resources:** [3.3 Rates of Change and Behavior of Graphs](#)

SUPPORTING SKILLS

- D** Calculate average rate of change over an interval.
SK-FUNC-AROC · first introduced in Math 21
- A** Calculate slope from two points.
SK-LIN-SLOPE-POINTS · first introduced in Math 12

M31-LIN-006

I can interpret interval notation and identify increasing or decreasing intervals.

OpenStax: **Primary resources:** [3.2 Domain and Range](#)

SUPPORTING SKILLS

- I** Describe increasing or decreasing behavior over intervals.
SK-FUNC-INTERVAL-BEHAVIOR
- D** Translate between inequality and interval notation.
SK-NOT-INTERVAL · first introduced in Math 12

M31-LIN-007

I can determine total change from an average rate of change over an interval.

OpenStax: **Primary resources:** [3.3 Rates of Change and Behavior of Graphs](#)

SUPPORTING SKILLS

- A** Calculate average rate of change over an interval.
SK-FUNC-AROC · first introduced in Math 21
- I** Use average rate of change and interval length to calculate total change.
SK-FUNC-AROC-TOTAL-CHANGE
- A** Add, subtract, multiply, and divide signed integers.
SK-NUM-INTEGGER-OPERATE · inherited from middle school

M31-LIN-008

I can determine and interpret the units of an average rate of change.

OpenStax: **Primary resources:** [3.3 Rates of Change and Behavior of Graphs](#)

SUPPORTING SKILLS

- I** Determine rate units from input- and output-axis units.
SK-FUNC-AROC-UNITS
- D** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

M31-LIN-009

I can identify a linear function from a table, graph, or formula.

OpenStax: **Primary resources:** [4.1 Linear Functions](#)

SUPPORTING SKILLS

- I** Use constant differences or ratios to distinguish linear and exponential patterns.
SK-EXP-DIFFERENCE-RATIO
- D** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21
- D** Use tabular patterns to predict graph shape and behavior.
SK-REP-TABLE-GRAPH · first introduced in Math 21

M31-LIN-010

I can interpret the slope and intercept of a linear model.

OpenStax: **Primary resources:** [4.1 Linear Functions](#)

SUPPORTING SKILLS

- A** Plot an intercept accurately on a coordinate plane.
SK-LIN-PLOT-INTERCEPT · first introduced in Math 12
- A** Determine slope from a graph.
SK-LIN-SLOPE-GRAPH · first introduced in Math 12
- D** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

M31-LIN-011

I can determine the intersection of two linear relationships graphically or algebraically.

OpenStax: **Primary resources:** [4.1 Solve Systems of Linear Equations with Two Variables](#)

SUPPORTING SKILLS

- A** Check an ordered pair in both equations of a system.
SK-SYS-CHECK · first introduced in Math 12
- A** Add equations to eliminate a variable.
SK-SYS-ELIMINATE · first introduced in Math 12
- A** Locate and interpret the intersection of two lines.
SK-SYS-GRAPH-INTERSECTION · first introduced in Math 12
- A** Substitute an equivalent expression for a variable.
SK-SYS-SUBSTITUTE · first introduced in Math 12

M31-LIN-012

I can write equations of horizontal, vertical, parallel, and perpendicular lines through a specified point.

OpenStax: **Primary resources:** [4.1 Linear Functions](#)

SUPPORTING SKILLS

- A** Translate quantities and relationships into an equation.
SK-ALG-TRANSLATE-EQUATION · first introduced in Math 12
- A** Convert among common forms of a linear equation.
SK-LIN-FORM-CONVERT · first introduced in Math 12
- D** Recognize that distinct parallel lines have equal slopes.
SK-LIN-PARALLEL-SLOPE · first introduced in Math 12
- D** Determine the negative reciprocal of a nonzero slope.
SK-LIN-PERPENDICULAR-SLOPE · first introduced in Math 12

Function Behavior and Structure

REQUIRED

OpenStax coverage: **Primary resources:** [3.1 Functions and Function Notation](#); [3.2 Domain and Range](#); [3.5 Transformation of Functions](#); [3.4 Composition of Functions](#); [3.7 Inverse Functions](#); [3.3 Rates of Change and Behavior of Graphs](#)

M31-FUN-001

I can determine whether a relation is a function.

OpenStax: **Primary resources:** [3.1 Functions and Function Notation](#)

SUPPORTING SKILLS

- D** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- A** Apply the vertical line test to a graph.
SK-FUNC-VERTICAL-LINE · first introduced in Math 12

M31-FUN-002

I can evaluate a function from a formula, graph, or table.

OpenStax: **Primary resources:** [3.1 Functions and Function Notation](#)

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- D** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- A** Solve for an input associated with a specified output.
SK-FUNC-SOLVE-INPUT · first introduced in Math 21

M31-FUN-003

I can determine domain and range from a formula, graph, or table.

OpenStax: **Primary resources:** [3.2 Domain and Range](#)

SUPPORTING SKILLS

- I** Determine domain restrictions from denominators, even roots, and logarithms.
SK-FUNC-DOMAIN-FORMULA
- D** Read domain and range from a graph.
SK-FUNC-DOMAIN-RANGE-GRAPH · first introduced in Math 12
- D** Read domain and range from a table.
SK-FUNC-DOMAIN-RANGE-TABLE · first introduced in Math 12

M31-FUN-004

I can determine algebraic domain restrictions caused by denominators and even roots.

OpenStax: **Primary resources:** [3.2 Domain and Range](#)

SUPPORTING SKILLS

- A** Factor a quadratic trinomial with a nonunit leading coefficient.
SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- A** Factor a monic quadratic trinomial.
SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- D** Require an even-root radicand to be nonnegative when determining domain.
SK-FUNC-DOMAIN-EVEN-ROOT · first introduced in Math 21
- D** Determine domain restrictions from denominators, even roots, and logarithms.
SK-FUNC-DOMAIN-FORMULA · first introduced in Math 31
- A** Determine excluded values from an original denominator.
SK-RAT-RESTRICTIONS · first introduced in Math 21

M31-FUN-005

I can graph a piecewise function from its formula.

OpenStax: **Primary resources:** [3.2 Domain and Range](#)

SUPPORTING SKILLS

- A** Describe increasing or decreasing behavior over intervals.
SK-FUNC-INTERVAL-BEHAVIOR · first introduced in Math 31
- I** Select the correct rule of a piecewise function for a given input.
SK-FUNC-PIECEWISE-EVALUATE
- I** Graph each rule on its stated domain with correct endpoints.
SK-FUNC-PIECEWISE-GRAPH

M31-FUN-006

I can write a piecewise formula that represents a graph.

OpenStax: **Primary resources:** [3.2 Domain and Range](#)

SUPPORTING SKILLS

- A** Describe increasing or decreasing behavior over intervals.
SK-FUNC-INTERVAL-BEHAVIOR · first introduced in Math 31
- I** Translate a segmented graph into a piecewise formula.
SK-FUNC-PIECEWISE-WRITE
- A** Convert among common forms of a linear equation.
SK-LIN-FORM-CONVERT · first introduced in Math 12

M31-FUN-007

I can graph a transformed function using shifts, stretches, compressions, and reflections.

OpenStax: **Primary resources:** [3.5 Transformation of Functions](#)

SUPPORTING SKILLS

- I** Interpret transformations applied to function inputs.
SK-FUNC-TRANSFORM-HORIZONTAL
- I** Distinguish stretches from compressions and horizontal effects from vertical effects.
SK-FUNC-TRANSFORM-SCALE
- I** Interpret transformations applied to function outputs.
SK-FUNC-TRANSFORM-VERTICAL
- A** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-FUN-008

I can describe how a transformation affects domain, range, or asymptotes.

OpenStax: **Primary resources:** [3.5 Transformation of Functions](#)

SUPPORTING SKILLS

- I** Exchange domain and range between a function and its inverse.
SK-FUNC-INVERSE-DOMAIN-RANGE
- D** Interpret transformations applied to function inputs.
SK-FUNC-TRANSFORM-HORIZONTAL · first introduced in Math 31
- D** Distinguish stretches from compressions and horizontal effects from vertical effects.
SK-FUNC-TRANSFORM-SCALE · first introduced in Math 31
- D** Interpret transformations applied to function outputs.
SK-FUNC-TRANSFORM-VERTICAL · first introduced in Math 31

M31-FUN-009

I can evaluate a composition from formulas, graphs, or tables.

OpenStax: **Primary resources:** [3.4 Composition of Functions](#)

SUPPORTING SKILLS

- I** Substitute one function formula into another.
SK-FUNC-COMPOSE-FORMULA
- I** Follow intermediate values to compose functions from tables.
SK-FUNC-COMPOSE-TABLE
- A** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- I** Read and write notation for function composition and inverse functions.
SK-FUNC-STRUCTURE-NOTATION

M31-FUN-010

I can write a formula for a composition of two functions.

OpenStax: **Primary resources:** [3.4 Composition of Functions](#)

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- D** Substitute one function formula into another.
SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31

M31-FUN-011

I can find an inverse function algebraically.

OpenStax: **Primary resources:** [3.7 Inverse Functions](#)

SUPPORTING SKILLS

- A** Isolate a specified variable in an equation or formula.
SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- I** Exchange variables and solve to obtain an inverse formula.
SK-FUNC-INVERSE-ALGEBRA
- I** Exchange input and output values when interpreting an inverse.
SK-FUNC-INVERSE-SWAP
- A** Read and write notation for function composition and inverse functions.
SK-FUNC-STRUCTURE-NOTATION · first introduced in Math 31

M31-FUN-012

I can evaluate and interpret an inverse from a graph, formula, or table.

OpenStax: **Primary resources:** [3.7 Inverse Functions](#)

SUPPORTING SKILLS

- A** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- A** Exchange variables and solve to obtain an inverse formula.
SK-FUNC-INVERSE-ALGEBRA · first introduced in Math 31
- D** Exchange input and output values when interpreting an inverse.
SK-FUNC-INVERSE-SWAP · first introduced in Math 31

M31-FUN-013

I can determine and interpret the units of an inverse function.

OpenStax: **Primary resources:** [3.7 Inverse Functions](#)

SUPPORTING SKILLS

- I** Reverse input and output units when interpreting an inverse.
SK-FUNC-INVERSE-UNITS
- A** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

M31-FUN-014

I can explain how the domain and range of a function relate to those of its inverse.

OpenStax: **Primary resources:** [3.7 Inverse Functions](#)

SUPPORTING SKILLS

- I** Exchange domain and range between a function and its inverse.
SK-FUNC-INVERSE-DOMAIN-RANGE
- A** Exchange input and output values when interpreting an inverse.
SK-FUNC-INVERSE-SWAP · first introduced in Math 31

M31-FUN-015

I can determine the range of a function by analyzing the domain of its inverse.

OpenStax: **Primary resources:** [3.7 Inverse Functions](#)

SUPPORTING SKILLS

- A** Determine domain restrictions from denominators, even roots, and logarithms.
SK-FUNC-DOMAIN-FORMULA · first introduced in Math 31
- D** Exchange domain and range between a function and its inverse.
SK-FUNC-INVERSE-DOMAIN-RANGE · first introduced in Math 31

M31-FUN-016

I can verify algebraically that two functions are inverses.

OpenStax: **Primary resources:** [3.7 Inverse Functions](#)

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- A** Substitute one function formula into another.
SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- I** Verify inverse functions using composition.
SK-FUNC-INVERSE-VERIFY

M31-FUN-017

I can interpret the meaning and units of a composition.

OpenStax: **Primary resources:** [3.4 Composition of Functions](#)

SUPPORTING SKILLS

- D** Substitute one function formula into another.
SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- D** Follow intermediate values to compose functions from tables.
SK-FUNC-COMPOSE-TABLE · first introduced in Math 31
- A** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- D** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

M31-FUN-018

I can classify a graph or table as concave up or concave down using changes in average rate of change.

OpenStax: **Primary resources:** [3.3 Rates of Change and Behavior of Graphs](#)

SUPPORTING SKILLS

- A** Calculate average rate of change over an interval.
SK-FUNC-AROC · first introduced in Math 21
- I** Use successive average rates of change to classify concavity.
SK-FUNC-CONCAVITY-AROC
- D** Use tabular patterns to predict graph shape and behavior.
SK-REP-TABLE-GRAPH · first introduced in Math 21

M31-FUN-019

I can determine whether a function is invertible using the horizontal line test.

OpenStax: **Primary resources:** [3.7 Inverse Functions](#)

SUPPORTING SKILLS

- I** Apply the horizontal line test for invertibility.
SK-FUNC-INVERSE-HLT
- I** Test whether a function is one-to-one on a domain.
SK-FUNC-ONE-TO-ONE

M31-FUN-020

I can graph an inverse function as a reflection across the line $y = x$.

OpenStax: **Primary resources:** [3.7 Inverse Functions](#)

SUPPORTING SKILLS

- A** Exchange domain and range between a function and its inverse.
SK-FUNC-INVERSE-DOMAIN-RANGE · first introduced in Math 31
- D** Exchange input and output values when interpreting an inverse.
SK-FUNC-INVERSE-SWAP · first introduced in Math 31
- D** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21

Quadratic Functions and Equations

REQUIRED

OpenStax coverage: **Primary resources:** [5.1 Quadratic Functions](#); [2.5 Quadratic Equations](#)

Supplemental resources: [5.3 Graphs of Polynomial Functions](#); [5.5 Zeros of Polynomial Functions](#)

M31-QUAD-001

I can identify the vertex, intercepts, axis of symmetry, and concavity of a quadratic function.

OpenStax: **Primary resources:** [5.1 Quadratic Functions](#)

SUPPORTING SKILLS

- A** Use successive average rates of change to classify concavity.
SK-FUNC-CONCAVITY-AROC · first introduced in Math 31
 - A** Identify standard, factored, and vertex forms.
SK-QUAD-FORM-IDENTIFY · first introduced in Math 21
 - D** Determine x- and y-intercepts of a quadratic function.
SK-QUAD-INTERCEPTS · first introduced in Math 21
 - D** Determine the vertex and axis of symmetry.
SK-QUAD-VERTEX · first introduced in Math 21
 - R** Recall and correctly use quadratic-function terms including parabola, vertex, axis of symmetry, intercept, zero, and solution.
SK-QUAD-VOCABULARY · first introduced in Math 21
-

M31-QUAD-002

I can connect standard, factored, and vertex forms to features of a quadratic graph.

OpenStax: **Primary resources:** [5.1 Quadratic Functions](#)

Supplemental resources: [5.3 Graphs of Polynomial Functions](#)

SUPPORTING SKILLS

- D** Identify standard, factored, and vertex forms.
SK-QUAD-FORM-IDENTIFY · first introduced in Math 21
 - A** Determine x- and y-intercepts of a quadratic function.
SK-QUAD-INTERCEPTS · first introduced in Math 21
 - A** Determine the vertex and axis of symmetry.
SK-QUAD-VERTEX · first introduced in Math 21
 - D** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21
-

M31-QUAD-003

I can rewrite a quadratic function in a form that reveals a requested feature.

OpenStax: **Primary resources:** [5.1 Quadratic Functions](#)

SUPPORTING SKILLS

- A** Factor a quadratic trinomial with a nonunit leading coefficient.
SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- A** Factor a monic quadratic trinomial.
SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- D** Create a perfect-square trinomial by completing the square.
SK-QUAD-COMPLETE-SQUARE · first introduced in Math 21
- D** Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.
SK-QUAD-FORM-CONVERT · first introduced in Math 21

M31-QUAD-004

I can solve a quadratic equation by factoring or using the quadratic formula.

OpenStax: **Primary resources:** [2.5 Quadratic Equations](#)

SUPPORTING SKILLS

- A** Apply the zero-product property.
SK-FACTOR-ZERO-PRODUCT · first introduced in Math 21
- A** Use the discriminant to predict the number and type of solutions.
SK-QUAD-DISCRIMINANT · first introduced in Math 21
- A** Substitute coefficients accurately into the quadratic formula.
SK-QUAD-FORMULA · first introduced in Math 21
- D** Select an efficient quadratic-solving method from equation structure.
SK-QUAD-METHOD-SELECT · first introduced in Math 21

M31-QUAD-005

I can graph a quadratic function using its algebraic structure and key features.

OpenStax: **Primary resources:** [5.1 Quadratic Functions](#)

SUPPORTING SKILLS

- A** Identify standard, factored, and vertex forms.
SK-QUAD-FORM-IDENTIFY · first introduced in Math 21
- D** Determine x- and y-intercepts of a quadratic function.
SK-QUAD-INTERCEPTS · first introduced in Math 21
- D** Determine the vertex and axis of symmetry.
SK-QUAD-VERTEX · first introduced in Math 21
- D** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-QUAD-006

I can construct a quadratic function from specified zeros, a vertex, or graphical information.

OpenStax: **Primary resources:** [5.1 Quadratic Functions](#)

Supplemental resources: [5.5 Zeros of Polynomial Functions](#)

SUPPORTING SKILLS

- A** Translate quantities and relationships into an equation.
SK-ALG-TRANSLATE-EQUATION · first introduced in Math 12
- I** Build a polynomial from zeros, multiplicities, and a point.
SK-POLY-CONSTRUCT
- A** Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.
SK-QUAD-FORM-CONVERT · first introduced in Math 21

M31-QUAD-007

I can create and interpret a quadratic trajectory model.

OpenStax: **Primary resources:** [5.1 Quadratic Functions](#)

SUPPORTING SKILLS

- D** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- D** Interpret vertex and zeros in an application.
SK-QUAD-MODEL-FEATURES · first introduced in Math 21
- A** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

Exponential Functions and Models

REQUIRED

OpenStax coverage: **Primary resources:** [6.1 Exponential Functions](#); [6.2 Graphs of Exponential Functions](#)

M31-EXP-001

I can distinguish linear change from exponential change using differences, ratios, or percent change.

OpenStax: **Primary resources:** [6.1 Exponential Functions](#)

SUPPORTING SKILLS

- D** Use constant differences or ratios to distinguish linear and exponential patterns.
SK-EXP-DIFFERENCE-RATIO · first introduced in Math 31
- D** Compare functions shown in different representations.
SK-FUNC-COMPARE-REP · first introduced in Math 21
- A** Calculate and interpret percent increase or decrease.
SK-NUM-PERCENT-CHANGE · inherited from middle school

M31-EXP-002

I can identify initial value and growth or decay factor in an exponential model.

OpenStax: **Primary resources:** [6.1 Exponential Functions](#)

SUPPORTING SKILLS

- I** Identify initial amount, rate, and time interval in context.
SK-EXP-MODEL-CONTEXT
- I** Convert a percent increase or decrease to a growth or decay factor.
SK-EXP-PERCENT-FACTOR
- I** Recall and correctly use exponential-function terms including initial value, growth or decay factor, growth or decay rate, and asymptote.
SK-EXP-VOCABULARY
- D** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21

M31-EXP-003

I can write an exponential function from a table, graph, or context.

OpenStax: **Primary resources:** [6.1 Exponential Functions](#)

SUPPORTING SKILLS

- D** Identify initial amount, rate, and time interval in context.
SK-EXP-MODEL-CONTEXT · first introduced in Math 31
- I** Determine an exponential model from tabular values.
SK-EXP-MODEL-TABLE
- I** Solve for an exponential model using two points.
SK-EXP-MODEL-TWO-POINTS
- D** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-EXP-004

I can evaluate and interpret an exponential model.

OpenStax: **Primary resources:** [6.1 Exponential Functions](#)

SUPPORTING SKILLS

- D** Identify initial amount, rate, and time interval in context.
SK-EXP-MODEL-CONTEXT · first introduced in Math 31
- A** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

M31-EXP-005

I can calculate growth under periodic compounding.

OpenStax: **Primary resources:** [6.1 Exponential Functions](#)

SUPPORTING SKILLS

- I** Substitute correctly into a periodic-compounding model.
SK-EXP-COMPOUND-PERIODIC
- A** Convert a percent increase or decrease to a growth or decay factor.
SK-EXP-PERCENT-FACTOR · first introduced in Math 31
- A** Calculate a percentage of a quantity or determine what percent one quantity is of another.
SK-NUM-PERCENT-CALCULATE · inherited from middle school

M31-EXP-006

I can calculate growth under continuous compounding.

OpenStax: **Primary resources:** [6.1 Exponential Functions](#)

SUPPORTING SKILLS

- I** Substitute correctly into a continuous-compounding model.
SK-EXP-COMPOUND-CONTINUOUS
- A** Identify initial amount, rate, and time interval in context.
SK-EXP-MODEL-CONTEXT · first introduced in Math 31

M31-EXP-007

I can determine an exponential formula from two points or other sufficient data.

OpenStax: **Primary resources:** [6.1 Exponential Functions](#)

SUPPORTING SKILLS

- A** Isolate a specified variable in an equation or formula.
SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- D** Solve for an exponential model using two points.
SK-EXP-MODEL-TWO-POINTS · first introduced in Math 31
- A** Interpret numerator and denominator roles in a rational exponent.
SK-EXP-RATIONAL · first introduced in Math 21

M31-EXP-008

I can determine domain, range, intercepts, and asymptotes of an exponential function.

OpenStax: **Primary resources:** [6.2 Graphs of Exponential Functions](#)

SUPPORTING SKILLS

- I** Identify horizontal asymptotic behavior of an exponential graph.
SK-EXP-ASYMPTOTE
- D** Determine domain restrictions from denominators, even roots, and logarithms.
SK-FUNC-DOMAIN-FORMULA · first introduced in Math 31
- D** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-EXP-009

I can compare the growth rates of two exponential functions.

OpenStax: **Primary resources:** [6.1 Exponential Functions](#)

SUPPORTING SKILLS

- I** Compare exponential growth rates expressed in different forms.
SK-EXP-COMPARE-GROWTH
- D** Compare functions shown in different representations.
SK-FUNC-COMPARE-REP · first introduced in Math 21

M31-EXP-010

I can determine the percent growth or decay represented by a continuous exponential model.

OpenStax: **Primary resources:** [6.1 Exponential Functions](#)

SUPPORTING SKILLS

- I** Convert between equivalent discrete- and continuous-growth forms.
SK-EXP-DISCRETE-CONTINUOUS
- D** Convert a percent increase or decrease to a growth or decay factor.
SK-EXP-PERCENT-FACTOR · first introduced in Math 31
- A** Calculate and interpret percent increase or decrease.
SK-NUM-PERCENT-CHANGE · inherited from middle school

Logarithmic Functions and Models

REQUIRED

OpenStax coverage: **Primary resources:** [6.3 Logarithmic Functions](#); [6.5 Logarithmic Properties](#); [6.6 Exponential and Logarithmic Equations](#); [6.7 Exponential and Logarithmic Models](#); [6.4 Graphs of Logarithmic Functions](#)
Supplemental resources: [6.2 Graphs of Exponential Functions](#)

M31-LOG-001

I can convert between logarithmic and exponential forms.

OpenStax: **Primary resources:** [6.3 Logarithmic Functions](#)

SUPPORTING SKILLS

- I** Convert between logarithmic and exponential forms.
SK-LOG-CONVERT
- I** Interpret a logarithm as an exponent.
SK-LOG-DEFINITION
- I** Read logarithmic notation and identify the base, argument, and corresponding exponential statement.
SK-LOG-NOTATION-READ

M31-LOG-002

I can evaluate a logarithm using its exponential meaning.

OpenStax: **Primary resources:** [6.3 Logarithmic Functions](#)

SUPPORTING SKILLS

- I** Rewrite exponential expressions using a common base when possible.
SK-EXP-REWRITE-BASE
- D** Interpret a logarithm as an exponent.
SK-LOG-DEFINITION · first introduced in Math 31

M31-LOG-003

I can expand or condense an expression using product, quotient, and power properties.

OpenStax: **Primary resources:** [6.5 Logarithmic Properties](#)

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I** Apply the logarithm power property in either direction.
SK-LOG-POWER
- I** Apply the logarithm product property in either direction.
SK-LOG-PRODUCT
- I** Apply the logarithm quotient property in either direction.
SK-LOG-QUOTIENT

M31-LOG-004

I can solve an exponential equation using logarithms.

OpenStax: **Primary resources:** [6.6 Exponential and Logarithmic Equations](#)

SUPPORTING SKILLS

- A** Isolate a specified variable in an equation or formula.
SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- A** Apply the logarithm power property in either direction.
SK-LOG-POWER · first introduced in Math 31
- I** Isolate an exponential expression and apply a logarithm.
SK-LOG-SOLVE-EXP

M31-LOG-005

I can solve a logarithmic equation using exponential form and check domain restrictions.

OpenStax: **Primary resources:** [6.6 Exponential and Logarithmic Equations](#)

SUPPORTING SKILLS

- A** Convert between logarithmic and exponential forms.
SK-LOG-CONVERT · first introduced in Math 31
- I** Reject solutions that make a logarithm argument nonpositive.
SK-LOG-DOMAIN-CHECK
- I** Combine or isolate logarithms before converting to exponential form.
SK-LOG-SOLVE-LOG

M31-LOG-006

I can calculate and interpret doubling time or half-life.

OpenStax: **Primary resources:** [6.7 Exponential and Logarithmic Models](#)

SUPPORTING SKILLS

- I** Translate a doubling-time or half-life question into an exponential equation.
SK-LOG-DOUBLING-HALF
- A** Isolate an exponential expression and apply a logarithm.
SK-LOG-SOLVE-EXP · first introduced in Math 31
- D** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

M31-LOG-007

I can determine domain, range, intercepts, and asymptotes of a logarithmic function.

OpenStax: **Primary resources:** [6.4 Graphs of Logarithmic Functions](#)

SUPPORTING SKILLS

- D** Determine domain restrictions from denominators, even roots, and logarithms.
SK-FUNC-DOMAIN-FORMULA · first introduced in Math 31
- I** Identify vertical asymptotic behavior of a logarithmic graph.
SK-LOG-ASYMPTOTE
- A** Reject solutions that make a logarithm argument nonpositive.
SK-LOG-DOMAIN-CHECK · first introduced in Math 31
- D** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-LOG-008

I can explain how exponential and logarithmic functions undo one another.

OpenStax: **Primary resources:** [6.3 Logarithmic Functions](#)

SUPPORTING SKILLS

- A** Exchange input and output values when interpreting an inverse.
SK-FUNC-INVERSE-SWAP · first introduced in Math 31
- A** Verify inverse functions using composition.
SK-FUNC-INVERSE-VERIFY · first introduced in Math 31
- D** Interpret a logarithm as an exponent.
SK-LOG-DEFINITION · first introduced in Math 31

I can solve a multi-step exponential or logarithmic equation.

OpenStax: **Primary resources:** [6.6 Exponential and Logarithmic Equations](#)

SUPPORTING SKILLS

- A** Reject solutions that make a logarithm argument nonpositive.
SK-LOG-DOMAIN-CHECK · first introduced in Math 31
- A** Apply the logarithm power property in either direction.
SK-LOG-POWER · first introduced in Math 31
- A** Apply the logarithm product property in either direction.
SK-LOG-PRODUCT · first introduced in Math 31
- A** Apply the logarithm quotient property in either direction.
SK-LOG-QUOTIENT · first introduced in Math 31
- D** Isolate an exponential expression and apply a logarithm.
SK-LOG-SOLVE-EXP · first introduced in Math 31
- D** Combine or isolate logarithms before converting to exponential form.
SK-LOG-SOLVE-LOG · first introduced in Math 31

I can determine an intersection of exponential functions graphically or algebraically.

OpenStax: **Primary resources:** [6.6 Exponential and Logarithmic Equations](#)

Supplemental resources: [6.2 Graphs of Exponential Functions](#)

SUPPORTING SKILLS

- A** Compare exponential growth rates expressed in different forms.
SK-EXP-COMPARE-GROWTH · first introduced in Math 31
- D** Isolate an exponential expression and apply a logarithm.
SK-LOG-SOLVE-EXP · first introduced in Math 31
- A** Check an ordered pair in both equations of a system.
SK-SYS-CHECK · first introduced in Math 12
- A** Locate and interpret the intersection of two lines.
SK-SYS-GRAPH-INTERSECTION · first introduced in Math 12

I can rewrite a discrete exponential model as an equivalent continuous exponential model.

OpenStax: **Primary resources:** [6.7 Exponential and Logarithmic Models](#)

SUPPORTING SKILLS

- D** Convert between equivalent discrete- and continuous-growth forms.
SK-EXP-DISCRETE-CONTINUOUS · first introduced in Math 31
- A** Convert a percent increase or decrease to a growth or decay factor.
SK-EXP-PERCENT-FACTOR · first introduced in Math 31
- A** Isolate an exponential expression and apply a logarithm.
SK-LOG-SOLVE-EXP · first introduced in Math 31

I can solve and interpret an exponential or logarithmic modeling problem.

OpenStax: **Primary resources:** [6.7 Exponential and Logarithmic Models](#)

SUPPORTING SKILLS

- D** Identify initial amount, rate, and time interval in context.
SK-EXP-MODEL-CONTEXT · first introduced in Math 31
- A** Isolate an exponential expression and apply a logarithm.
SK-LOG-SOLVE-EXP · first introduced in Math 31
- A** Combine or isolate logarithms before converting to exponential form.
SK-LOG-SOLVE-LOG · first introduced in Math 31
- D** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

Extension Content

EXTENSION

OpenStax coverage: **Primary resources:** [3.6 Absolute Value Functions](#); [3.5 Transformation of Functions](#); [6.7 Exponential and Logarithmic Models](#)

M31-EXT-001

I can analyze an absolute-value function using transformations.

OpenStax: **Primary resources:** [3.6 Absolute Value Functions](#)

SUPPORTING SKILLS

- A** Interpret transformations applied to function inputs.
SK-FUNC-TRANSFORM-HORIZONTAL · first introduced in Math 31
- A** Distinguish stretches from compressions and horizontal effects from vertical effects.
SK-FUNC-TRANSFORM-SCALE · first introduced in Math 31
- A** Interpret transformations applied to function outputs.
SK-FUNC-TRANSFORM-VERTICAL · first introduced in Math 31
- A** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-EXT-002

I can determine even or odd symmetry from a graph or formula.

OpenStax: **Primary resources:** [3.5 Transformation of Functions](#)

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I** Test a formula or graph for even or odd symmetry.
SK-FUNC-SYMMETRY

M31-EXT-003

I can interpret a contextual logarithmic scale.

OpenStax: **Primary resources:** [6.7 Exponential and Logarithmic Models](#)

SUPPORTING SKILLS

- A** Interpret a logarithm as an exponent.
SK-LOG-DEFINITION · first introduced in Math 31
- I** Interpret equal intervals on a logarithmic scale as equal multiplicative changes.
SK-LOG-SCALE-INTERPRET
- A** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21

Math 32 Curriculum

Precalculus: Trigonometry | Middlesex School Mathematics | Resource: Math 32 Workbook

COURSE AT A GLANCE

Trigonometric Foundations

REQUIRED

pp. 3-6 Pythagorean Theorem; Special Right Triangles; pp. 29-33 Review Questions; pp. 7-12 Introduction to Radians; Related Angles; Radians and Degrees; pp. 13-14 Arc Length; pp. 15-21 Unit Circle Coordinates; Sine, Cosine, Tangent; pp. 24-28 Trig Functions Beyond Acute Angles; Non-Unit Circles; pp. 34-45 Multiple Angle Solutions; Reciprocal Functions; pp. 22-23 Right Triangle Applications

UNIT LEARNING OBJECTIVES

- I can use the Pythagorean theorem to determine an unknown side of a right triangle.
- I can determine exact side lengths in 30-60-90 and 45-45-90 triangles.
- I can convert angle measures between degrees and radians.
- I can determine coterminal and reference angles for a given angle.
- I can calculate arc length from a central angle and radius.
- I can determine exact coordinates on the unit circle.
- I can evaluate all six trigonometric functions exactly for common angles.
- I can solve right triangles using trigonometric ratios.
- I can apply right-triangle trigonometry to contextual problems.
- I can evaluate trigonometric functions from a point on a circle or terminal side.
- I can solve a basic trigonometric equation for all angles in a specified interval.

Periodic Functions and Modeling

REQUIRED

pp. 46-49 Graphing Sine and Cosine; pp. 50-62 Vertical and Horizontal Transformations; pp. 63-68 Graphing All Transformations; pp. 69-76 Modeling Problems; pp. 77-85 Graphing Cosecant, Secant, and Tangent

UNIT LEARNING OBJECTIVES

- I can graph basic sine and cosine functions using key features.
- I can identify amplitude, midline, period, and phase shift from an equation or graph.
- I can graph transformed sine and cosine functions.
- I can write sine and cosine equations that represent a periodic graph.
- I can construct a sinusoidal model from contextual information or data.
- I can interpret the parameters and outputs of a sinusoidal model in context.
- I can graph tangent, secant, and cosecant functions with their asymptotes.
- I can determine whether a function is even, odd, or neither from its graph or formula.

Inverse Functions and Trigonometric Equations

REQUIRED

pp. 86-91 Calculating Angles; Inverse Trig Functions; pp. 92-98 Solving Trig Equations, Part 2; pp. 99-100 Compositions of Trig and Inverse Trig Functions

UNIT LEARNING OBJECTIVES

- I can read inverse-trigonometric notation and distinguish it from reciprocal notation.
- I can evaluate inverse sine, inverse cosine, and inverse tangent expressions.
- I can explain how restricted domains make inverse trigonometric functions possible.
- I can solve trigonometric equations exactly or approximately on a specified interval.
- I can express all solutions of a trigonometric equation in general form.
- I can evaluate compositions of trigonometric and inverse trigonometric functions.

Non-Right Triangles

REQUIRED

pp. 102-110 Law of Sines; Ambiguous Case; pp. 111-115 Law of Cosines; Mixed Practice

UNIT LEARNING OBJECTIVES

- I can solve a non-right triangle using the Law of Sines.
- I can determine the number of possible triangles in the ambiguous SSA case.
- I can solve a non-right triangle using the Law of Cosines.
- I can select and apply an appropriate method to solve a triangle in context.

Trigonometric Identities

REQUIRED

pp. 116, 119-123 Pythagorean, reciprocal/quotient, even/odd, cofunction, and required practice

UNIT LEARNING OBJECTIVES

I can rewrite trigonometric expressions using reciprocal and quotient identities, including $\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}$.

I can rewrite trigonometric expressions using the Pythagorean identities.

I can use even/odd and cofunction identities to rewrite trigonometric expressions.

I can verify a trigonometric identity using the required identity families.

Additional Trigonometric Identities

EXTENSION

pp. 117-118, 124-126 Double-angle and sum/difference identities

UNIT LEARNING OBJECTIVES

I can apply double-angle identities.

I can apply half-angle identities.

I can apply angle sum and difference identities.

I can determine exact trigonometric values using extension identities.

Triangle Area from SAS

EXTENSION

pp. 101 New Area Formula

UNIT LEARNING OBJECTIVES

I can calculate the area of a triangle from two sides and the included angle.

Polar Coordinates

REQUIRED

pp. 127-128 Polar Coordinates and Coordinate Conversion

UNIT LEARNING OBJECTIVES

I can plot and identify points in polar coordinates.

I can represent a polar point using equivalent coordinate pairs.

I can convert points between polar and rectangular coordinates.

Polar Functions

EXTENSION

pp. 129-132 Polar Functions;
pp. 133-135 Polar/Rectangular Function Conversion

UNIT LEARNING OBJECTIVES

I can graph and interpret basic polar functions.

I can convert equations between polar and rectangular forms.

Vectors

EXTENSION

pp. 136-140 Vector Concepts and Operations; pp. 141-144 Components, Magnitude, Direction, and 3D Vectors

UNIT LEARNING OBJECTIVES

I can represent a vector geometrically and in component form.

I can add, subtract, and multiply vectors by scalars.

I can determine vector magnitude, direction, and unit vector.

I can resolve a vector into horizontal and vertical components.

I can perform basic operations with vectors in three dimensions.

Parametric Equations

EXTENSION

pp. 145-147 Introduction to Parametrics

UNIT LEARNING OBJECTIVES

I can graph a plane curve defined by parametric equations with its orientation.

I can eliminate a parameter to obtain a rectangular equation.

I can create or interpret parametric equations for motion in the plane.

IN-DEPTH CURRICULUM

SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Trigonometric Foundations

REQUIRED

Resource coverage: pp. 3-6 Pythagorean Theorem; Special Right Triangles; pp. 29-33 Review Questions; pp. 7-12 Introduction to Radians; Related Angles; Radians and Degrees; pp. 13-14 Arc Length; pp. 15-21 Unit Circle Coordinates; Sine, Cosine, Tangent; pp. 24-28 Trig Functions Beyond Acute Angles; Non-Unit Circles; pp. 34-45 Multiple Angle Solutions; Reciprocal Functions; pp. 22-23 Right Triangle Applications

M32-FND-001

I can use the Pythagorean theorem to determine an unknown side of a right triangle.

Resource: pp. 3-6 Pythagorean Theorem; Special Right Triangles; pp. 29-33 Review Questions

SUPPORTING SKILLS



Extract perfect-power factors from a radical.

SK-RAD-FACTOR-PERFECT · first introduced in Math 21



Apply the Pythagorean theorem or its converse.

SK-TRI-PYTHAGOREAN · first introduced in Math 22

M32-FND-002

I can determine exact side lengths in 30-60-90 and 45-45-90 triangles.

Resource: pp. 3-6 Pythagorean Theorem; Special Right Triangles; pp. 29-33 Review Questions

SUPPORTING SKILLS



Distinguish exact values from measured or rounded approximations.

SK-GE0-EXACT-APPROX · first introduced in Math 22



Extract perfect-power factors from a radical.

SK-RAD-FACTOR-PERFECT · first introduced in Math 21



Recall special-right-triangle ratios.

SK-TRI-SPECIAL-RATIOS · first introduced in Math 22

M32-FND-003

I can convert angle measures between degrees and radians.

Resource: pp. 7-12 Introduction to Radians; Related Angles; Radians and Degrees; pp. 29-33 Review Questions

SUPPORTING SKILLS

- A** Add, subtract, multiply, and divide fractions.
SK-NUM-FRACTION-OPERATE · inherited from middle school
- I** Convert between degree and radian measure.
SK-TRIG-ANGLE-CONVERT

M32-FND-004

I can determine coterminal and reference angles for a given angle.

Resource: pp. 7-12 Introduction to Radians; Related Angles; Radians and Degrees; pp. 29-33 Review Questions

SUPPORTING SKILLS

- I** Generate and recognize coterminal angles.
SK-TRIG-ANGLE-COTERMINAL
- I** Determine a reference angle and its quadrant.
SK-TRIG-ANGLE-REFERENCE
- I** Determine the sign of a trigonometric function by quadrant.
SK-TRIG-SIGN-QUADRANT

M32-FND-005

I can calculate arc length from a central angle and radius.

Resource: pp. 13-14 Arc Length; pp. 29-33 Review Questions

SUPPORTING SKILLS

- D** Calculate arc length from radius and central angle.
SK-CIR-ARC-LENGTH · first introduced in Math 22
- A** Select and report appropriate linear, square, or angular units.
SK-GEO-UNITS · first introduced in Math 22
- A** Convert between degree and radian measure.
SK-TRIG-ANGLE-CONVERT · first introduced in Math 32

M32-FND-006

I can determine exact coordinates on the unit circle.

Resource: pp. 15-21 Unit Circle Coordinates; Sine, Cosine, Tangent; pp. 29-33 Review Questions

SUPPORTING SKILLS

- A** Recall special-right-triangle ratios.
SK-TRI-SPECIAL-RATIOS · first introduced in Math 22
- A** Determine a reference angle and its quadrant.
SK-TRIG-ANGLE-REFERENCE · first introduced in Math 32
- I** Determine unit-circle coordinates for common angles.
SK-TRIG-UNIT-CIRCLE-COORD

I can evaluate all six trigonometric functions exactly for common angles.

Resource: pp. 15-21 Unit Circle Coordinates; Sine, Cosine, Tangent; pp. 24-28 Trig Functions Beyond Acute Angles; Non-Unit Circles; pp. 29-33 Review Questions; pp. 34-45 Multiple Angle Solutions; Reciprocal Functions

SUPPORTING SKILLS

- I** Express common trigonometric values exactly.
SK-TRIG-EXACT-VALUE
- I** Recall the six trigonometric functions, their reciprocal relationships, and their notation.
SK-TRIG-NOTATION-VOCAB
- I** Relate sine/cosecant, cosine/secant, and tangent/cotangent.
SK-TRIG-RECIPROCAL
- A** Determine the sign of a trigonometric function by quadrant.
SK-TRIG-SIGN-QUADRANT · first introduced in Math 32
- A** Determine unit-circle coordinates for common angles.
SK-TRIG-UNIT-CIRCLE-COORD · first introduced in Math 32

I can solve right triangles using trigonometric ratios.

Resource: pp. 22-23 Right Triangle Applications; pp. 29-33 Review Questions

SUPPORTING SKILLS

- D** Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
SK-TRI-TRIG-RATIO · first introduced in Math 22
- D** Use an inverse trigonometric function to determine a missing angle.
SK-TRI-TRIG-SOLVE-ANGLE · first introduced in Math 22
- D** Use a trigonometric ratio to determine a missing side.
SK-TRI-TRIG-SOLVE-SIDE · first introduced in Math 22
- R** Label sides relative to a selected angle.
SK-TRIG-TRIANGLE-LABEL · first introduced in Math 22

I can apply right-triangle trigonometry to contextual problems.

Resource: pp. 22-23 Right Triangle Applications; pp. 29-33 Review Questions

SUPPORTING SKILLS

- A** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12
- D** Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
SK-TRI-TRIG-RATIO · first introduced in Math 22
- A** Use an inverse trigonometric function to determine a missing angle.
SK-TRI-TRIG-SOLVE-ANGLE · first introduced in Math 22
- A** Use a trigonometric ratio to determine a missing side.
SK-TRI-TRIG-SOLVE-SIDE · first introduced in Math 22

I can evaluate trigonometric functions from a point on a circle or terminal side.

Resource: pp. 24-28 Trig Functions Beyond Acute Angles; Non-Unit Circles; pp. 29-33 Review Questions

SUPPORTING SKILLS

- A** Apply the Pythagorean theorem or its converse.
SK-TRI-PYTHAGOREAN · first introduced in Math 22
- I** Use coordinates and radius to evaluate trigonometric functions.
SK-TRIG-CIRCLE-POINT
- A** Relate sine/cosecant, cosine/secant, and tangent/cotangent.
SK-TRIG-RECIPROCAL · first introduced in Math 32
- A** Determine the sign of a trigonometric function by quadrant.
SK-TRIG-SIGN-QUADRANT · first introduced in Math 32

I can solve a basic trigonometric equation for all angles in a specified interval.

Resource: pp. 34-45 Multiple Angle Solutions; Reciprocal Functions

SUPPORTING SKILLS

- A** Translate between inequality and interval notation.
SK-NOT-INTERVAL · first introduced in Math 12
- A** Determine a reference angle and its quadrant.
SK-TRIG-ANGLE-REFERENCE · first introduced in Math 32
- I** Use reference angles and periodicity to find every solution in an interval.
SK-TRIG-EQUATION-INTERVAL
- I** Isolate a trigonometric expression before solving.
SK-TRIG-EQUATION-ISOLATE
- A** Determine the sign of a trigonometric function by quadrant.
SK-TRIG-SIGN-QUADRANT · first introduced in Math 32

Periodic Functions and Modeling

REQUIRED

Resource coverage: pp. 46-49 Graphing Sine and Cosine; pp. 50-62 Vertical and Horizontal Transformations; pp. 63-68 Graphing All Transformations; pp. 69-76 Modeling Problems; pp. 77-85 Graphing Cosecant, Secant, and Tangent

M32-GRF-001

I can graph basic sine and cosine functions using key features.

Resource: pp. 46-49 Graphing Sine and Cosine

SUPPORTING SKILLS

- D** Use tabular patterns to predict graph shape and behavior.
SK-REP-TABLE-GRAPH · first introduced in Math 21
- I** Locate quarter-period key points of a periodic function.
SK-TRIG-GRAPH-KEYPOINTS
- A** Determine unit-circle coordinates for common angles.
SK-TRIG-UNIT-CIRCLE-COORD · first introduced in Math 32

M32-GRF-002

I can identify amplitude, midline, period, and phase shift from an equation or graph.

Resource: pp. 50-62 Vertical and Horizontal Transformations; pp. 63-68 Graphing All Transformations

SUPPORTING SKILLS

- I** Determine amplitude and vertical reflection.
SK-TRIG-GRAPH-AMPLITUDE
- I** Determine a sinusoid's midline and vertical shift.
SK-TRIG-GRAPH-MIDLINE
- I** Determine period from a graph or equation.
SK-TRIG-GRAPH-PERIOD
- I** Determine horizontal shift from a graph or equation.
SK-TRIG-GRAPH-PHASE
- I** Recall and correctly use the terms amplitude, midline, period, phase shift, vertical shift, and asymptote.
SK-TRIG-GRAPH-VOCAB

M32-GRF-003

I can graph transformed sine and cosine functions.

Resource: pp. 50-62 Vertical and Horizontal Transformations; pp. 63-68 Graphing All Transformations

SUPPORTING SKILLS

- A** Interpret transformations applied to function inputs.
SK-FUNC-TRANSFORM-HORIZONTAL · first introduced in Math 31
- D** Distinguish stretches from compressions and horizontal effects from vertical effects.
SK-FUNC-TRANSFORM-SCALE · first introduced in Math 31
- A** Interpret transformations applied to function outputs.
SK-FUNC-TRANSFORM-VERTICAL · first introduced in Math 31
- D** Locate quarter-period key points of a periodic function.
SK-TRIG-GRAPH-KEYPOINTS · first introduced in Math 32

M32-GRF-004

I can write sine and cosine equations that represent a periodic graph.

Resource: pp. 63-68 Graphing All Transformations

SUPPORTING SKILLS

- D** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21
- D** Determine amplitude and vertical reflection.
SK-TRIG-GRAPH-AMPLITUDE · first introduced in Math 32
- D** Determine a sinusoid's midline and vertical shift.
SK-TRIG-GRAPH-MIDLINE · first introduced in Math 32
- D** Determine period from a graph or equation.
SK-TRIG-GRAPH-PERIOD · first introduced in Math 32
- D** Determine horizontal shift from a graph or equation.
SK-TRIG-GRAPH-PHASE · first introduced in Math 32

M32-GRF-005

I can construct a sinusoidal model from contextual information or data.

Resource: pp. 69-76 Modeling Problems

SUPPORTING SKILLS

- A** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A** Determine amplitude and vertical reflection.
SK-TRIG-GRAPH-AMPLITUDE · first introduced in Math 32
- A** Determine a sinusoid's midline and vertical shift.
SK-TRIG-GRAPH-MIDLINE · first introduced in Math 32
- A** Determine period from a graph or equation.
SK-TRIG-GRAPH-PERIOD · first introduced in Math 32
- I** Translate contextual maximum, minimum, cycle, and starting position into model parameters.
SK-TRIG-MODEL-PARAMETERS

M32-GRF-006

I can interpret the parameters and outputs of a sinusoidal model in context.

Resource: pp. 69-76 Modeling Problems

SUPPORTING SKILLS

- A** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12
- D** Translate contextual maximum, minimum, cycle, and starting position into model parameters.
SK-TRIG-MODEL-PARAMETERS · first introduced in Math 32

M32-GRF-007

I can graph tangent, secant, and cosecant functions with their asymptotes.

Resource: pp. 77-85 Graphing Cosecant, Secant, and Tangent

SUPPORTING SKILLS

- A** Interpret transformations applied to function inputs.
SK-FUNC-TRANSFORM-HORIZONTAL · first introduced in Math 31
- I** Locate vertical asymptotes of tangent, secant, and cosecant.
SK-TRIG-GRAPH-ASYMPTOTE
- A** Locate quarter-period key points of a periodic function.
SK-TRIG-GRAPH-KEYPOINTS · first introduced in Math 32
- D** Relate sine/cosecant, cosine/secant, and tangent/cotangent.
SK-TRIG-RECIPROCAL · first introduced in Math 32

M32-GRF-008

I can determine whether a function is even, odd, or neither from its graph or formula.

Resource: pp. 46-49 Graphing Sine and Cosine

SUPPORTING SKILLS

- D** Test a formula or graph for even or odd symmetry.
SK-FUNC-SYMMETRY · first introduced in Math 31

Inverse Functions and Trigonometric Equations

REQUIRED

Resource coverage: pp. 86-91 Calculating Angles; Inverse Trig Functions; pp. 92-98 Solving Trig Equations, Part 2; pp. 99-100 Compositions of Trig and Inverse Trig Functions

M32-INV-006

I can read inverse-trigonometric notation and distinguish it from reciprocal notation.

Resource: No mapped OpenStax section

SUPPORTING SKILLS

I Read inverse-trigonometric notation and distinguish it from reciprocal notation.

SK-TRIG-INVERSE-NOTATION

A Relate sine/cosecant, cosine/secant, and tangent/cotangent.

SK-TRIG-RECIPROCAL · first introduced in Math 32

M32-INV-001

I can evaluate inverse sine, inverse cosine, and inverse tangent expressions.

Resource: pp. 86-91 Calculating Angles; Inverse Trig Functions

SUPPORTING SKILLS

A Express common trigonometric values exactly.

SK-TRIG-EXACT-VALUE · first introduced in Math 32

I Evaluate an inverse trigonometric expression exactly or approximately.

SK-TRIG-INVERSE-EVALUATE

A Read inverse-trigonometric notation and distinguish it from reciprocal notation.

SK-TRIG-INVERSE-NOTATION · first introduced in Math 32

I Recall the principal-value ranges of inverse trigonometric functions.

SK-TRIG-INVERSE-RANGE

M32-INV-002

I can explain how restricted domains make inverse trigonometric functions possible.

Resource: pp. 86-91 Calculating Angles; Inverse Trig Functions

SUPPORTING SKILLS

A Exchange domain and range between a function and its inverse.

SK-FUNC-INVERSE-DOMAIN-RANGE · first introduced in Math 31

A Apply the horizontal line test for invertibility.

SK-FUNC-INVERSE-HLT · first introduced in Math 31

A Test whether a function is one-to-one on a domain.

SK-FUNC-ONE-TO-ONE · first introduced in Math 31

D Recall the principal-value ranges of inverse trigonometric functions.

SK-TRIG-INVERSE-RANGE · first introduced in Math 32

M32-INV-003

I can solve trigonometric equations exactly or approximately on a specified interval.

Resource: pp. 92-98 Solving Trig Equations, Part 2

SUPPORTING SKILLS

- A** Check a proposed solution in the original equation.
SK-ALG-CHECK-SOLUTION · first introduced in Math 12
- A** Determine a reference angle and its quadrant.
SK-TRIG-ANGLE-REFERENCE · first introduced in Math 32
- D** Use reference angles and periodicity to find every solution in an interval.
SK-TRIG-EQUATION-INTERVAL · first introduced in Math 32
- D** Isolate a trigonometric expression before solving.
SK-TRIG-EQUATION-ISOLATE · first introduced in Math 32
- A** Evaluate an inverse trigonometric expression exactly or approximately.
SK-TRIG-INVERSE-EVALUATE · first introduced in Math 32

M32-INV-004

I can express all solutions of a trigonometric equation in general form.

Resource: pp. 92-98 Solving Trig Equations, Part 2

SUPPORTING SKILLS

- D** Generate and recognize coterminal angles.
SK-TRIG-ANGLE-COTERMINAL · first introduced in Math 32
- I** Express a periodic solution family in general form.
SK-TRIG-EQUATION-GENERAL
- A** Use reference angles and periodicity to find every solution in an interval.
SK-TRIG-EQUATION-INTERVAL · first introduced in Math 32

M32-INV-005

I can evaluate compositions of trigonometric and inverse trigonometric functions.

Resource: pp. 99-100 Compositions of Trig and Inverse Trig Functions

SUPPORTING SKILLS

- A** Substitute one function formula into another.
SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- A** Verify inverse functions using composition.
SK-FUNC-INVERSE-VERIFY · first introduced in Math 31
- D** Evaluate an inverse trigonometric expression exactly or approximately.
SK-TRIG-INVERSE-EVALUATE · first introduced in Math 32
- A** Recall the principal-value ranges of inverse trigonometric functions.
SK-TRIG-INVERSE-RANGE · first introduced in Math 32

Non-Right Triangles

REQUIRED

Resource coverage: pp. 102-110 Law of Sines; Ambiguous Case; pp. 111-115 Law of Cosines; Mixed Practice

M32-TRI-002

I can solve a non-right triangle using the Law of Sines.

Resource: pp. 102-110 Law of Sines; Ambiguous Case

SUPPORTING SKILLS

- A** Evaluate an inverse trigonometric expression exactly or approximately.
SK-TRIG-INVERSE-EVALUATE · first introduced in Math 32
- I** Apply the Law of Sines to determine missing sides or angles.
SK-TRIG-LAW-SINES
- D** Classify triangle data as SSS, SAS, ASA, AAS, or SSA.
SK-TRIG-TRIANGLE-DATA · first introduced in Math 32

M32-TRI-003

I can determine the number of possible triangles in the ambiguous SSA case.

Resource: pp. 102-110 Law of Sines; Ambiguous Case

SUPPORTING SKILLS

- I** Test and interpret the possible outcomes of SSA data.
SK-TRIG-AMBIGUOUS-CASE
- D** Apply the Law of Sines to determine missing sides or angles.
SK-TRIG-LAW-SINES · first introduced in Math 32
- A** Classify triangle data as SSS, SAS, ASA, AAS, or SSA.
SK-TRIG-TRIANGLE-DATA · first introduced in Math 32

M32-TRI-004

I can solve a non-right triangle using the Law of Cosines.

Resource: pp. 111-115 Law of Cosines; Mixed Practice

SUPPORTING SKILLS

- A** Evaluate an inverse trigonometric expression exactly or approximately.
SK-TRIG-INVERSE-EVALUATE · first introduced in Math 32
- I** Apply the Law of Cosines to determine missing sides or angles.
SK-TRIG-LAW-COSINES
- D** Classify triangle data as SSS, SAS, ASA, AAS, or SSA.
SK-TRIG-TRIANGLE-DATA · first introduced in Math 32

I can select and apply an appropriate method to solve a triangle in context.

Resource: pp. 111-115 Law of Cosines; Mixed Practice

SUPPORTING SKILLS

- A** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12
- A** Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
SK-TRI-TRIG-RATIO · first introduced in Math 22
- A** Apply the Law of Cosines to determine missing sides or angles.
SK-TRIG-LAW-COSINES · first introduced in Math 32
- A** Apply the Law of Sines to determine missing sides or angles.
SK-TRIG-LAW-SINES · first introduced in Math 32
- D** Classify triangle data as SSS, SAS, ASA, AAS, or SSA.
SK-TRIG-TRIANGLE-DATA · first introduced in Math 32
- I** Select right-triangle ratios, Law of Sines, or Law of Cosines from the given data.
SK-TRIG-TRIANGLE-METHOD

Trigonometric Identities

REQUIRED

Resource coverage: pp. 116, 119-123 Pythagorean, reciprocal/quotient, even/odd, cofunction, and required practice

M32-IDN-001

I can rewrite trigonometric expressions using reciprocal and quotient identities, including $\tan(\theta) = \sin(\theta)/\cos(\theta)$.

Resource: pp. 116, 119-123 Pythagorean, reciprocal/quotient, even/odd, cofunction, and required practice

SUPPORTING SKILLS

- I** Apply reciprocal and quotient identities.
SK-TRIG-IDENTITY-RECIPROCAL-QUOTIENT
- I** Substitute an identity to rewrite an expression.
SK-TRIG-IDENTITY-SUBSTITUTE

M32-IDN-002

I can rewrite trigonometric expressions using the Pythagorean identities.

Resource: pp. 116, 119-123 Pythagorean, reciprocal/quotient, even/odd, cofunction, and required practice

SUPPORTING SKILLS

- I** Apply the Pythagorean trigonometric identities.
SK-TRIG-IDENTITY-PYTHAGOREAN
- D** Substitute an identity to rewrite an expression.
SK-TRIG-IDENTITY-SUBSTITUTE · first introduced in Math 32

M32-IDN-003

I can use even/odd and cofunction identities to rewrite trigonometric expressions.

Resource: pp. 116, 119-123 Pythagorean, reciprocal/quotient, even/odd, cofunction, and required practice

SUPPORTING SKILLS

- D** Test a formula or graph for even or odd symmetry.
SK-FUNC-SYMMETRY · first introduced in Math 31
- I** Apply sine/cosine cofunction identities.
SK-TRIG-IDENTITY-COFUNCTION
- I** Apply even and odd trigonometric identities.
SK-TRIG-IDENTITY-EVEN-ODD
- D** Substitute an identity to rewrite an expression.
SK-TRIG-IDENTITY-SUBSTITUTE · first introduced in Math 32

M32-IDN-004

I can verify a trigonometric identity using the required identity families.

Resource: pp. 116, 119-123 Pythagorean, reciprocal/quotient, even/odd, cofunction, and required practice

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I** Choose a productive identity or algebraic manipulation.
SK-TRIG-IDENTITY-STRATEGY
- D** Substitute an identity to rewrite an expression.
SK-TRIG-IDENTITY-SUBSTITUTE · first introduced in Math 32
- I** Maintain equivalence while transforming one side of an identity.
SK-TRIG-IDENTITY-VERIFY

Additional Trigonometric Identities

EXTENSION

Resource coverage: pp. 117-118, 124-126 Double-angle and sum/difference identities

M32-IDX-001

I can apply double-angle identities.

Resource: pp. 117-118, 124-126 Double-angle and sum/difference identities

SUPPORTING SKILLS

I Apply double-angle identities.

SK-TRIG-IDENTITY-DOUBLE-ANGLE

A Substitute an identity to rewrite an expression.

SK-TRIG-IDENTITY-SUBSTITUTE · first introduced in Math 32

M32-IDX-002

I can apply half-angle identities.

Resource: pp. 117-118, 124-126 Double-angle and sum/difference identities

SUPPORTING SKILLS

A Apply double-angle identities.

SK-TRIG-IDENTITY-DOUBLE-ANGLE · first introduced in Math 32

I Apply half-angle identities.

SK-TRIG-IDENTITY-HALF-ANGLE

M32-IDX-003

I can apply angle sum and difference identities.

Resource: pp. 117-118, 124-126 Double-angle and sum/difference identities

SUPPORTING SKILLS

A Substitute an identity to rewrite an expression.

SK-TRIG-IDENTITY-SUBSTITUTE · first introduced in Math 32

I Apply angle sum and difference identities.

SK-TRIG-IDENTITY-SUM-DIFFERENCE

M32-IDX-004

I can determine exact trigonometric values using extension identities.

Resource: pp. 117-118, 124-126 Double-angle and sum/difference identities

SUPPORTING SKILLS

D Express common trigonometric values exactly.

SK-TRIG-EXACT-VALUE · first introduced in Math 32

A Apply double-angle identities.

SK-TRIG-IDENTITY-DOUBLE-ANGLE · first introduced in Math 32

A Apply half-angle identities.

SK-TRIG-IDENTITY-HALF-ANGLE · first introduced in Math 32

A Apply angle sum and difference identities.

SK-TRIG-IDENTITY-SUM-DIFFERENCE · first introduced in Math 32

Triangle Area from SAS

EXTENSION

Resource coverage: pp. 101 New Area Formula

M32-TRX-001

I can calculate the area of a triangle from two sides and the included angle.

Resource: pp. 101 New Area Formula

SUPPORTING SKILLS

- A** Express common trigonometric values exactly.
SK-TRIG-EXACT-VALUE · first introduced in Math 32
- I** Calculate triangle area from two sides and the included angle.
SK-TRIG-TRIANGLE-AREA-SAS
- I** Classify triangle data as SSS, SAS, ASA, AAS, or SSA.
SK-TRIG-TRIANGLE-DATA

Polar Coordinates

REQUIRED

Resource coverage: pp. 127-128 Polar Coordinates and Coordinate Conversion

M32-POL-001

I can plot and identify points in polar coordinates.

Resource: pp. 127-128 Polar Coordinates and Coordinate Conversion

SUPPORTING SKILLS

- I** Read and write polar-coordinate notation and identify the radius, angle, pole, and polar axis.
SK-POL-NOTATION-READ
- I** Plot a point from radius and angle.
SK-POL-PLOT
- A** Determine a reference angle and its quadrant.
SK-TRIG-ANGLE-REFERENCE · first introduced in Math 32

M32-POL-002

I can represent a polar point using equivalent coordinate pairs.

Resource: pp. 127-128 Polar Coordinates and Coordinate Conversion

SUPPORTING SKILLS

- I** Generate equivalent polar coordinate pairs.
SK-POL-EQUIVALENT
- A** Generate and recognize coterminal angles.
SK-TRIG-ANGLE-COTERMINAL · first introduced in Math 32

M32-POL-003

I can convert points between polar and rectangular coordinates.

Resource: pp. 127-128 Polar Coordinates and Coordinate Conversion

SUPPORTING SKILLS

- I** Use coordinate relationships to convert a point between systems.
SK-POL-CONVERT-POINT
- A** Use coordinates and radius to evaluate trigonometric functions.
SK-TRIG-CIRCLE-POINT · first introduced in Math 32
- A** Evaluate an inverse trigonometric expression exactly or approximately.
SK-TRIG-INVERSE-EVALUATE · first introduced in Math 32

Polar Functions

EXTENSION

Resource coverage: pp. 129-132 Polar Functions; pp. 133-135 Polar/Rectangular Function Conversion

M32-PLX-001

I can graph and interpret basic polar functions.

Resource: pp. 129-132 Polar Functions

SUPPORTING SKILLS

- A** Identify input and output values in a formula, graph, table, or context.

SK-FUNC-INPUT-OUTPUT · first introduced in Math 12

- D** Plot a point from radius and angle.

SK-POL-PLOT · first introduced in Math 32

- A** Locate quarter-period key points of a periodic function.

SK-TRIG-GRAPH-KEYPOINTS · first introduced in Math 32

M32-PLX-002

I can convert equations between polar and rectangular forms.

Resource: pp. 133-135 Polar/Rectangular Function Conversion

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.

SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school

- I** Substitute coordinate identities to convert an equation between systems.

SK-POL-CONVERT-EQUATION

- A** Apply the Pythagorean trigonometric identities.

SK-TRIG-IDENTITY-PYTHAGOREAN · first introduced in Math 32

Vectors

EXTENSION

Resource coverage: pp. 136-140 Vector Concepts and Operations; pp. 141-144 Components, Magnitude, Direction, and 3D Vectors

M32-VEC-001

I can represent a vector geometrically and in component form.

Resource: pp. 136-140 Vector Concepts and Operations

SUPPORTING SKILLS

- A** Add congruence, parallel, perpendicular, and angle markings to a diagram.
SK-GE0-DIAGRAM-MARK · first introduced in Math 22
- I** Write a vector in component form.
SK-VEC-COMPONENTS
- I** Read and write vector notation and correctly use the terms magnitude, direction, component, and unit vector.
SK-VEC-NOTATION-VOCAB

M32-VEC-002

I can add, subtract, and multiply vectors by scalars.

Resource: pp. 136-140 Vector Concepts and Operations

SUPPORTING SKILLS

- A** Add, subtract, multiply, and divide signed integers.
SK-NUM-INTEGER-OPERATE · inherited from middle school
- I** Perform component-wise vector operations.
SK-VEC-OPERATE

M32-VEC-003

I can determine vector magnitude, direction, and unit vector.

Resource: pp. 136-140 Vector Concepts and Operations; pp. 141-144 Components, Magnitude, Direction, and 3D Vectors

SUPPORTING SKILLS

- A** Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula.
SK-COORD-DISTANCE-PYTHAGOREAN · first introduced in Math 22
- I** Determine a vector's direction angle with correct quadrant.
SK-VEC-DIRECTION
- I** Calculate vector magnitude using the distance formula.
SK-VEC-MAGNITUDE
- I** Normalize a nonzero vector to obtain a unit vector.
SK-VEC-UNIT

M32-VEC-004

I can resolve a vector into horizontal and vertical components.

Resource: pp. 141-144 Components, Magnitude, Direction, and 3D Vectors

SUPPORTING SKILLS

- A** Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
SK-TRI-TRIG-RATIO · first introduced in Math 22
- D** Write a vector in component form.
SK-VEC-COMPONENTS · first introduced in Math 32
- A** Determine a vector's direction angle with correct quadrant.
SK-VEC-DIRECTION · first introduced in Math 32

I can perform basic operations with vectors in three dimensions.

Resource: pp. 141-144 Components, Magnitude, Direction, and 3D Vectors

SUPPORTING SKILLS

- I** Represent and operate on vectors with three components.

SK-VEC-3D

- D** Calculate vector magnitude using the distance formula.

SK-VEC-MAGNITUDE · first introduced in Math 32

- D** Perform component-wise vector operations.

SK-VEC-OPERATE · first introduced in Math 32

Parametric Equations

EXTENSION

Resource coverage: pp. 145-147 Introduction to Parametrics

M32-PAR-001

I can graph a plane curve defined by parametric equations with its orientation.

Resource: pp. 145-147 Introduction to Parametrics

SUPPORTING SKILLS

- A** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- I** Graph a parametric curve with its direction of motion.
SK-PAR-GRAPH
- I** Read parametric notation and correctly use the terms parameter, coordinate equation, orientation, and parameter interval.
SK-PAR-NOTATION-VOCAB
- I** Generate ordered pairs from parametric equations.
SK-PAR-TABLE

M32-PAR-002

I can eliminate a parameter to obtain a rectangular equation.

Resource: pp. 145-147 Introduction to Parametrics

SUPPORTING SKILLS

- A** Isolate a specified variable in an equation or formula.
SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- I** Eliminate a parameter algebraically.
SK-PAR-ELIMINATE
- A** Substitute an equivalent expression for a variable.
SK-SYS-SUBSTITUTE · first introduced in Math 12

M32-PAR-003

I can create or interpret parametric equations for motion in the plane.

Resource: pp. 145-147 Introduction to Parametrics

SUPPORTING SKILLS

- A** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- I** Interpret or construct parametric equations for planar motion.
SK-PAR-MODEL
- D** Generate ordered pairs from parametric equations.
SK-PAR-TABLE · first introduced in Math 32
- A** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

Math 39 Curriculum

Precalculus with Data Analysis | Middlesex School Mathematics

COURSE AT A GLANCE

Linear Modeling and Residuals

REQUIRED

Supplemental resources: [3.3 Rates of Change and Behavior of Graphs](#); [4.1 Linear Functions](#); [4.3 Fitting Linear Models to Data](#)

UNIT LEARNING OBJECTIVES

- I can calculate and interpret average rate of change with appropriate units.
- I can construct a linear function from two data points, a table, a graph, or a context.
- I can fit a linear regression model to data using technology and interpret its parameters.
- I can interpret a correlation coefficient as evidence about the direction and strength of a linear relationship.
- I can calculate residuals, construct a residual plot, and use its pattern to evaluate a model.
- I can use a linear model to make and interpret predictions, including solving for either variable.

Exponential Modeling

REQUIRED

Supplemental resources: [College Algebra 2e 6.1 Exponential Functions](#); [Precalculus 2e 4.1 Exponential Functions](#); [Precalculus 2e 4.7 Exponential and Logarithmic Models](#); [Precalculus 2e 4.6 Exponential and Logarithmic Equations](#); [Precalculus 2e 4.8 Fitting Exponential Models to Data](#); [College Algebra 2e 4.3 Fitting Linear Models to Data](#)

UNIT LEARNING OBJECTIVES

- I can distinguish linear and exponential change using rates, multipliers, tables, graphs, or contexts.
- I can construct an exponential function from two data points, a table, or a context.
- I can interpret the initial value, growth or decay factor, and growth or decay rate of an exponential model.
- I can determine and interpret doubling time or half-life from an exponential model.
- I can solve exponential and logarithmic equations in modeling contexts.
- I can fit an exponential regression model using technology and evaluate its fit.
- I can compare linear and exponential models and justify which is more appropriate for a data set or context.

Power Functions and Function Comparison

REQUIRED

Supplemental resources: [Precalculus 2e 3.3 Power Functions and Polynomial Functions](#); [Precalculus 2e 3.9 Modeling Using Variation](#); [College Algebra 2e 2.6 Other Types of Equations](#); [Precalculus 2e 4.2 Graphs of Exponential Functions](#); [Precalculus 2e 4.4 Graphs of Logarithmic Functions](#)

UNIT LEARNING OBJECTIVES

- I can identify a power function and interpret the parameters in $y = kx^p$.
- I can construct a power function from two data points or a proportional relationship.
- I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.
- I can recognize and model direct, inverse, and other power-variation relationships.
- I can solve equations involving integer or rational powers.
- I can compare the long-run dominance of logarithmic, power, polynomial, and exponential functions.
- I can fit a power regression model and interpret its parameters and correlation evidence.

Function Combinations and Polynomials

REQUIRED

Supplemental resources: [1.4 Composition of Functions](#); [3.4 Graphs of Polynomial Functions](#); [3.6 Zeros of Polynomial Functions](#)

UNIT LEARNING OBJECTIVES

- I can add, subtract, multiply, and divide functions represented by tables, graphs, or formulas.
- I can determine where a combination of functions is positive, zero, or undefined.
- I can determine polynomial end behavior from degree and leading coefficient.
- I can identify zeros and multiplicities and relate them to polynomial graph behavior.
- I can estimate a polynomial's possible degree from its zeros, turning points, inflection points, and end behavior.
- I can construct a polynomial function from specified zeros, multiplicities, and a point.
- I can match polynomial formulas and graphs using intercepts, multiplicity, degree, and end behavior.
- I can build and optimize a polynomial model in a geometric or economic context.

Describing Data

REQUIRED

No mapped OpenStax section

UNIT LEARNING OBJECTIVES

- I can distinguish categorical and quantitative data.
- I can construct and interpret frequency and relative-frequency tables.
- I can construct and interpret bar charts, pie charts, and histograms using appropriate technology.
- I can identify misleading features in a data display.
- I can describe a distribution's modality, symmetry, skewness, and outliers.
- I can calculate and interpret mean, median, and mode and select an appropriate measure of center.
- I can calculate and interpret range, standard deviation, quartiles, and interquartile range.
- I can construct and interpret a five-number summary and boxplot.
- I can use percentiles and z-scores to locate and compare values within distributions.
- I can compare distributions using their shape, center, spread, and unusual features.

Probability

REQUIRED

Supplemental resources: [11.7 Probability](#)

UNIT LEARNING OBJECTIVES

- I can interpret and write notation for compound and conditional probabilities.
- I can organize categorical data in a contingency table or Venn diagram.
- I can calculate and interpret empirical, marginal, joint, and conditional probabilities from a contingency table.
- I can identify an experiment, outcomes, events, and a sample space.
- I can calculate theoretical probabilities when outcomes are equally likely.
- I can use complements to calculate probabilities.
- I can calculate probabilities of compound 'and' events and distinguish independent from dependent events.
- I can calculate probabilities of compound 'or' events and distinguish overlapping from disjoint events.
- I can compare experimental and theoretical probability and explain the law of large numbers.

Expected Value

EXTENSION

No mapped OpenStax section

UNIT LEARNING OBJECTIVES

- I can calculate the expected value of a discrete probability situation.
- I can interpret expected value to evaluate a game, risk, or long-run decision.

Linearization of Nonlinear Data

EXTENSION

No mapped OpenStax section

UNIT LEARNING OBJECTIVES

- I can linearize power or exponential data and translate between the linearized and original models.

IN-DEPTH CURRICULUM

SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Linear Modeling and Residuals

REQUIRED

OpenStax coverage: **Supplemental resources:** [3.3 Rates of Change and Behavior of Graphs](#); [4.1 Linear Functions](#); [4.3 Fitting Linear Models to Data](#)

M39-LIN-001

I can calculate and interpret average rate of change with appropriate units.

OpenStax: **Supplemental resources:** [3.3 Rates of Change and Behavior of Graphs](#)

SUPPORTING SKILLS

- Calculate average rate of change over an interval.
SK-FUNC-AROC · first introduced in Math 21
- Interpret average rate of change with contextual units.
SK-FUNC-AROC-INTERPRET · first introduced in Math 21
- Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

M39-LIN-002

I can construct a linear function from two data points, a table, a graph, or a context.

OpenStax: **Supplemental resources:** [4.1 Linear Functions](#)

SUPPORTING SKILLS

- Convert among common forms of a linear equation.
SK-LIN-FORM-CONVERT · first introduced in Math 12
- Calculate slope from two points.
SK-LIN-SLOPE-POINTS · first introduced in Math 12
- Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21

M39-LIN-003

I can fit a linear regression model to data using technology and interpret its parameters.

OpenStax: **Supplemental resources:** [4.3 Fitting Linear Models to Data](#)

SUPPORTING SKILLS

- D** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- I** Use technology to determine a least-squares linear regression model.
SK-REG-LINEAR-FIT
- I** Recall and correctly use regression terms including explanatory variable, response variable, correlation, residual, and residual plot.
SK-REG-VOCABULARY

M39-LIN-004

I can interpret a correlation coefficient as evidence about the direction and strength of a linear relationship.

OpenStax: **Supplemental resources:** [4.3 Fitting Linear Models to Data](#)

SUPPORTING SKILLS

- I** Interpret the sign and magnitude of a correlation coefficient in context.
SK-REG-CORRELATION-INTERPRET

M39-LIN-005

I can calculate residuals, construct a residual plot, and use its pattern to evaluate a model.

OpenStax: **Supplemental resources:** [4.3 Fitting Linear Models to Data](#)

SUPPORTING SKILLS

- I** Use residual patterns and contextual evidence to judge whether a model is appropriate.
SK-REG-MODEL-ASSESS
- I** Calculate an observed value minus its model-predicted value.
SK-REG-RESIDUAL-CALCULATE
- I** Construct and interpret a residual plot.
SK-REG-RESIDUAL-PLOT

M39-LIN-006

I can use a linear model to make and interpret predictions, including solving for either variable.

OpenStax: **Supplemental resources:** [4.3 Fitting Linear Models to Data](#)

SUPPORTING SKILLS

- A** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D** Solve for an input associated with a specified output.
SK-FUNC-SOLVE-INPUT · first introduced in Math 21
- A** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12

Exponential Modeling

REQUIRED

OpenStax coverage: **Supplemental resources:** [College Algebra 2e 6.1 Exponential Functions](#); [Precalculus 2e 4.1 Exponential Functions](#); [Precalculus 2e 4.7 Exponential and Logarithmic Models](#); [Precalculus 2e 4.6 Exponential and Logarithmic Equations](#); [Precalculus 2e 4.8 Fitting Exponential Models to Data](#); [College Algebra 2e 4.3 Fitting Linear Models to Data](#)

M39-EXP-001

I can distinguish linear and exponential change using rates, multipliers, tables, graphs, or contexts.

OpenStax: **Supplemental resources:** [6.1 Exponential Functions](#)

SUPPORTING SKILLS

D Use constant differences or ratios to distinguish linear and exponential patterns.

SK-EXP-DIFFERENCE-RATIO · first introduced in Math 31

D Compare functions shown in different representations.

SK-FUNC-COMPARE-REP · first introduced in Math 21

M39-EXP-002

I can construct an exponential function from two data points, a table, or a context.

OpenStax: **Supplemental resources:** [4.1 Exponential Functions](#)

SUPPORTING SKILLS

D Identify initial amount, rate, and time interval in context.

SK-EXP-MODEL-CONTEXT · first introduced in Math 31

D Determine an exponential model from tabular values.

SK-EXP-MODEL-TABLE · first introduced in Math 31

D Solve for an exponential model using two points.

SK-EXP-MODEL-TWO-POINTS · first introduced in Math 31

M39-EXP-003

I can interpret the initial value, growth or decay factor, and growth or decay rate of an exponential model.

OpenStax: **Supplemental resources:** [4.1 Exponential Functions](#)

SUPPORTING SKILLS

D Convert a percent increase or decrease to a growth or decay factor.

SK-EXP-PERCENT-FACTOR · first introduced in Math 31

R Recall and correctly use exponential-function terms including initial value, growth or decay factor, growth or decay rate, and asymptote.

SK-EXP-VOCABULARY · first introduced in Math 31

D Explain a model parameter in the language of its context.

SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21

M39-EXP-004

I can determine and interpret doubling time or half-life from an exponential model.

OpenStax: **Supplemental resources:** [4.7 Exponential and Logarithmic Models](#)

SUPPORTING SKILLS

A Identify initial amount, rate, and time interval in context.

SK-EXP-MODEL-CONTEXT · first introduced in Math 31

D Translate a doubling-time or half-life question into an exponential equation.

SK-LOG-DOUBLING-HALF · first introduced in Math 31

M39-EXP-005

I can solve exponential and logarithmic equations in modeling contexts.

OpenStax: **Supplemental resources:** [4.6 Exponential and Logarithmic Equations](#)

SUPPORTING SKILLS

- A** Reject solutions that make a logarithm argument nonpositive.
SK-LOG-DOMAIN-CHECK · first introduced in Math 31
- R** Isolate an exponential expression and apply a logarithm.
SK-LOG-SOLVE-EXP · first introduced in Math 31
- R** Combine or isolate logarithms before converting to exponential form.
SK-LOG-SOLVE-LOG · first introduced in Math 31

M39-EXP-006

I can fit an exponential regression model using technology and evaluate its fit.

OpenStax: **Supplemental resources:** [4.8 Fitting Exponential Models to Data](#)

SUPPORTING SKILLS

- A** Interpret the sign and magnitude of a correlation coefficient in context.
SK-REG-CORRELATION-INTERPRET · first introduced in Math 39
- I** Use technology to determine an exponential regression model.
SK-REG-EXPONENTIAL-FIT
- D** Use residual patterns and contextual evidence to judge whether a model is appropriate.
SK-REG-MODEL-ASSESS · first introduced in Math 39

M39-EXP-007

I can compare linear and exponential models and justify which is more appropriate for a data set or context.

OpenStax: **Supplemental resources:** [Precalculus 2e 4.7 Exponential and Logarithmic Models](#); [College Algebra 2e 4.3 Fitting Linear Models to Data](#); [Precalculus 2e 4.8 Fitting Exponential Models to Data](#)

SUPPORTING SKILLS

- A** Compare functions shown in different representations.
SK-FUNC-COMPARE-REP · first introduced in Math 21
- I** Select a function family whose behavior fits a table, graph, or context.
SK-MODEL-FAMILY-SELECT
- D** Use residual patterns and contextual evidence to judge whether a model is appropriate.
SK-REG-MODEL-ASSESS · first introduced in Math 39

Power Functions and Function Comparison

REQUIRED

OpenStax coverage: **Supplemental resources:** [Precalculus 2e 3.3 Power Functions and Polynomial Functions](#); [Precalculus 2e 3.9 Modeling Using Variation](#); [College Algebra 2e 2.6 Other Types of Equations](#); [Precalculus 2e 4.2 Graphs of Exponential Functions](#); [Precalculus 2e 4.4 Graphs of Logarithmic Functions](#)

M39-POW-001

I can identify a power function and interpret the parameters in $y = kx^p$.

OpenStax: **Supplemental resources:** [3.3 Power Functions and Polynomial Functions](#)

SUPPORTING SKILLS

- I** Interpret the coefficient and exponent in a power function.
SK-POWER-PARAMETER-INTERPRET
- I** Recall and correctly use power-function terms including coefficient, exponent, direct variation, and inverse variation.
SK-POWER-VOCABULARY
- I** Represent and interpret a power-variation relationship.
SK-VAR-POWER

M39-POW-002

I can construct a power function from two data points or a proportional relationship.

OpenStax: **Supplemental resources:** [3.9 Modeling Using Variation](#)

SUPPORTING SKILLS

- A** Isolate a specified variable in an equation or formula.
SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- I** Determine a power function from two points or a proportional relationship.
SK-POWER-CONSTRUCT

M39-POW-003

I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.

OpenStax: **Supplemental resources:** [3.3 Power Functions and Polynomial Functions](#)

SUPPORTING SKILLS

- I** Determine power-function behavior as the input approaches zero or positive or negative infinity.
SK-POWER-END-BEHAVIOR
- I** Relate the coefficient and exponent of a power function to its graph shape.
SK-POWER-SHAPE
- D** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21

M39-POW-004

I can recognize and model direct, inverse, and other power-variation relationships.

OpenStax: **Supplemental resources:** [3.9 Modeling Using Variation](#)

SUPPORTING SKILLS

- I** Represent and interpret a direct-variation relationship.
SK-VAR-DIRECT
- I** Represent and interpret an inverse-variation relationship.
SK-VAR-INVERSE
- D** Represent and interpret a power-variation relationship.
SK-VAR-POWER · first introduced in Math 39

M39-POW-005

I can solve equations involving integer or rational powers.

OpenStax: **Supplemental resources:** [2.6 Other Types of Equations](#)

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- A** Interpret numerator and denominator roles in a rational exponent.
SK-EXP-RATIONAL · first introduced in Math 21

M39-POW-006

I can compare the long-run dominance of logarithmic, power, polynomial, and exponential functions.

OpenStax: **Supplemental resources:** [3.3 Power Functions and Polynomial Functions](#); [4.2 Graphs of Exponential Functions](#); [4.4 Graphs of Logarithmic Functions](#)

SUPPORTING SKILLS

- D** Compare functions shown in different representations.
SK-FUNC-COMPARE-REP · first introduced in Math 21
- I** Compare the long-run growth or decay of functions from different families.
SK-FUNC-DOMINANCE

M39-POW-007

I can fit a power regression model and interpret its parameters and correlation evidence.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- D** Interpret the coefficient and exponent in a power function.
SK-POWER-PARAMETER-INTERPRET · first introduced in Math 39
- A** Interpret the sign and magnitude of a correlation coefficient in context.
SK-REG-CORRELATION-INTERPRET · first introduced in Math 39
- I** Use technology to determine a power regression model.
SK-REG-POWER-FIT

Function Combinations and Polynomials

REQUIRED

OpenStax coverage: **Supplemental resources:** [1.4 Composition of Functions](#); [3.4 Graphs of Polynomial Functions](#); [3.6 Zeros of Polynomial Functions](#)

M39-POL-001

I can add, subtract, multiply, and divide functions represented by tables, graphs, or formulas.

OpenStax: **Supplemental resources:** [1.4 Composition of Functions](#)

SUPPORTING SKILLS

A Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12

I Add, subtract, multiply, and divide function outputs.
SK-FUNC-OPERATIONS

M39-POL-002

I can determine where a combination of functions is positive, zero, or undefined.

OpenStax: **Supplemental resources:** [1.4 Composition of Functions](#); [3.4 Graphs of Polynomial Functions](#)

SUPPORTING SKILLS

I Determine the domain of a sum, difference, product, quotient, or composition.
SK-FUNC-DOMAIN-COMBINE

D Add, subtract, multiply, and divide function outputs.
SK-FUNC-OPERATIONS · first introduced in Math 39

M39-POL-003

I can determine polynomial end behavior from degree and leading coefficient.

OpenStax: **Supplemental resources:** [3.4 Graphs of Polynomial Functions](#)

SUPPORTING SKILLS

I Infer polynomial end behavior from degree and leading coefficient.
SK-POLY-END-BEHAVIOR

R Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M39-POL-004

I can identify zeros and multiplicities and relate them to polynomial graph behavior.

OpenStax: **Supplemental resources:** [3.4 Graphs of Polynomial Functions](#)

SUPPORTING SKILLS

I Relate zero multiplicity to local graph behavior.
SK-POLY-MULTIPLICITY

A Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

D Determine real and complex zeros using an appropriate method.
SK-POLY-ZEROS · first introduced in Math 21

M39-POL-005

I can estimate a polynomial's possible degree from its zeros, turning points, inflection points, and end behavior.

OpenStax: **Supplemental resources:** [3.4 Graphs of Polynomial Functions](#)

SUPPORTING SKILLS

- I** Use inflection points to establish a minimum possible polynomial degree.
SK-POLY-DEGREE-INFLECTION
- I** Use turning points to establish a minimum possible polynomial degree.
SK-POLY-DEGREE-TURNING
- I** Use zeros and their multiplicities to establish a minimum possible polynomial degree.
SK-POLY-DEGREE-ZEROS
- D** Infer polynomial end behavior from degree and leading coefficient.
SK-POLY-END-BEHAVIOR · first introduced in Math 39
- D** Relate zero multiplicity to local graph behavior.
SK-POLY-MULTIPLICITY · first introduced in Math 39

M39-POL-006

I can construct a polynomial function from specified zeros, multiplicities, and a point.

OpenStax: **Supplemental resources:** [3.6 Zeros of Polynomial Functions](#)

SUPPORTING SKILLS

- D** Build a polynomial from zeros, multiplicities, and a point.
SK-POLY-CONSTRUCT · first introduced in Math 31
- A** Relate zero multiplicity to local graph behavior.
SK-POLY-MULTIPLICITY · first introduced in Math 39
- A** Determine real and complex zeros using an appropriate method.
SK-POLY-ZEROS · first introduced in Math 21

M39-POL-007

I can match polynomial formulas and graphs using intercepts, multiplicity, degree, and end behavior.

OpenStax: **Supplemental resources:** [3.4 Graphs of Polynomial Functions](#)

SUPPORTING SKILLS

- D** Infer polynomial end behavior from degree and leading coefficient.
SK-POLY-END-BEHAVIOR · first introduced in Math 39
- A** Relate zero multiplicity to local graph behavior.
SK-POLY-MULTIPLICITY · first introduced in Math 39
- A** Determine real and complex zeros using an appropriate method.
SK-POLY-ZEROS · first introduced in Math 21

M39-POL-008

I can build and optimize a polynomial model in a geometric or economic context.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- A** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- D** Interpret vertex and zeros in an application.
SK-QUAD-MODEL-FEATURES · first introduced in Math 21

Describing Data

REQUIRED

OpenStax coverage: No mapped OpenStax section

M39-STA-001

I can distinguish categorical and quantitative data.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I** Classify data as categorical or quantitative.
SK-STAT-DATA-TYPE

M39-STA-002

I can construct and interpret frequency and relative-frequency tables.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I** Construct and interpret a frequency table.
SK-STAT-FREQUENCY-TABLE
- I** Calculate and interpret relative frequency.
SK-STAT-RELATIVE-FREQUENCY

M39-STA-003

I can construct and interpret bar charts, pie charts, and histograms using appropriate technology.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I** Construct and interpret bar charts and pie charts.
SK-STAT-CATEGORICAL-DISPLAY
- I** Construct and interpret a histogram for quantitative data.
SK-STAT-HISTOGRAM

M39-STA-004

I can identify misleading features in a data display.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I** Identify perceptual distortion, misleading scales, and other misleading features in data displays.
SK-STAT-GRAPH-CRITIQUE

M39-STA-005

I can describe a distribution's modality, symmetry, skewness, and outliers.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I** Describe a distribution by modality, symmetry, and skewness.
SK-STAT-DISTRIBUTION-SHAPE
- I** Identify and interpret unusual observations or outliers.
SK-STAT-OUTLIER-IDENTIFY
- I** Recall and correctly use terms for data type, distribution shape, center, spread, percentile, and outlier.
SK-STAT-VOCABULARY

M39-STA-006

I can calculate and interpret mean, median, and mode and select an appropriate measure of center.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I** Select an appropriate measure of center based on distribution shape and context.
SK-STAT-CENTER-SELECT
- I** Calculate and interpret the arithmetic mean.
SK-STAT-MEAN
- I** Calculate and interpret the median.
SK-STAT-MEDIAN
- I** Identify and interpret the mode or modes.
SK-STAT-MODE

M39-STA-007

I can calculate and interpret range, standard deviation, quartiles, and interquartile range.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I** Determine quartiles and calculate and interpret the interquartile range.
SK-STAT-QUARTILES-IQR
- I** Calculate and interpret the range.
SK-STAT-RANGE
- I** Calculate and interpret standard deviation as a measure of spread.
SK-STAT-STANDARD-DEVIATION

M39-STA-008

I can construct and interpret a five-number summary and boxplot.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I** Construct and interpret a boxplot.
SK-STAT-BOXPLOT
- I** Determine and interpret a five-number summary.
SK-STAT-FIVE-NUMBER

M39-STA-009

I can use percentiles and z-scores to locate and compare values within distributions.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I** Determine and interpret the percentile rank of a value.
SK-STAT-PERCENTILE
- I** Calculate and interpret a z-score.
SK-STAT-ZSCORE

I can compare distributions using their shape, center, spread, and unusual features.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- A** Select an appropriate measure of center based on distribution shape and context.
SK-STAT-CENTER-SELECT · first introduced in Math 39
- I** Compare distributions using shape, center, spread, and unusual features.
SK-STAT-DISTRIBUTION-COMPARE
- A** Describe a distribution by modality, symmetry, and skewness.
SK-STAT-DISTRIBUTION-SHAPE · first introduced in Math 39

Probability

REQUIRED

OpenStax coverage: Supplemental resources: [11.7 Probability](#)

M39-PRO-009

I can interpret and write notation for compound and conditional probabilities.

OpenStax: Supplemental resources: [11.7 Probability](#)

SUPPORTING SKILLS

- I** Read and write probability notation and correctly use the terms outcome, event, sample space, complement, intersection, union, and conditional probability.

SK-PROB-NOTATION-VOCAB

M39-PRO-001

I can organize categorical data in a contingency table or Venn diagram.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I** Construct and interpret a contingency table.
SK-PROB-CONTINGENCY-TABLE
 - I** Represent categorical events and their overlap with a Venn diagram.
SK-PROB-VENN-DIAGRAM
-

M39-PRO-002

I can calculate and interpret empirical, marginal, joint, and conditional probabilities from a contingency table.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I** Calculate and interpret a conditional probability.
SK-PROB-CONDITIONAL
 - I** Calculate probability from observed relative frequency.
SK-PROB-EMPIRICAL
 - I** Calculate and interpret marginal and joint probabilities.
SK-PROB-MARGINAL-JOINT
-

M39-PRO-003

I can identify an experiment, outcomes, events, and a sample space.

OpenStax: Supplemental resources: [11.7 Probability](#)

SUPPORTING SKILLS

- A** Read and write probability notation and correctly use the terms outcome, event, sample space, complement, intersection, union, and conditional probability.

SK-PROB-NOTATION-VOCAB · first introduced in Math 39

- I** Identify outcomes, events, and the sample space of an experiment.

SK-PROB-SAMPLE-SPACE

M39-PRO-004

I can calculate theoretical probabilities when outcomes are equally likely.

OpenStax: **Supplemental resources:** [11.7 Probability](#)

SUPPORTING SKILLS

- A** Identify outcomes, events, and the sample space of an experiment.
SK-PROB-SAMPLE-SPACE · first introduced in Math 39
- I** Calculate theoretical probability for equally likely outcomes.
SK-PROB-THEORETICAL

M39-PRO-005

I can use complements to calculate probabilities.

OpenStax: **Supplemental resources:** [11.7 Probability](#)

SUPPORTING SKILLS

- I** Use the complement rule to calculate probability.
SK-PROB-COMPLEMENT

M39-PRO-006

I can calculate probabilities of compound 'and' events and distinguish independent from dependent events.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I** Calculate the probability of a compound and event.
SK-PROB-AND
- D** Calculate and interpret a conditional probability.
SK-PROB-CONDITIONAL · first introduced in Math 39
- I** Determine whether events are independent or dependent.
SK-PROB-INDEPENDENCE

M39-PRO-007

I can calculate probabilities of compound 'or' events and distinguish overlapping from disjoint events.

OpenStax: **Supplemental resources:** [11.7 Probability](#)

SUPPORTING SKILLS

- I** Determine whether events are overlapping or disjoint.
SK-PROB-DISJOINT
- I** Calculate the probability of a compound or event.
SK-PROB-OR

M39-PRO-008

I can compare experimental and theoretical probability and explain the law of large numbers.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- D** Calculate probability from observed relative frequency.
SK-PROB-EMPIRICAL · first introduced in Math 39
- I** Explain how experimental probability stabilizes with many trials.
SK-PROB-LARGE-NUMBERS
- D** Calculate theoretical probability for equally likely outcomes.
SK-PROB-THEORETICAL · first introduced in Math 39

Expected Value

EXTENSION

OpenStax coverage: No mapped OpenStax section

M39-EV-001

I can calculate the expected value of a discrete probability situation.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I** Read expected-value notation and recall the meanings of outcome, probability, payoff, and long-run average.
SK-PROB-EXPECTED-NOTATION
 - I** Calculate the expected value of a discrete probability situation.
SK-PROB-EXPECTED-VALUE
-

M39-EV-002

I can interpret expected value to evaluate a game, risk, or long-run decision.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I** Interpret expected value as a long-run average for decision-making.
SK-PROB-EXPECTED-INTERPRET
- D** Calculate the expected value of a discrete probability situation.
SK-PROB-EXPECTED-VALUE · first introduced in Math 39

Linearization of Nonlinear Data

EXTENSION

OpenStax coverage: No mapped OpenStax section

M39-POW-008

I can linearize power or exponential data and translate between the linearized and original models.

OpenStax: No mapped OpenStax section

SUPPORTING SKILLS

- I** Transform exponential data so that a linear model can be fit to its logarithms.
SK-REG-LOG-LINEARIZE
- I** Transform power data so that a linear model can be fit on logarithmic axes.
SK-REG-LOGLOG-LINEARIZE
- I** Translate a linearized regression equation back to its original nonlinear form.
SK-REG-UNLINEARIZE

Math 49 Curriculum

Precalculus with Limits | Middlesex School Mathematics

COURSE AT A GLANCE

Review Content

REVIEW

Primary resources: [1.2 Domain and Range](#); [1.5 Transformation of Functions](#); [4.1 Exponential Functions](#); [4.3 Logarithmic Functions](#); [4.6 Exponential and Logarithmic Equations](#)
Supplemental resources: [4.2 Graphs of Exponential Functions](#); [4.7 Exponential and Logarithmic Models](#); [4.4 Graphs of Logarithmic Functions](#)

UNIT LEARNING OBJECTIVES

- I can determine the domain and range of a function from a formula, graph, or table.
- I can describe and apply transformations of a parent function.
- I can analyze exponential functions using their equations, graphs, and contexts.
- I can analyze logarithmic functions using their equations, graphs, and contexts.
- I can solve exponential and logarithmic equations using equivalent forms and logarithm properties.

Function Operations, Composition, and Inverses

REQUIRED

Primary resources: [1.5 Transformation of Functions](#); [1.4 Composition of Functions](#); [1.7 Inverse Functions](#)

UNIT LEARNING OBJECTIVES

- I can determine whether a function is even, odd, or neither.
- I can add, subtract, multiply, and divide functions and state the resulting domain.
- I can evaluate a composition from formulas, graphs, or tables.
- I can write a formula for the composition of two functions and determine its domain.
- I can decompose a function into a composition of simpler functions.
- I can determine whether a function has an inverse on a given domain.
- I can find and verify an inverse function algebraically.
- I can interpret the relationship between a function and its inverse from graphs or tables.

Power Functions

REQUIRED

Primary resources: [Precalculus 2e 3.3 Power Functions and Polynomial Functions](#); [Precalculus 2e 3.9 Modeling Using Variation](#); [College Algebra 2e 2.6 Other Types of Equations](#); [Precalculus 2e 3.8 Inverses and Radical Functions](#)
Supplemental resources: [Precalculus 2e 3.3 Power Functions and Polynomial Functions](#)

UNIT LEARNING OBJECTIVES

- I can identify a power function and interpret the parameters in $y = kx^p$.
- I can construct a power function from two data points or a proportional relationship.
- I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.
- I can recognize and model direct, inverse, and other power-variation relationships.
- I can solve equations involving integer or rational powers.
- I can build or select a power-function model for a context and interpret its parameters.
- I can analyze power functions with fractional exponents, including their domains, graph shapes, and behavior near zero.

Polynomial Functions

REQUIRED

Primary resources: [3.4 Graphs of Polynomial Functions](#); [3.6 Zeros of Polynomial Functions](#)
Supplemental resources: [3.6 Zeros of Polynomial Functions](#)

UNIT LEARNING OBJECTIVES

- I can predict the end behavior of a polynomial from its degree and leading coefficient.
- I can determine polynomial zeros and their multiplicities.
- I can construct a polynomial function from specified zeros or graphical features.
- I can sketch and analyze a polynomial function using intercepts, multiplicity, and end behavior.
- I can determine the minimum possible degree of a polynomial function from the zeros, turning points, inflection points, and end behavior of its graph.
- I can build or select a polynomial model for a context and interpret its features.

Rational Functions and Limits

REQUIRED

Primary resources: [Precalculus 2e 3.7 Rational Functions](#); [Calculus Volume 1 2.2 The Limit of a Function](#); [Calculus Volume 1 2.3 The Limit Laws](#); [Precalculus 2e 3.3 Power Functions and Polynomial Functions](#)
Supplemental resources: [Calculus Volume 1 2.2 The Limit of a Function](#); [Precalculus 2e 3.4 Graphs of Polynomial Functions](#); [Precalculus 2e 3.7 Rational Functions](#); [Precalculus 2e 4.4 Graphs of Logarithmic Functions](#); [Precalculus 2e 3.9 Modeling Using Variation](#)

UNIT LEARNING OBJECTIVES

- I can identify holes and vertical, horizontal, or slant asymptotes of a rational function.
- I can sketch and analyze a rational function using its algebraic and asymptotic features.
- I can interpret limit notation as a statement about function behavior near an input.
- I can estimate a limit from a graph or table.
- I can evaluate a limit algebraically using direct substitution or simplification.
- I can interpret an infinite limit as vertical-asymptotic behavior.
- I can use limit notation to describe behavior at infinity.
- I can compare the long-run growth of polynomial, exponential, logarithmic, and rational functions, and combinations thereof.
- I can build or select a rational or rational-adjacent model for a context and interpret its long-run behavior.

Sequences and Series

REQUIRED

Primary resources: [11.1 Sequences and Their Notations](#); [11.2 Arithmetic Sequences](#); [11.3 Geometric Sequences](#); [11.4 Series and Their Notations](#)
Supplemental resources: [11.1 Sequences and Their Notations](#); [11.3 Geometric Sequences](#); [11.2 Arithmetic Sequences](#)

UNIT LEARNING OBJECTIVES

- I can evaluate and graph a sequence defined explicitly or recursively.
- I can write explicit and recursive formulas for arithmetic sequences.
- I can write explicit and recursive formulas for geometric sequences.
- I can distinguish arithmetic, geometric, and other sequences from formulas, tables, or contexts.
- I can create and interpret a sequence model for discrete change.
- I can interpret and use summation notation.
- I can calculate a finite arithmetic series.
- I can calculate a finite geometric series.
- I can determine whether an infinite geometric series converges and find its sum when it does.
- I can create and interpret a series model for accumulated change.

Additional Limit Concepts

EXTENSION

Primary resources: [2.2 The Limit of a Function](#); [2.4 Continuity](#)

UNIT LEARNING OBJECTIVES

- I can determine one-sided limits and decide whether a two-sided limit exists.
- I can distinguish a function value from a limit and determine continuity at a point.

Parametric Equations

EXTENSION

Primary resources: [8.7 Parametric Equations: Graphs](#); [8.6 Parametric Equations](#)

UNIT LEARNING OBJECTIVES

- I can graph a plane curve defined by parametric equations with its orientation.
- I can eliminate a parameter to obtain a rectangular equation.
- I can create or interpret parametric equations for motion in the plane.

Conic Sections

EXTENSION

Primary resources: [10.1 The Ellipse](#); [10.2 The Hyperbola](#); [10.3 The Parabola](#)
Supplemental resources: [10.2 The Hyperbola](#); [10.3 The Parabola](#); [8.6 Parametric Equations](#); [10.1 The Ellipse](#)

UNIT LEARNING OBJECTIVES

- I can classify a circle, ellipse, or hyperbola from its equation or defining geometric relationship.
- I can use completing the square to rewrite a general-form conic equation in standard form.
- I can write, interpret, and graph implicit and parametric equations of a circle.
- I can write, interpret, and graph implicit and parametric equations of an ellipse.
- I can write, interpret, and graph implicit and parametric equations of a hyperbola.

Conic Sections (Continued)

EXTENSION

UNIT LEARNING OBJECTIVES - CONTINUED

I can determine the center, vertices, foci, axes, or asymptotes needed to describe a circle, ellipse, or hyperbola.

I can explain the geometric definitions of circles, ellipses, and hyperbolas using foci and distance relationships.

I can explain a parabola as the set of points equidistant from a focus and directrix.

IN-DEPTH CURRICULUM

SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Review Content

REVIEW

OpenStax coverage: **Primary resources:** [1.2 Domain and Range](#); [1.5 Transformation of Functions](#); [4.1 Exponential Functions](#); [4.3 Logarithmic Functions](#); [4.6 Exponential and Logarithmic Equations](#)

Supplemental resources: [4.2 Graphs of Exponential Functions](#); [4.7 Exponential and Logarithmic Models](#); [4.4 Graphs of Logarithmic Functions](#)

M49-FND-001

I can determine the domain and range of a function from a formula, graph, or table.

OpenStax: **Primary resources:** [1.2 Domain and Range](#)

SUPPORTING SKILLS

D Determine domain restrictions from denominators, even roots, and logarithms.
SK-FUNC-DOMAIN-FORMULA · first introduced in Math 31

R Read domain and range from a graph.
SK-FUNC-DOMAIN-RANGE-GRAPH · first introduced in Math 12

R Read domain and range from a table.
SK-FUNC-DOMAIN-RANGE-TABLE · first introduced in Math 12

M49-FND-002

I can describe and apply transformations of a parent function.

OpenStax: **Primary resources:** [1.5 Transformation of Functions](#)

SUPPORTING SKILLS

R Interpret transformations applied to function inputs.
SK-FUNC-TRANSFORM-HORIZONTAL · first introduced in Math 31

R Distinguish stretches from compressions and horizontal effects from vertical effects.
SK-FUNC-TRANSFORM-SCALE · first introduced in Math 31

R Interpret transformations applied to function outputs.
SK-FUNC-TRANSFORM-VERTICAL · first introduced in Math 31

M49-FND-004

I can analyze exponential functions using their equations, graphs, and contexts.

OpenStax: **Primary resources:** [4.1 Exponential Functions](#)

Supplemental resources: [4.2 Graphs of Exponential Functions](#); [4.7 Exponential and Logarithmic Models](#)

SUPPORTING SKILLS

- R** Identify horizontal asymptotic behavior of an exponential graph.
SK-EXP-ASYMPTOTE · first introduced in Math 31
- R** Identify initial amount, rate, and time interval in context.
SK-EXP-MODEL-CONTEXT · first introduced in Math 31
- A** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21

M49-FND-005

I can analyze logarithmic functions using their equations, graphs, and contexts.

OpenStax: **Primary resources:** [4.3 Logarithmic Functions](#)

Supplemental resources: [4.4 Graphs of Logarithmic Functions](#); [4.7 Exponential and Logarithmic Models](#)

SUPPORTING SKILLS

- R** Identify vertical asymptotic behavior of a logarithmic graph.
SK-LOG-ASYMPTOTE · first introduced in Math 31
- R** Interpret a logarithm as an exponent.
SK-LOG-DEFINITION · first introduced in Math 31
- A** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21

M49-FND-006

I can solve exponential and logarithmic equations using equivalent forms and logarithm properties.

OpenStax: **Primary resources:** [4.6 Exponential and Logarithmic Equations](#)

SUPPORTING SKILLS

- R** Rewrite exponential expressions using a common base when possible.
SK-EXP-REWRITE-BASE · first introduced in Math 31
- A** Reject solutions that make a logarithm argument nonpositive.
SK-LOG-DOMAIN-CHECK · first introduced in Math 31
- A** Apply the logarithm power property in either direction.
SK-LOG-POWER · first introduced in Math 31
- A** Apply the logarithm product property in either direction.
SK-LOG-PRODUCT · first introduced in Math 31
- A** Apply the logarithm quotient property in either direction.
SK-LOG-QUOTIENT · first introduced in Math 31
- R** Isolate an exponential expression and apply a logarithm.
SK-LOG-SOLVE-EXP · first introduced in Math 31
- R** Combine or isolate logarithms before converting to exponential form.
SK-LOG-SOLVE-LOG · first introduced in Math 31

Function Operations, Composition, and Inverses

REQUIRED

OpenStax coverage: **Primary resources:** [1.5 Transformation of Functions](#); [1.4 Composition of Functions](#); [1.7 Inverse Functions](#)

M49-FND-003

I can determine whether a function is even, odd, or neither.

OpenStax: **Primary resources:** [1.5 Transformation of Functions](#)

SUPPORTING SKILLS

- D** Test a formula or graph for even or odd symmetry.
SK-FUNC-SYMMETRY · first introduced in Math 31

M49-OPS-001

I can add, subtract, multiply, and divide functions and state the resulting domain.

OpenStax: **Primary resources:** [1.4 Composition of Functions](#)

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I** Determine the domain of a sum, difference, product, quotient, or composition.
SK-FUNC-DOMAIN-COMBINE
- I** Add, subtract, multiply, and divide function outputs.
SK-FUNC-OPERATIONS

M49-OPS-002

I can evaluate a composition from formulas, graphs, or tables.

OpenStax: **Primary resources:** [1.4 Composition of Functions](#)

SUPPORTING SKILLS

- D** Substitute one function formula into another.
SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- D** Follow intermediate values to compose functions from tables.
SK-FUNC-COMPOSE-TABLE · first introduced in Math 31
- A** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- R** Read and write notation for function composition and inverse functions.
SK-FUNC-STRUCTURE-NOTATION · first introduced in Math 31

M49-OPS-003

I can write a formula for the composition of two functions and determine its domain.

OpenStax: **Primary resources:** [1.4 Composition of Functions](#)

SUPPORTING SKILLS

- D** Substitute one function formula into another.
SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- D** Determine the domain of a sum, difference, product, quotient, or composition.
SK-FUNC-DOMAIN-COMBINE · first introduced in Math 39

M49-OPS-004

I can decompose a function into a composition of simpler functions.

OpenStax: **Primary resources:** [1.4 Composition of Functions](#)

SUPPORTING SKILLS

- A** Substitute one function formula into another.
SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- I** Identify simpler inner and outer functions in a composition.
SK-FUNC-DECOMPOSE

M49-OPS-005

I can determine whether a function has an inverse on a given domain.

OpenStax: **Primary resources:** [1.7 Inverse Functions](#)

SUPPORTING SKILLS

- D** Apply the horizontal line test for invertibility.
SK-FUNC-INVERSE-HLT · first introduced in Math 31
- R** Test whether a function is one-to-one on a domain.
SK-FUNC-ONE-TO-ONE · first introduced in Math 31

M49-OPS-006

I can find and verify an inverse function algebraically.

OpenStax: **Primary resources:** [1.7 Inverse Functions](#)

SUPPORTING SKILLS

- A** Substitute one function formula into another.
SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- D** Exchange variables and solve to obtain an inverse formula.
SK-FUNC-INVERSE-ALGEBRA · first introduced in Math 31
- D** Verify inverse functions using composition.
SK-FUNC-INVERSE-VERIFY · first introduced in Math 31

M49-OPS-007

I can interpret the relationship between a function and its inverse from graphs or tables.

OpenStax: **Primary resources:** [1.7 Inverse Functions](#)

SUPPORTING SKILLS

- D** Exchange domain and range between a function and its inverse.
SK-FUNC-INVERSE-DOMAIN-RANGE · first introduced in Math 31
- D** Exchange input and output values when interpreting an inverse.
SK-FUNC-INVERSE-SWAP · first introduced in Math 31
- D** Reverse input and output units when interpreting an inverse.
SK-FUNC-INVERSE-UNITS · first introduced in Math 31

Power Functions

REQUIRED

OpenStax coverage: **Primary resources:** [Precalculus 2e 3.3 Power Functions and Polynomial Functions](#); [Precalculus 2e 3.9 Modeling Using Variation](#); [College Algebra 2e 2.6 Other Types of Equations](#); [Precalculus 2e 3.8 Inverses and Radical Functions](#)
Supplemental resources: [Precalculus 2e 3.3 Power Functions and Polynomial Functions](#)

M49-PWR-001

I can identify a power function and interpret the parameters in $y = kx^p$.

OpenStax: **Primary resources:** [3.3 Power Functions and Polynomial Functions](#)

SUPPORTING SKILLS

- D** Interpret the coefficient and exponent in a power function.
SK-POWER-PARAMETER-INTERPRET · first introduced in Math 39
- R** Recall and correctly use power-function terms including coefficient, exponent, direct variation, and inverse variation.
SK-POWER-VOCABULARY · first introduced in Math 39
- D** Represent and interpret a power-variation relationship.
SK-VAR-POWER · first introduced in Math 39

M49-PWR-002

I can construct a power function from two data points or a proportional relationship.

OpenStax: **Primary resources:** [3.9 Modeling Using Variation](#)

SUPPORTING SKILLS

- A** Isolate a specified variable in an equation or formula.
SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- D** Determine a power function from two points or a proportional relationship.
SK-POWER-CONSTRUCT · first introduced in Math 39

M49-PWR-003

I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.

OpenStax: **Primary resources:** [3.3 Power Functions and Polynomial Functions](#)

SUPPORTING SKILLS

- D** Determine power-function behavior as the input approaches zero or positive or negative infinity.
SK-POWER-END-BEHAVIOR · first introduced in Math 39
- D** Relate the coefficient and exponent of a power function to its graph shape.
SK-POWER-SHAPE · first introduced in Math 39
- A** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21

M49-PWR-004

I can recognize and model direct, inverse, and other power-variation relationships.

OpenStax: **Primary resources:** [3.9 Modeling Using Variation](#)

SUPPORTING SKILLS

- D** Represent and interpret a direct-variation relationship.
SK-VAR-DIRECT · first introduced in Math 39
- D** Represent and interpret an inverse-variation relationship.
SK-VAR-INVERSE · first introduced in Math 39
- D** Represent and interpret a power-variation relationship.
SK-VAR-POWER · first introduced in Math 39

I can solve equations involving integer or rational powers.

OpenStax: **Primary resources:** [2.6 Other Types of Equations](#)

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- D** Interpret numerator and denominator roles in a rational exponent.
SK-EXP-RATIONAL · first introduced in Math 21

I can build or select a power-function model for a context and interpret its parameters.

OpenStax: **Primary resources:** [3.9 Modeling Using Variation](#)

SUPPORTING SKILLS

- I** Select a function family whose behavior fits a table, graph, or context.
SK-MODEL-FAMILY-SELECT
- D** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A** Determine a power function from two points or a proportional relationship.
SK-POWER-CONSTRUCT · first introduced in Math 39

I can analyze power functions with fractional exponents, including their domains, graph shapes, and behavior near zero.

OpenStax: **Primary resources:** [3.8 Inverses and Radical Functions](#)

Supplemental resources: [3.3 Power Functions and Polynomial Functions](#)

SUPPORTING SKILLS

- A** Interpret numerator and denominator roles in a rational exponent.
SK-EXP-RATIONAL · first introduced in Math 21
- A** Require an even-root radicand to be nonnegative when determining domain.
SK-FUNC-DOMAIN-EVEN-ROOT · first introduced in Math 21
- I** Analyze the domain, graph shape, and behavior near zero of a power function with a fractional exponent.
SK-POWER-FRACTIONAL-ANALYZE

Polynomial Functions

REQUIRED

OpenStax coverage: **Primary resources:** [3.4 Graphs of Polynomial Functions](#); [3.6 Zeros of Polynomial Functions](#)
Supplemental resources: [3.6 Zeros of Polynomial Functions](#)

M49-ALG-002

I can predict the end behavior of a polynomial from its degree and leading coefficient.

OpenStax: **Primary resources:** [3.4 Graphs of Polynomial Functions](#)

SUPPORTING SKILLS

- I** Infer polynomial end behavior from degree and leading coefficient.
SK-POLY-END-BEHAVIOR
- R** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M49-ALG-003

I can determine polynomial zeros and their multiplicities.

OpenStax: **Primary resources:** [3.4 Graphs of Polynomial Functions](#)

SUPPORTING SKILLS

- I** Relate zero multiplicity to local graph behavior.
SK-POLY-MULTIPLICITY
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12
- D** Determine real and complex zeros using an appropriate method.
SK-POLY-ZEROS · first introduced in Math 21

M49-ALG-004

I can construct a polynomial function from specified zeros or graphical features.

OpenStax: **Primary resources:** [3.6 Zeros of Polynomial Functions](#)

SUPPORTING SKILLS

- D** Build a polynomial from zeros, multiplicities, and a point.
SK-POLY-CONSTRUCT · first introduced in Math 31
- A** Relate zero multiplicity to local graph behavior.
SK-POLY-MULTIPLICITY · first introduced in Math 39
- A** Determine real and complex zeros using an appropriate method.
SK-POLY-ZEROS · first introduced in Math 21

M49-ALG-005

I can sketch and analyze a polynomial function using intercepts, multiplicity, and end behavior.

OpenStax: **Primary resources:** [3.4 Graphs of Polynomial Functions](#)

SUPPORTING SKILLS

- D** Infer polynomial end behavior from degree and leading coefficient.
SK-POLY-END-BEHAVIOR · first introduced in Math 39
- D** Relate zero multiplicity to local graph behavior.
SK-POLY-MULTIPLICITY · first introduced in Math 39
- A** Determine real and complex zeros using an appropriate method.
SK-POLY-ZEROS · first introduced in Math 21

I can determine the minimum possible degree of a polynomial function from the zeros, turning points, inflection points, and end behavior of its graph.

OpenStax: **Primary resources:** [3.4 Graphs of Polynomial Functions](#)

SUPPORTING SKILLS

- I** Use inflection points to establish a minimum possible polynomial degree.
SK-POLY-DEGREE-INFLECTION
- I** Use turning points to establish a minimum possible polynomial degree.
SK-POLY-DEGREE-TURNING
- I** Use zeros and their multiplicities to establish a minimum possible polynomial degree.
SK-POLY-DEGREE-ZEROS
- A** Infer polynomial end behavior from degree and leading coefficient.
SK-POLY-END-BEHAVIOR · first introduced in Math 39

I can build or select a polynomial model for a context and interpret its features.

OpenStax: **Primary resources:** [3.4 Graphs of Polynomial Functions](#)

Supplemental resources: [3.6 Zeros of Polynomial Functions](#)

SUPPORTING SKILLS

- D** Select a function family whose behavior fits a table, graph, or context.
SK-MODEL-FAMILY-SELECT · first introduced in Math 39
- D** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A** Build a polynomial from zeros, multiplicities, and a point.
SK-POLY-CONSTRUCT · first introduced in Math 31

Rational Functions and Limits

REQUIRED

OpenStax coverage: **Primary resources:** [Precalculus 2e 3.7 Rational Functions](#); [Calculus Volume 1 2.2 The Limit of a Function](#); [Calculus Volume 1 2.3 The Limit Laws](#); [Precalculus 2e 3.3 Power Functions and Polynomial Functions](#)
Supplemental resources: [Calculus Volume 1 2.2 The Limit of a Function](#); [Precalculus 2e 3.4 Graphs of Polynomial Functions](#); [Precalculus 2e 3.7 Rational Functions](#); [Precalculus 2e 4.4 Graphs of Logarithmic Functions](#); [Precalculus 2e 3.9 Modeling Using Variation](#)

M49-ALG-006

I can identify holes and vertical, horizontal, or slant asymptotes of a rational function.

OpenStax: **Primary resources:** [3.7 Rational Functions](#)

SUPPORTING SKILLS

- A** Cancel common factors rather than individual terms.
SK-RAT-CANCEL-FACTORS · first introduced in Math 21
- I** Determine horizontal or slant asymptotic behavior.
SK-RAT-END-BEHAVIOR
- A** Factor numerators and denominators completely.
SK-RAT-FACTOR · first introduced in Math 21
- I** Recall and correctly use rational-function terms including restriction, hole, asymptote, and end behavior.
SK-RAT-FUNCTION-VOCAB
- I** Identify a removable discontinuity and its coordinates.
SK-RAT-HOLE
- I** Determine vertical asymptotes from noncanceling denominator zeros.
SK-RAT-VERTICAL-ASYMPTOTE
- R** Recall and correctly use rational-expression terms including numerator, denominator, restriction, and excluded value.
SK-RAT-VOCABULARY · first introduced in Math 21

M49-ALG-007

I can sketch and analyze a rational function using its algebraic and asymptotic features.

OpenStax: **Primary resources:** [3.7 Rational Functions](#)

SUPPORTING SKILLS

- D** Determine horizontal or slant asymptotic behavior.
SK-RAT-END-BEHAVIOR · first introduced in Math 49
- D** Identify a removable discontinuity and its coordinates.
SK-RAT-HOLE · first introduced in Math 49
- A** Determine excluded values from an original denominator.
SK-RAT-RESTRICTIONS · first introduced in Math 21
- I** Determine the sign of a rational function on intervals.
SK-RAT-SIGN-INTERVAL
- D** Determine vertical asymptotes from noncanceling denominator zeros.
SK-RAT-VERTICAL-ASYMPTOTE · first introduced in Math 49

M49-LIM-001

I can interpret limit notation as a statement about function behavior near an input.

OpenStax: **Primary resources:** [2.2 The Limit of a Function](#)

SUPPORTING SKILLS

- I** Read and write limit notation with correct approach direction.
SK-LIM-NOTATION

M49-LIM-002

I can estimate a limit from a graph or table.

OpenStax: **Primary resources:** [2.2 The Limit of a Function](#)

SUPPORTING SKILLS

- I** Estimate one-sided and two-sided limits from a graph.
SK-LIM-GRAPH
- I** Estimate a limit from input-output values approaching from both sides.
SK-LIM-TABLE

M49-LIM-003

I can evaluate a limit algebraically using direct substitution or simplification.

OpenStax: **Primary resources:** [2.3 The Limit Laws](#)

SUPPORTING SKILLS

- I** Resolve an indeterminate algebraic form by factoring or rationalizing.
SK-LIM-SIMPLIFY
- I** Test a limit using direct substitution.
SK-LIM-SUBSTITUTE
- A** Factor numerators and denominators completely.
SK-RAT-FACTOR · first introduced in Math 21

M49-LIM-005

I can interpret an infinite limit as vertical-asymptotic behavior.

OpenStax: **Primary resources:** [2.2 The Limit of a Function](#)

SUPPORTING SKILLS

- I** Interpret unbounded output near a finite input as an infinite limit.
SK-LIM-INFINITE
- D** Determine vertical asymptotes from noncanceling denominator zeros.
SK-RAT-VERTICAL-ASYMPTOTE · first introduced in Math 49

M49-LIM-006

I can use limit notation to describe behavior at infinity.

OpenStax: **Primary resources:** [Precalculus 2e 3.7 Rational Functions](#)
Supplemental resources: [Calculus Volume 1 2.2 The Limit of a Function](#)

SUPPORTING SKILLS

- I** Compare dominant terms to determine behavior at infinity.
SK-LIM-INFINITY
- D** Determine horizontal or slant asymptotic behavior.
SK-RAT-END-BEHAVIOR · first introduced in Math 49

I can compare the long-run growth of polynomial, exponential, logarithmic, and rational functions, and combinations thereof.

OpenStax: **Primary resources:** [3.3 Power Functions and Polynomial Functions](#)

Supplemental resources: [3.4 Graphs of Polynomial Functions](#); [3.7 Rational Functions](#); [4.4 Graphs of Logarithmic Functions](#)

SUPPORTING SKILLS

- A** Compare functions shown in different representations.
SK-FUNC-COMPARE-REP · first introduced in Math 21
- D** Compare the long-run growth or decay of functions from different families.
SK-FUNC-DOMINANCE · first introduced in Math 39
- I** Apply the long-run growth hierarchy among logarithmic, polynomial, and exponential functions.
SK-LIM-GROWTH-ORDER
- A** Compare dominant terms to determine behavior at infinity.
SK-LIM-INFINITY · first introduced in Math 49

I can build or select a rational or rational-adjacent model for a context and interpret its long-run behavior.

OpenStax: **Primary resources:** [3.7 Rational Functions](#)

Supplemental resources: [3.9 Modeling Using Variation](#)

SUPPORTING SKILLS

- D** Select a function family whose behavior fits a table, graph, or context.
SK-MODEL-FAMILY-SELECT · first introduced in Math 39
- D** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A** Determine horizontal or slant asymptotic behavior.
SK-RAT-END-BEHAVIOR · first introduced in Math 49

Sequences and Series

REQUIRED

OpenStax coverage: **Primary resources:** [11.1 Sequences and Their Notations](#); [11.2 Arithmetic Sequences](#); [11.3 Geometric Sequences](#); [11.4 Series and Their Notations](#)

Supplemental resources: [11.1 Sequences and Their Notations](#); [11.3 Geometric Sequences](#); [11.2 Arithmetic Sequences](#)

M49-SEQ-001

I can evaluate and graph a sequence defined explicitly or recursively.

OpenStax: **Primary resources:** [11.1 Sequences and Their Notations](#)

SUPPORTING SKILLS

- I** Generate an explicit formula from a sequence pattern.
SK-SEQ-EXPLICIT
- I** Graph a sequence as discrete ordered pairs.
SK-SEQ-GRAPH
- I** Interpret subscripts and evaluate terms of a sequence.
SK-SEQ-NOTATION
- I** Generate a recursive formula with an initial condition.
SK-SEQ-RECURSIVE
- I** Recall and correctly use the terms sequence, term, index, explicit formula, recursive formula, arithmetic sequence, and geometric sequence.
SK-SEQ-VOCABULARY

M49-SEQ-002

I can write explicit and recursive formulas for arithmetic sequences.

OpenStax: **Primary resources:** [11.2 Arithmetic Sequences](#)

SUPPORTING SKILLS

- I** Recognize constant difference in an arithmetic sequence.
SK-SEQ-DIFFERENCE
- D** Generate an explicit formula from a sequence pattern.
SK-SEQ-EXPLICIT · first introduced in Math 49
- D** Generate a recursive formula with an initial condition.
SK-SEQ-RECURSIVE · first introduced in Math 49

M49-SEQ-003

I can write explicit and recursive formulas for geometric sequences.

OpenStax: **Primary resources:** [11.3 Geometric Sequences](#)

SUPPORTING SKILLS

- D** Generate an explicit formula from a sequence pattern.
SK-SEQ-EXPLICIT · first introduced in Math 49
- I** Recognize constant ratio in a geometric sequence.
SK-SEQ-RATIO
- D** Generate a recursive formula with an initial condition.
SK-SEQ-RECURSIVE · first introduced in Math 49

M49-SEQ-004

I can distinguish arithmetic, geometric, and other sequences from formulas, tables, or contexts.

OpenStax: **Primary resources:** [11.2 Arithmetic Sequences](#)

Supplemental resources: [11.1 Sequences and Their Notations](#); [11.3 Geometric Sequences](#)

SUPPORTING SKILLS

- A** Compare functions shown in different representations.
SK-FUNC-COMPARE-REP · first introduced in Math 21
- D** Recognize constant difference in an arithmetic sequence.
SK-SEQ-DIFFERENCE · first introduced in Math 49
- D** Recognize constant ratio in a geometric sequence.
SK-SEQ-RATIO · first introduced in Math 49

M49-SEQ-005

I can create and interpret a sequence model for discrete change.

OpenStax: **Primary resources:** [11.3 Geometric Sequences](#)

Supplemental resources: [11.2 Arithmetic Sequences](#)

SUPPORTING SKILLS

- A** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Generate an explicit formula from a sequence pattern.
SK-SEQ-EXPLICIT · first introduced in Math 49
- I** Identify initial value and pattern of change in a discrete context.
SK-SEQ-MODEL
- A** Generate a recursive formula with an initial condition.
SK-SEQ-RECURSIVE · first introduced in Math 49

M49-SER-001

I can interpret and use summation notation.

OpenStax: **Primary resources:** [11.4 Series and Their Notations](#)

SUPPORTING SKILLS

- A** Interpret subscripts and evaluate terms of a sequence.
SK-SEQ-NOTATION · first introduced in Math 49
- I** Expand and evaluate a finite sum written in sigma notation.
SK-SER-SIGMA
- I** Recall and correctly use the terms series, partial sum, sigma notation, finite series, infinite series, and convergence.
SK-SER-VOCABULARY

M49-SER-002

I can calculate a finite arithmetic series.

OpenStax: **Primary resources:** [11.4 Series and Their Notations](#)

SUPPORTING SKILLS

- A** Recognize constant difference in an arithmetic sequence.
SK-SEQ-DIFFERENCE · first introduced in Math 49
- I** Apply an arithmetic-series sum formula.
SK-SER-ARITHMETIC

M49-SER-003

I can calculate a finite geometric series.

OpenStax: **Primary resources:** [11.4 Series and Their Notations](#)

SUPPORTING SKILLS

- A** Recognize constant ratio in a geometric sequence.
SK-SEQ-RATIO · first introduced in Math 49
- I** Apply a finite geometric-series sum formula.
SK-SER-GEOMETRIC-FINITE

M49-SER-004

I can determine whether an infinite geometric series converges and find its sum when it does.

OpenStax: **Primary resources:** [11.4 Series and Their Notations](#)

SUPPORTING SKILLS

- A** Recognize constant ratio in a geometric sequence.
SK-SEQ-RATIO · first introduced in Math 49
- I** Test convergence and sum an infinite geometric series.
SK-SER-GEOMETRIC-INFINITE

M49-SER-005

I can create and interpret a series model for accumulated change.

OpenStax: **Primary resources:** [11.4 Series and Their Notations](#)

SUPPORTING SKILLS

- A** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Apply an arithmetic-series sum formula.
SK-SER-ARITHMETIC · first introduced in Math 49
- A** Apply a finite geometric-series sum formula.
SK-SER-GEOMETRIC-FINITE · first introduced in Math 49
- A** Test convergence and sum an infinite geometric series.
SK-SER-GEOMETRIC-INFINITE · first introduced in Math 49
- I** Identify the initial term, change factor, and number of terms in an application.
SK-SER-MODEL

Additional Limit Concepts

EXTENSION

OpenStax coverage: **Primary resources:** [2.2 The Limit of a Function](#); [2.4 Continuity](#)

M49-LIM-004

I can determine one-sided limits and decide whether a two-sided limit exists.

OpenStax: **Primary resources:** [2.2 The Limit of a Function](#)

SUPPORTING SKILLS

- D** Estimate one-sided and two-sided limits from a graph.
SK-LIM-GRAPH · first introduced in Math 49
- D** Read and write limit notation with correct approach direction.
SK-LIM-NOTATION · first introduced in Math 49
- A** Estimate a limit from input-output values approaching from both sides.
SK-LIM-TABLE · first introduced in Math 49

M49-LIM-008

I can distinguish a function value from a limit and determine continuity at a point.

OpenStax: **Primary resources:** [2.4 Continuity](#)

SUPPORTING SKILLS

- I** Check the three conditions for continuity at a point.
SK-LIM-CONTINUITY
- A** Estimate one-sided and two-sided limits from a graph.
SK-LIM-GRAPH · first introduced in Math 49
- D** Test a limit using direct substitution.
SK-LIM-SUBSTITUTE · first introduced in Math 49

Parametric Equations

EXTENSION

OpenStax coverage: **Primary resources:** [8.7 Parametric Equations: Graphs](#); [8.6 Parametric Equations](#)

M49-PAR-001

I can graph a plane curve defined by parametric equations with its orientation.

OpenStax: **Primary resources:** [8.7 Parametric Equations: Graphs](#)

SUPPORTING SKILLS

- A** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D** Graph a parametric curve with its direction of motion.
SK-PAR-GRAPH · first introduced in Math 32
- D** Generate ordered pairs from parametric equations.
SK-PAR-TABLE · first introduced in Math 32

M49-PAR-002

I can eliminate a parameter to obtain a rectangular equation.

OpenStax: **Primary resources:** [8.6 Parametric Equations](#)

SUPPORTING SKILLS

- A** Isolate a specified variable in an equation or formula.
SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- D** Eliminate a parameter algebraically.
SK-PAR-ELIMINATE · first introduced in Math 32
- A** Substitute an equivalent expression for a variable.
SK-SYS-SUBSTITUTE · first introduced in Math 12

M49-PAR-003

I can create or interpret parametric equations for motion in the plane.

OpenStax: **Primary resources:** [8.7 Parametric Equations: Graphs](#)

SUPPORTING SKILLS

- D** Interpret or construct parametric equations for planar motion.
SK-PAR-MODEL · first introduced in Math 32
- A** Generate ordered pairs from parametric equations.
SK-PAR-TABLE · first introduced in Math 32
- A** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

Conic Sections

EXTENSION

OpenStax coverage: **Primary resources:** [10.1 The Ellipse](#); [10.2 The Hyperbola](#); [10.3 The Parabola](#)

Supplemental resources: [10.2 The Hyperbola](#); [10.3 The Parabola](#); [8.6 Parametric Equations](#); [10.1 The Ellipse](#)

M49-CON-001

I can classify a circle, ellipse, or hyperbola from its equation or defining geometric relationship.

OpenStax: **Primary resources:** [10.1 The Ellipse](#)

Supplemental resources: [10.2 The Hyperbola](#)

SUPPORTING SKILLS

- I** Classify a circle, ellipse, hyperbola, or parabola from an equation or geometric definition.
SK-CONIC-CLASSIFY
 - I** Use fixed-distance and focal-distance relationships to explain circles, ellipses, and hyperbolas.
SK-CONIC-DISTANCE-DEFINITION
 - I** Recall the defining terms and standard notation for circles, parabolas, ellipses, and hyperbolas.
SK-CONIC-VOCABULARY
-

M49-CON-002

I can use completing the square to rewrite a general-form conic equation in standard form.

OpenStax: **Primary resources:** [10.1 The Ellipse](#)

Supplemental resources: [10.2 The Hyperbola](#); [10.3 The Parabola](#)

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
 - I** Complete squares to convert a general conic equation to standard form.
SK-CONIC-GENERAL-STANDARD
 - D** Create a perfect-square trinomial by completing the square.
SK-QUAD-COMPLETE-SQUARE · first introduced in Math 21
-

M49-CON-003

I can write, interpret, and graph implicit and parametric equations of a circle.

OpenStax: **Primary resources:** [10.1 The Ellipse](#)

Supplemental resources: [8.6 Parametric Equations](#)

SUPPORTING SKILLS

- I** Connect implicit, standard, and parametric equations of a circle.
SK-CONIC-CIRCLE-EQUATIONS
- I** Determine and use a circle's center and radius.
SK-CONIC-CIRCLE-FEATURES
- A** Graph a parametric curve with its direction of motion.
SK-PAR-GRAPH · first introduced in Math 32
- A** Generate ordered pairs from parametric equations.
SK-PAR-TABLE · first introduced in Math 32

I can write, interpret, and graph implicit and parametric equations of an ellipse.

OpenStax: **Primary resources:** [10.1 The Ellipse](#)

Supplemental resources: [8.6 Parametric Equations](#)

SUPPORTING SKILLS

- I** Connect implicit, standard, and parametric equations of an ellipse.
SK-CONIC-ELLIPSE-EQUATIONS
- I** Determine and use an ellipse's center, axes, vertices, and foci.
SK-CONIC-ELLIPSE-FEATURES
- A** Graph a parametric curve with its direction of motion.
SK-PAR-GRAPH · first introduced in Math 32
- A** Generate ordered pairs from parametric equations.
SK-PAR-TABLE · first introduced in Math 32

I can write, interpret, and graph implicit and parametric equations of a hyperbola.

OpenStax: **Primary resources:** [10.2 The Hyperbola](#)

Supplemental resources: [8.6 Parametric Equations](#)

SUPPORTING SKILLS

- I** Connect implicit, standard, and parametric equations of a hyperbola.
SK-CONIC-HYPERBOLA-EQUATIONS
- I** Determine and use a hyperbola's center, axes, vertices, foci, and asymptotes.
SK-CONIC-HYPERBOLA-FEATURES
- A** Graph a parametric curve with its direction of motion.
SK-PAR-GRAPH · first introduced in Math 32
- A** Generate ordered pairs from parametric equations.
SK-PAR-TABLE · first introduced in Math 32

I can determine the center, vertices, foci, axes, or asymptotes needed to describe a circle, ellipse, or hyperbola.

OpenStax: **Primary resources:** [10.2 The Hyperbola](#)

Supplemental resources: [10.1 The Ellipse](#)

SUPPORTING SKILLS

- D** Determine and use a circle's center and radius.
SK-CONIC-CIRCLE-FEATURES · first introduced in Math 49
- A** Classify a circle, ellipse, hyperbola, or parabola from an equation or geometric definition.
SK-CONIC-CLASSIFY · first introduced in Math 49
- D** Determine and use an ellipse's center, axes, vertices, and foci.
SK-CONIC-ELLIPSE-FEATURES · first introduced in Math 49
- D** Determine and use a hyperbola's center, axes, vertices, foci, and asymptotes.
SK-CONIC-HYPERBOLA-FEATURES · first introduced in Math 49

M49-CON-007

I can explain the geometric definitions of circles, ellipses, and hyperbolas using foci and distance relationships.

OpenStax: Primary resources: [10.1 The Ellipse](#)

Supplemental resources: [10.2 The Hyperbola](#)

SUPPORTING SKILLS

- D** Use fixed-distance and focal-distance relationships to explain circles, ellipses, and hyperbolas.

SK-CONIC-DISTANCE-DEFINITION · first introduced in Math 49

- A** Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula.

SK-COORD-DISTANCE-PYTHAGOREAN · first introduced in Math 22

M49-CON-008

I can explain a parabola as the set of points equidistant from a focus and directrix.

OpenStax: Primary resources: [10.3 The Parabola](#)

SUPPORTING SKILLS

- I** Explain a parabola using its focus-directrix distance relationship.

SK-CONIC-PARABOLA-DEFINITION

- A** Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula.

SK-COORD-DISTANCE-PYTHAGOREAN · first introduced in Math 22